KINGSTON
REV 1.01
FAB G RETAIL

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RULES: (APPLIED WHEN POSSIBLE)
1. MSB TO LSB IS TOP TO BOTTOM
2. WHEN POSSIBLE: INPUTS ON LEFT, OUTPUTS ON RIGHT
3. ORDER OF PAGES=CHIP INTERFACES, TERMINATION, POWER, DECOUPLING
4. AVOID USING OFF PAGE CONNECTORS FOR ON PAGE CONNECTIONS
5. LANED SIGNALS ARE GROUPED ON SYMBOLS
6. TRANSIMITTER NAME USED AS PREFIX WITH RX AND TX CONNECTIONS
7. SUFFIX _V IS USED FOR VOLTAGE RAIL SIGNAL NAMES
8. SUFFIX _DP AND _DN ARE USED FOR DIFFERIENTAL PAIRS
9. UNNAMED NETS ARE NAMED WITH /2 TEXT SIZE
10. SUFFIX _N FOR ACTIVE LOW OR N JUNCTION
12. SUFFIX _P FOR P JUNCTION
13. SUFFIX _EN FOR ENABLE
14. 'CLK' FOR CLOCKS, 'RST' FOR RESETS
15. PWRGD FOR POWER GOOD
16. REV AND FAB ARE SET USING CUSTOM VARIABLES
    TOOLS>OPTIONS>VARIABLES

DRAWING
Mon Jan 18 12:03:57 2016

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	1/72	1/72	G	1.01

SOC:PCIEX,CLOCKS,VIDEO

NOTES:
1.TO SUPPORT A PCIE WIFI INTERFACE (J3C1), POPULATE C6P6 AND C6P7
2.SEE PAGE 32 FOR 100MHZ CLOCK ROUTING INFORMATION

SOC_BASE

U7E1 IC

PCIE
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36	IN	PEX_WIFI_SOC_TP	AR19	P_UMI_RX3_P	P_UMI_TX3_P	AP20	PEX_SOC_WIFI_TP	OUT	36
36	IN	PEX_WIFI_SOC_TN	AR20	P_UMI_RX3_N	P_UMI_TX3_N	AP21	PEX_SOC_WIFI_TN	OUT	36
32	IN	PEX_ENET_SOC_TP_C	AN21	P_UMI_RX2_P	P_UMI_TX2_P	AP23	PEX_SOC_ENET_TP	OUT	32
32	IN	PEX_ENET_SOC_TN_C	AN22	P_UMI_RX2_N	P_UMI_TX2_N	AP24	PEX_SOC_ENET_TN	OUT	32
25	IN	PEX_L1_SB_SOC_TP_C	AV22	P_UMI_RX1_P	P_UMI_TX1_P	AU23	PEX_L1_SOC_SB_TP	OUT	25
25	IN	PEX_L1_SB_SOC_TN_C	AV23	P_UMI_RX1_N	P_UMI_TX1_N	AU24	PEX_L1_SOC_SB_TN	OUT	25
25	IN	PEX_L0_SB_SOC_TP_C	AT24	P_UMI_RX0_P	P_UMI_TX0_P	AV25	PEX_L0_SOC_SB_TP	OUT	25
25	IN	PEX_L0_SB_SOC_TN_C	AT25	P_UMI_RX0_N	P_UMI_TX0_N	AV26	PEX_L0_SOC_SB_TN	OUT	25
					P_RX_ZVDD_095	AN19	RX_ZVDD_095		
					P_TX_ZVDD_095	AN18	TX_ZVDD_095		

C6P7 1 2 0.1 UF 10% 6.3 V EMPTY 402
C6P6 1 2 0.1 UF 10% 6.3 V EMPTY 402
C5T40 1 2 0.1 UF 10% 16 V XR5 402
C5T39 1 2 0.1 UF 10% 16 V XR5 402
C7E1 1 2 0.1 UF 10% 16 V XR5 402
C7E2 1 2 0.1 UF 10% 16 V XR5 402
C7E3 1 2 0.1 UF 10% 16 V XR5 402
C7E4 1 2 0.1 UF 10% 16 V XR5 402

R3T5 1 2 200 OHM 1% CH 402
R3T6 1 2 0 OHM 5% EMPTY 402
V_SOCPHY

V_SOCPHY
R4T21 1 2 510 OHM 5% CH 402
2 IN TEST25_N
USED BY AMD FOR SOC PLL CHARACTERIZATION
2 IN TEST25_P
R4T22 1 2 510 OHM 5% CH 402

U7E1 IC

VIDEO
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	BI	DP_CR_CLK	AP14	DP_CR_CLK
25	IN	DP_L0_SB_SOC_TP_C	AU18	DP1_RX0_P
25	IN	DP_L0_SB_SOC_TN_C	AU17	DP1_RX0_N
38	OUT	TMDS_TX_DP0	AU15	DP0_TX2_P
38	OUT	TMDS_TX_DN0	AU14	DP0_TX2_N
38	OUT	TMDS_TX_DP2	AV17	DP0_TX0_P
38	OUT	TMDS_TX_DN2	AV16	DP0_TX0_N
25	IN	DP_L1_SB_SOC_TP_C	AT19	DP1_RX1_P
25	IN	DP_L1_SB_SOC_TN_C	AT18	DP1_RX1_N
38	OUT	TMDS_TX_DP1	AT16	DP0_TX1_P
38	OUT	TMDS_TX_DN1	AT15	DP0_TX1_N
		DP_AUX_SOC_SB_TP	AT13	DP1_AUX_P
		DP_AUX_SOC_SB_TN	AT12	DP1_AUX_N
38	OUT	SOC_DDC_CLK	AR14	DP0_AUX_P
38	BI	SOC_DDC_DATA	AR13	DP0_AUX_N
38	OUT	TMDS_TX_CLKP	AV14	DP0_TX3_P
38	OUT	TMDS_TX_CLKN	AV13	DP0_TX3_N
			AP15	DP1_HPD
34	OUT	SPDIF_OUT	AP12	SPDIF_OUT
			AU12	DP0_HPD
		DP_TX_CALR	AM12	DP_TX_ZVSS
		DP_AUX_CALR	AR12	DP_AUX_ZVSS
		DP_RX_CALR	AN13	DP_RX_ZVSS

C3R25 1 2 0.1 UF 10% 16 V XR5 402
C3R23 1 2 0.1 UF 10% 16 V XR5 402

25 IN DP_AUX_SB_SOC_TP_C
25 IN DP_AUX_SB_SOC_TN_C
25 OUT DP1_HPD
38 IN DP0_HPD

R5D15 1 2 100 KOHM 1% CH 402
R3R6 1 2 0 OHM 5% EMPTY 402
R3R11 1 2 150 OHMS 1% CH 402
R3R13 1 2 200 OHM 1% CH 402

U7E1 IC

CLOCKS
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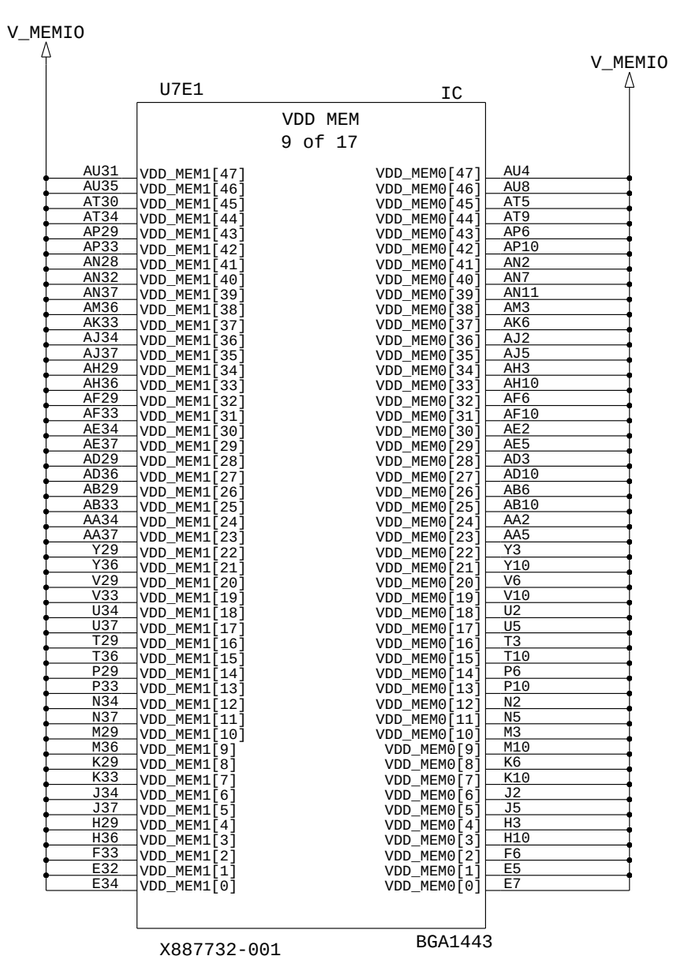
29	IN	SOC_NB_PEX_SS_100M_CLKP	AU26	CLKIN_NB_P
29	IN	SOC_NB_PEX_SS_100M_CLKN	AU27	CLKIN_NB_N
29	IN	SOC_SYNC_NS_100M_CLKP	AT21	DISP_CLKIN_P
29	IN	SOC_SYNC_NS_100M_CLKN	AT22	DISP_CLKIN_N
29	IN	SOC_AV_NS_100M_CLKP	AV19	AV_CLKIN_P
29	IN	SOC_AV_NS_100M_CLKN	AV20	AV_CLKIN_N
29	IN	SOC_PEX_SS_100M_CLKP	AU20	CLKIN_P
29	IN	SOC_PEX_SS_100M_CLKN	AU21	CLKIN_N
2	OUT	TEST25_P	AR23	TEST25_P
2	OUT	TEST25_N	AR22	TEST25_N

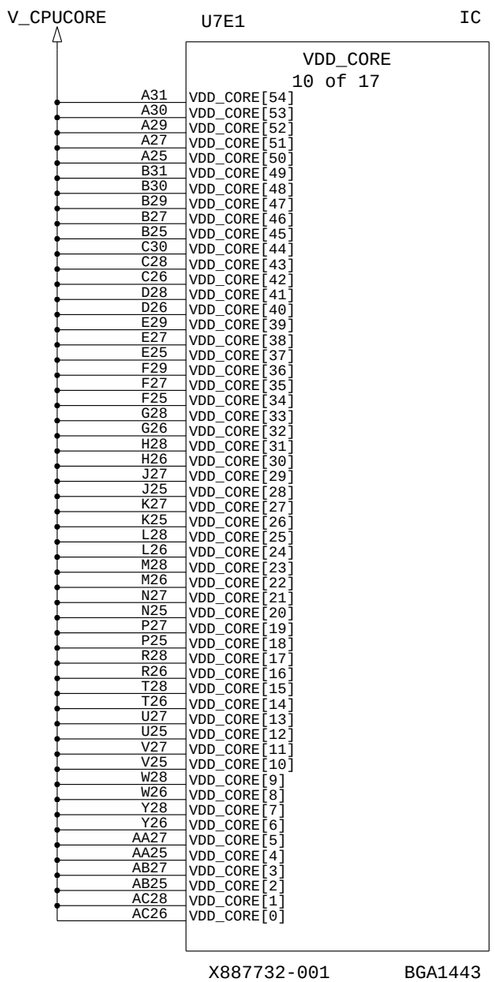
DVI PCB ROUTING ORDERING	DP PCB ROUTING ORDERING	PIN NAME
TMDS CLOCK -	DP LANE 3 -	DP0_TX3_N
TMDS CLOCK +	DP LANE 3 +	DP0_TX3_P
TMDS DATA0 -	DP LANE 2 -	DP0_TX2_N
TMDS DATA0 +	DP LANE 2 +	DP0_TX2_P
TMDS DATA1 -	DP LANE 1 -	DP0_TX1_N
TMDS DATA1 +	DP LANE 1 +	DP0_TX1_P
TMDS DATA2 -	DP LANE 0 -	DP0_TX0_N
TMDS DATA2 +	DP LANE 0 +	DP0_TX0_P

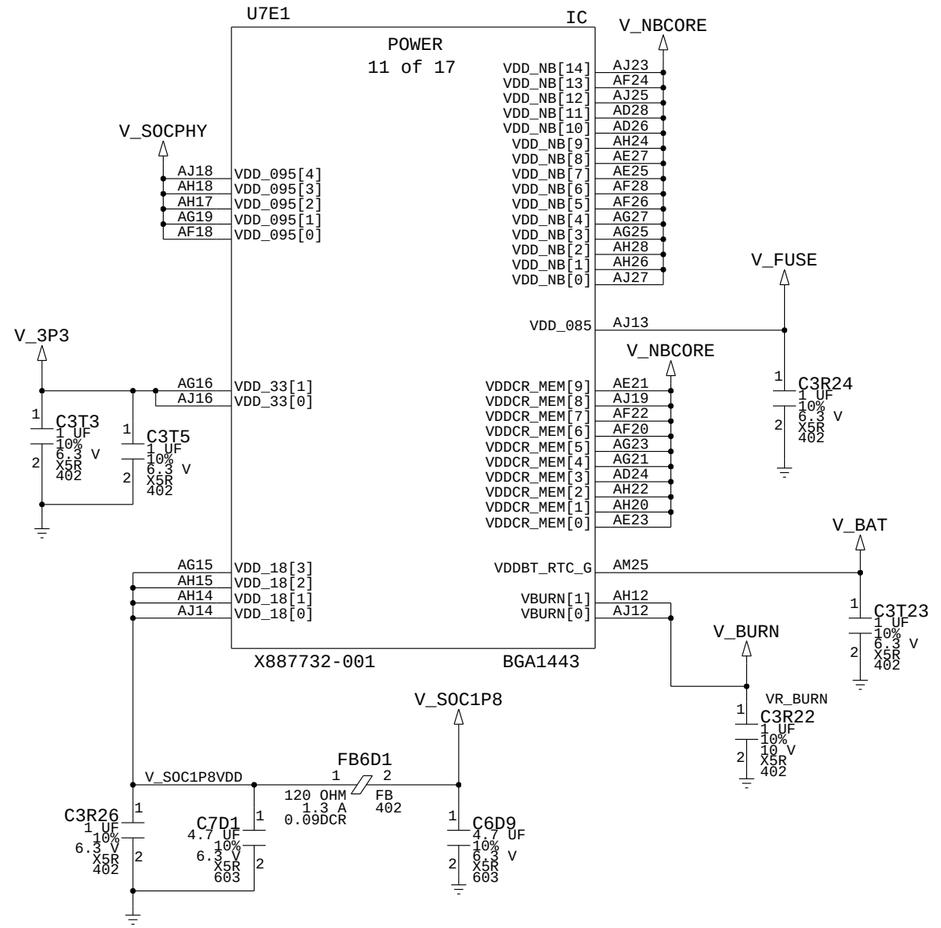
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X941293-001	IC	U7E1	ARLENE,TSMC,SHINKO,SPIL,AMD PENANG,TYPICAL,79W TDP+,A0,AKA ROUTE 1	ARLENE_SOC_R1
X941294-001	IC	U7E1	ARLENE,TSMC,IBIDEN,SPIL,AMD PENANG,TYPICAL,79W TDP+,A0,AKA ROUTE 2	ARLENE_SOC_R2
X941295-001	IC	U7E1	ARLENE,TSMC,UMTC,SPIL,AMD PENANG,TYPICAL,79W TDP+,A0,AKA ROUTE 3	ARLENE_SOC_R3
X941296-001	IC	U7E1	ARLENE,TSMC,NOT DEFINED,SPIL,AMD PENANG,FF,79W TDP+,A0	ARLENE_SOC_FF
X941297-001	IC	U7E1	ARLENE,TSMC,NOT DEFINED,SPIL,AMD PENANG,FS,79W TDP+,A0	ARLENE_SOC_FS
X941298-001	IC	U7E1	ARLENE,TSMC,NOT DEFINED,SPIL,AMD PENANG,SF,79W TDP+,A0	ARLENE_SOC_SF
X941299-001	IC	U7E1	ARLENE,TSMC,NOT DEFINED,SPIL,AMD PENANG,SS,79W TDP+,A0	ARLENE_SOC_SS

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	2/72	2/72	G	1.01

SOC: POWER: MEMIO,MEMCORE,CPUCORE,NBCORE,MISC







MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	3/72	3/72	G	1.01

SOC: POWER: V_GFXCORE,VSS

V_GFXCORE

U7E1 — IC

VDD_GFX
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U15	VDD_GFX[60]	VDD_GFX[120]	J23
U13	VDD_GFX[59]	VDD_GFX[119]	J21
U11	VDD_GFX[58]	VDD_GFX[118]	J19
V23	VDD_GFX[57]	VDD_GFX[117]	J17
V21	VDD_GFX[56]	VDD_GFX[116]	J15
V19	VDD_GFX[55]	VDD_GFX[115]	J13
V17	VDD_GFX[54]	VDD_GFX[114]	J11
V15	VDD_GFX[53]	VDD_GFX[113]	K23
V13	VDD_GFX[52]	VDD_GFX[112]	K21
V11	VDD_GFX[51]	VDD_GFX[111]	K19
W24	VDD_GFX[50]	VDD_GFX[110]	K17
W22	VDD_GFX[49]	VDD_GFX[109]	K15
W20	VDD_GFX[48]	VDD_GFX[108]	K13
W18	VDD_GFX[47]	VDD_GFX[107]	K11
W16	VDD_GFX[46]	VDD_GFX[106]	L24
W14	VDD_GFX[45]	VDD_GFX[105]	L22
W12	VDD_GFX[44]	VDD_GFX[104]	L20
Y24	VDD_GFX[43]	VDD_GFX[103]	L18
Y22	VDD_GFX[42]	VDD_GFX[102]	L16
Y20	VDD_GFX[41]	VDD_GFX[101]	L14
Y18	VDD_GFX[40]	VDD_GFX[100]	L12
Y16	VDD_GFX[39]	VDD_GFX[99]	M24
Y14	VDD_GFX[38]	VDD_GFX[98]	M22
Y12	VDD_GFX[37]	VDD_GFX[97]	M20
AA23	VDD_GFX[36]	VDD_GFX[96]	M18
AA21	VDD_GFX[35]	VDD_GFX[95]	M16
AA19	VDD_GFX[34]	VDD_GFX[94]	M14
AA17	VDD_GFX[33]	VDD_GFX[93]	M12
AA15	VDD_GFX[32]	VDD_GFX[92]	N23
AA13	VDD_GFX[31]	VDD_GFX[91]	N21
AA11	VDD_GFX[30]	VDD_GFX[90]	N19
AB23	VDD_GFX[29]	VDD_GFX[89]	N17
AB21	VDD_GFX[28]	VDD_GFX[88]	N15
AB19	VDD_GFX[27]	VDD_GFX[87]	N13
AB17	VDD_GFX[26]	VDD_GFX[86]	N11
AB15	VDD_GFX[25]	VDD_GFX[85]	P23
AB13	VDD_GFX[24]	VDD_GFX[84]	P21
AB11	VDD_GFX[23]	VDD_GFX[83]	P19
AC24	VDD_GFX[22]	VDD_GFX[82]	P17
AC22	VDD_GFX[21]	VDD_GFX[81]	P15
AC20	VDD_GFX[20]	VDD_GFX[80]	P13
AC18	VDD_GFX[19]	VDD_GFX[79]	P11
AC16	VDD_GFX[18]	VDD_GFX[78]	R24
AC14	VDD_GFX[17]	VDD_GFX[77]	R22
AC12	VDD_GFX[16]	VDD_GFX[76]	R20
AD22	VDD_GFX[15]	VDD_GFX[75]	R18
AD20	VDD_GFX[14]	VDD_GFX[74]	R16
AD18	VDD_GFX[13]	VDD_GFX[73]	R14
AD16	VDD_GFX[12]	VDD_GFX[72]	R12
AD14	VDD_GFX[11]	VDD_GFX[71]	T24
AD12	VDD_GFX[10]	VDD_GFX[70]	T22
AE17	VDD_GFX[9]	VDD_GFX[69]	T20
AE15	VDD_GFX[8]	VDD_GFX[68]	T18
AE13	VDD_GFX[7]	VDD_GFX[67]	T16
AE11	VDD_GFX[6]	VDD_GFX[66]	T14
AF17	VDD_GFX[5]	VDD_GFX[65]	T12
AF15	VDD_GFX[4]	VDD_GFX[64]	U23
AF13	VDD_GFX[3]	VDD_GFX[63]	U21
AF11	VDD_GFX[2]	VDD_GFX[62]	U19
AG14	VDD_GFX[1]	VDD_GFX[61]	U17
AG12	VDD_GFX[0]		

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V_GFXCORE

V_GFXCORE

U7E1 — IC

VDD_GFX
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A23	VDD_GFX[181]		
A21	VDD_GFX[180]		
A19	VDD_GFX[179]		
A17	VDD_GFX[178]		
A15	VDD_GFX[177]		
A13	VDD_GFX[176]		
A11	VDD_GFX[175]		
A10	VDD_GFX[174]		
A9	VDD_GFX[173]		
B23	VDD_GFX[172]		
B21	VDD_GFX[171]		
B19	VDD_GFX[170]		
B17	VDD_GFX[169]		
B15	VDD_GFX[168]		
B13	VDD_GFX[167]		
B11	VDD_GFX[166]		
B10	VDD_GFX[165]		
C24	VDD_GFX[164]		
C22	VDD_GFX[163]		
C20	VDD_GFX[162]		
C18	VDD_GFX[161]		
C16	VDD_GFX[160]		
C14	VDD_GFX[159]		
C12	VDD_GFX[158]		
C10	VDD_GFX[157]		
D24	VDD_GFX[156]		
D22	VDD_GFX[155]		
D20	VDD_GFX[154]		
D18	VDD_GFX[153]		
D16	VDD_GFX[152]		
D14	VDD_GFX[151]		
D12	VDD_GFX[150]		
D10	VDD_GFX[149]		
E23	VDD_GFX[148]		
E21	VDD_GFX[147]		
E19	VDD_GFX[146]		
E17	VDD_GFX[145]		
E15	VDD_GFX[144]		
E13	VDD_GFX[143]		
E11	VDD_GFX[142]		
F23	VDD_GFX[141]		
F21	VDD_GFX[140]		
F19	VDD_GFX[139]		
F17	VDD_GFX[138]		
F15	VDD_GFX[137]		
F13	VDD_GFX[136]		
F11	VDD_GFX[135]		
G24	VDD_GFX[134]		
G22	VDD_GFX[133]		
G20	VDD_GFX[132]		
G18	VDD_GFX[131]		
G16	VDD_GFX[130]		
G14	VDD_GFX[129]		
G12	VDD_GFX[128]		
H24	VDD_GFX[127]		
H22	VDD_GFX[126]		
H20	VDD_GFX[125]		
H18	VDD_GFX[124]		
H16	VDD_GFX[123]		
H14	VDD_GFX[122]		
H12	VDD_GFX[121]		

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U7E1 — IC

VSS
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AN30	VSS[60]	VSS[120]	AF8
AN26	VSS[59]	VSS[119]	AF3
AN23	VSS[58]	VSS[118]	AG37
AN20	VSS[57]	VSS[117]	AG34
AN17	VSS[56]	VSS[116]	AG30
AN14	VSS[55]	VSS[115]	AG28
AN12	VSS[54]	VSS[114]	AG26
AN9	VSS[53]	VSS[113]	AG24
AN5	VSS[52]	VSS[112]	AG22
AP36	VSS[51]	VSS[111]	AG20
AP35	VSS[50]	VSS[110]	AG18
AP31	VSS[49]	VSS[109]	AG9
AP25	VSS[48]	VSS[108]	AG5
AP22	VSS[47]	VSS[107]	AG2
AP19	VSS[46]	VSS[106]	AH33
AP16	VSS[45]	VSS[105]	AH31
AP13	VSS[44]	VSS[104]	AH27
AP8	VSS[43]	VSS[103]	AH25
AP4	VSS[42]	VSS[102]	AH23
AP3	VSS[41]	VSS[101]	AH21
AR37	VSS[40]	VSS[100]	AH19
AR27	VSS[39]	VSS[99]	AH16
AR24	VSS[38]	VSS[98]	AH13
AR21	VSS[37]	VSS[97]	AH8
AR18	VSS[36]	VSS[96]	AH6
AR15	VSS[35]	VSS[95]	AJ30
AR2	VSS[34]	VSS[94]	AJ26
AT36	VSS[33]	VSS[93]	AJ24
AT32	VSS[32]	VSS[92]	AJ20
AT28	VSS[31]	VSS[91]	AJ17
AT26	VSS[30]	VSS[90]	AJ15
AT23	VSS[29]	VSS[89]	AJ9
AT20	VSS[28]	VSS[88]	AK36
AT17	VSS[27]	VSS[87]	AK31
AT14	VSS[26]	VSS[86]	AK29
AT11	VSS[25]	VSS[85]	AK23
AT7	VSS[24]	VSS[84]	AK10
AT3	VSS[23]	VSS[83]	AK8
AU38	VSS[22]	VSS[82]	AK3
AU37	VSS[21]	VSS[81]	AL37
AU33	VSS[20]	VSS[80]	AL34
AU29	VSS[19]	VSS[79]	AL30
AU25	VSS[18]	VSS[78]	AL28
AU22	VSS[17]	VSS[77]	AL25
AU19	VSS[16]	VSS[76]	AL22
AU16	VSS[15]	VSS[75]	AL19
AU13	VSS[14]	VSS[74]	AL16
AU10	VSS[13]	VSS[73]	AL12
AU6	VSS[12]	VSS[72]	AL9
AU2	VSS[11]	VSS[71]	AL5
AU1	VSS[10]	VSS[70]	AL2
AV38	VSS[9]	VSS[69]	AM33
AV37	VSS[8]	VSS[68]	AM27
AV27	VSS[7]	VSS[67]	AM24
AV24	VSS[6]	VSS[66]	AM21
AV21	VSS[5]	VSS[65]	AM18
AV18	VSS[4]	VSS[64]	AM15
AV15	VSS[3]	VSS[63]	AM13
AV12	VSS[2]	VSS[62]	AM6
AV2	VSS[1]	VSS[61]	AN34
AV1	VSS[0]		

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VSS
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AB31	VSS[180]	VSS[240]	U22
AB28	VSS[179]	VSS[239]	U20
AB26	VSS[178]	VSS[238]	U18
AB24	VSS[177]	VSS[237]	U16
AB22	VSS[176]	VSS[236]	U14
AB20	VSS[175]	VSS[235]	U12
AB18	VSS[174]	VSS[234]	U9
AB16	VSS[173]	VSS[233]	V36
AB14	VSS[172]	VSS[232]	V31
AB12	VSS[171]	VSS[231]	V28
AB8	VSS[170]	VSS[230]	V26
AB3	VSS[169]	VSS[229]	V24
AC37	VSS[168]	VSS[228]	V22
AC34	VSS[167]	VSS[227]	V20
AC30	VSS[166]	VSS[226]	V18
AC27	VSS[165]	VSS[225]	V16
AC23	VSS[164]	VSS[224]	V14
AC21	VSS[163]	VSS[223]	V12
AC19	VSS[162]	VSS[222]	V8
AC17	VSS[161]	VSS[221]	V3
AC15	VSS[160]	VSS[220]	W37
AC13	VSS[159]	VSS[219]	W34
AC11	VSS[158]	VSS[218]	W30
AC9	VSS[157]	VSS[217]	W27
AC5	VSS[156]	VSS[216]	W25
AC2	VSS[155]	VSS[215]	W23
AD33	VSS[154]	VSS[214]	W21
AD31	VSS[153]	VSS[213]	W19
AD27	VSS[152]	VSS[212]	W17
AD25	VSS[151]	VSS[211]	W15
AD23	VSS[150]	VSS[210]	W13
AD21	VSS[149]	VSS[209]	W11
AD19	VSS[148]	VSS[208]	W9
AD17	VSS[147]	VSS[207]	W5
AD15	VSS[146]	VSS[206]	W2
AD13	VSS[145]	VSS[205]	Y33
AD11	VSS[144]	VSS[204]	Y31
AD8	VSS[143]	VSS[203]	Y27
AD6	VSS[142]	VSS[202]	Y25
AE30	VSS[141]	VSS[201]	Y23
AE28	VSS[140]	VSS[200]	Y21
AE26	VSS[139]	VSS[199]	Y19
AE24	VSS[138]	VSS[198]	Y17
AE22	VSS[137]	VSS[197]	Y15
AE20	VSS[136]	VSS[196]	Y13
AE18	VSS[135]	VSS[195]	Y11
AE16	VSS[134]	VSS[194]	Y8
AE14	VSS[133]	VSS[193]	Y6
AE12	VSS[132]	VSS[192]	AA30
AE9	VSS[131]	VSS[191]	AA28
AF36	VSS[130]	VSS[190]	AA26
AF31	VSS[129]	VSS[189]	AA24
AF27	VSS[128]	VSS[188]	AA22
AF25	VSS[127]	VSS[187]	AA20
AF23	VSS[126]	VSS[186]	AA18
AF21	VSS[125]	VSS[185]	AA16
AF19	VSS[124]	VSS[184]	AA14
AF16	VSS[123]	VSS[183]	AA12
AF14	VSS[122]	VSS[182]	AA9
AF12	VSS[121]	VSS[181]	AB36

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MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	4/72	4/72	G	1.01

SOC: MEMORY PARTITION C & D

U7E1 — IC — MEM CHANNEL D 6 of 17 — X887732-001 — BGA1443

MD_DQ63	P2	MD_DQ63
MD_DQ62	W1	MD_DQ62
MD_DQ61	N4	MD_DQ61
MD_DQ60	V2	MD_DQ60
MD_DQ59	N1	MD_DQ59
MD_DQ58	W3	MD_DQ58
MD_DQ57	P1	MD_DQ57
MD_DQ56	W4	MD_DQ56
MD_DQ55	Y2	MD_DQ55
MD_DQ54	M2	MD_DQ54
MD_DQ53	Y1	MD_DQ53
MD_DQ52	M1	MD_DQ52
MD_DQ51	T1	MD_DQ51
MD_DQ50	N3	MD_DQ50
MD_DQ49	U3	MD_DQ49
MD_DQ48	R4	MD_DQ48
MD_DQ47	M7	MD_DQ47
MD_DQ46	R6	MD_DQ46
MD_DQ45	M5	MD_DQ45
MD_DQ44	P7	MD_DQ44
MD_DQ43	M4	MD_DQ43
MD_DQ42	R8	MD_DQ42
MD_DQ41	N6	MD_DQ41
MD_DQ40	T4	MD_DQ40
MD_DQ39	L4	MD_DQ39
MD_DQ38	H1	MD_DQ38
MD_DQ37	L3	MD_DQ37
MD_DQ36	G4	MD_DQ36
MD_DQ35	L1	MD_DQ35
MD_DQ34	H2	MD_DQ34
MD_DQ33	K1	MD_DQ33
MD_DQ32	J1	MD_DQ32
MD_DQ31	H5	MD_DQ31
MD_DQ30	K5	MD_DQ30
MD_DQ29	H4	MD_DQ29
MD_DQ28	K4	MD_DQ28
MD_DQ27	G8	MD_DQ27
MD_DQ26	L6	MD_DQ26
MD_DQ25	H7	MD_DQ25
MD_DQ24	L8	MD_DQ24
MD_DQ23	G3	MD_DQ23
MD_DQ22	C1	MD_DQ22
MD_DQ21	G1	MD_DQ21
MD_DQ20	C3	MD_DQ20
MD_DQ19	F1	MD_DQ19
MD_DQ18	D2	MD_DQ18
MD_DQ17	E1	MD_DQ17
MD_DQ16	D1	MD_DQ16
MD_DQ15	C8	MD_DQ15
MD_DQ14	D5	MD_DQ14
MD_DQ13	E8	MD_DQ13
MD_DQ12	G6	MD_DQ12
MD_DQ11	D9	MD_DQ11
MD_DQ10	F5	MD_DQ10
MD_DQ9	B7	MD_DQ9
MD_DQ8	F4	MD_DQ8
MD_DQ7	A3	MD_DQ7
MD_DQ6	D6	MD_DQ6
MD_DQ5	B3	MD_DQ5
MD_DQ4	A7	MD_DQ4
MD_DQ3	D4	MD_DQ3
MD_DQ2	C6	MD_DQ2
MD_DQ1	C4	MD_DQ1
MD_DQ0	A6	MD_DQ0
MD_DQSL_0_P	B5	MD_DQSL_0_P
MD_DQSL_0_N	A5	MD_DQSL_0_N
MD_DQSL_1_P	E3	MD_DQSL_1_P
MD_DQSL_1_N	E4	MD_DQSL_1_N
MD_DQSL_2_P	J3	MD_DQSL_2_P
MD_DQSL_2_N	J4	MD_DQSL_2_N
MD_DQSL_3_P	R1	MD_DQSL_3_P
MD_DQSL_3_N	T2	MD_DQSL_3_N
MD_DQSU_0_P	E6	MD_DQSU_0_P
MD_DQSU_0_N	F7	MD_DQSU_0_N
MD_DQSU_1_P	J8	MD_DQSU_1_P
MD_DQSU_1_N	K7	MD_DQSU_1_N
MD_DQSU_2_P	P4	MD_DQSU_2_P
MD_DQSU_2_N	P5	MD_DQSU_2_N
MD_DQSU_3_P	U1	MD_DQSU_3_P
MD_DQSU_3_N	V1	MD_DQSU_3_N
MD_DML_0	A4	MD_DML_0
MD_DMU_0	D7	MD_DMU_0
MD_DML_1	F2	MD_DML_1
MD_DMU_1	J6	MD_DMU_1
MD_DML_2	K2	MD_DML_2
MD_DMU_2	N8	MD_DMU_2
MD_DML_3	U4	MD_DML_3
MD_DMU_3	R3	MD_DMU_3

V_MEMIO

R3R10 40.2 OHM 1% — 2 EMPTY 402

R3R5 40.2 OHM 1% — 2 EMPTY 402

R2R4 40.2 OHM 1% — 2 CH 402

R2R3 0 OHM 5% 402 EMPTY

MD_TEST	AG11	MD_TEST	
MD_CLK0_DP	K9	MD_CLK0_DP	OUT 12 13
MD_CLK0_DN	J10	MD_CLK0_DN	OUT 12 13
MD_A15	V7	MD_A<15>	B1 12 13 14
MD_A14	W10	MD_A<14>	B1 12 13 14
MD_A13	U10	MD_A<13>	B1 12 13 14
MD_A12	W6	MD_A<12>	B1 12 13 14
MD_A11	Y5	MD_A<11>	B1 12 13 14
MD_A10	N10	MD_A<10>	B1 12 13 14
MD_A9	V9	MD_A<9>	B1 12 13 14
MD_A8	W7	MD_A<8>	B1 12 13 14
MD_A7	U7	MD_A<7>	B1 12 13 14
MD_A6	Y7	MD_A<6>	B1 12 13 14
MD_A5	V4	MD_A<5>	B1 12 13 14
MD_A4	R10	MD_A<4>	B1 12 13 14
MD_A3	U8	MD_A<3>	B1 12 13 14
MD_A2	Y4	MD_A<2>	B1 12 13 14
MD_A1	W8	MD_A<1>	B1 12 13 14
MD_A0	T9	MD_A<0>	B1 12 13 14
MD_BA0	T7	MD_BA<0>	B1 12 13 14
MD_BA1	V5	MD_BA<1>	B1 12 13 14
MD_BA2	U6	MD_BA<2>	B1 12 13 14
MD_CKE0	N7	MD_CKE0	OUT 12 13 14
MD_CAS_N	L10	MD_CAS_N	OUT 12 13 14
MD_RAS_N	M9	MD_RAS_N	OUT 12 13 14
MD_CS0_N	R7	MD_CS0_N	OUT 12 13 14
MD_ODT0	T5	MD_ODT0	OUT 12 13 14
MD_WE_N	P9	MD_WE_N	OUT 12 13 14
MD_RESET	Y9	MD_RESET_N	OUT 12 13 14
MD_CKE1	H9		
MD_CS1_N	F9		
MD_ODT1	G10		
MD_VREFDQ	G7	MD_VREFDQ_MM	OUT 12 13
MD_ZVDDIO	J7	MD_ZVDDIO	

U7E1 — IC — MEM CHANNEL C 5 of 17 — X887732-001 — BGA1443

MC_DQ63	AN8	MC_DQ63
MC_DQ62	AM11	MC_DQ62
MC_DQ61	AM8	MC_DQ61
MC_DQ60	AN10	MC_DQ60
MC_DQ59	AR7	MC_DQ59
MC_DQ58	AP11	MC_DQ58
MC_DQ57	AP9	MC_DQ57
MC_DQ56	AR11	MC_DQ56
MC_DQ55	AV11	MC_DQ55
MC_DQ54	AR8	MC_DQ54
MC_DQ53	AU11	MC_DQ53
MC_DQ52	AV7	MC_DQ52
MC_DQ51	AV10	MC_DQ51
MC_DQ50	AT8	MC_DQ50
MC_DQ49	AR10	MC_DQ49
MC_DQ48	AV8	MC_DQ48
MC_DQ47	AT2	MC_DQ47
MC_DQ46	AN6	MC_DQ46
MC_DQ45	AP2	MC_DQ45
MC_DQ44	AN4	MC_DQ44
MC_DQ43	AT1	MC_DQ43
MC_DQ42	AM7	MC_DQ42
MC_DQ41	AN3	MC_DQ41
MC_DQ40	AP7	MC_DQ40
MC_DQ39	AU7	MC_DQ39
MC_DQ38	AV3	MC_DQ38
MC_DQ37	AV6	MC_DQ37
MC_DQ36	AU3	MC_DQ36
MC_DQ35	AT6	MC_DQ35
MC_DQ34	AR4	MC_DQ34
MC_DQ33	AV5	MC_DQ33
MC_DQ32	AT4	MC_DQ32
MC_DQ31	AJ8	MC_DQ31
MC_DQ30	AL6	MC_DQ30
MC_DQ29	AH4	MC_DQ29
MC_DQ28	AL8	MC_DQ28
MC_DQ27	AH5	MC_DQ27
MC_DQ26	AM5	MC_DQ26
MC_DQ25	AJ6	MC_DQ25
MC_DQ24	AM4	MC_DQ24
MC_DQ23	AR1	MC_DQ23
MC_DQ22	AK1	MC_DQ22
MC_DQ21	AP1	MC_DQ21
MC_DQ20	AJ4	MC_DQ20
MC_DQ19	AN1	MC_DQ19
MC_DQ18	AK2	MC_DQ18
MC_DQ17	AM1	MC_DQ17
MC_DQ16	AL1	MC_DQ16
MC_DQ15	AC1	MC_DQ15
MC_DQ14	AG4	MC_DQ14
MC_DQ13	AB2	MC_DQ13
MC_DQ12	AG3	MC_DQ12
MC_DQ11	AB1	MC_DQ11
MC_DQ10	AH1	MC_DQ10
MC_DQ9	AC3	MC_DQ9
MC_DQ8	AH2	MC_DQ8
MC_DQ7	AJ3	MC_DQ7
MC_DQ6	AA3	MC_DQ6
MC_DQ5	AJ1	MC_DQ5
MC_DQ4	AA1	MC_DQ4
MC_DQ3	AE3	MC_DQ3
MC_DQ2	AA4	MC_DQ2
MC_DQ1	AE4	MC_DQ1
MC_DQ0	AD1	MC_DQ0
MC_DQSL_0_P	AD2	MC_DQSL_0_P
MC_DQSL_0_N	AE1	MC_DQSL_0_N
MC_DQSL_1_P	AL3	MC_DQSL_1_P
MC_DQSL_1_N	AL4	MC_DQSL_1_N
MC_DQSL_2_P	AV4	MC_DQSL_2_P
MC_DQSL_2_N	AU5	MC_DQSL_2_N
MC_DQSL_3_P	AU9	MC_DQSL_3_P
MC_DQSL_3_N	AV9	MC_DQSL_3_N
MC_DQSU_0_P	AF2	MC_DQSU_0_P
MC_DQSU_0_N	AG1	MC_DQSU_0_N
MC_DQSU_1_P	AK4	MC_DQSU_1_P
MC_DQSU_1_N	AK5	MC_DQSU_1_N
MC_DQSU_2_P	AP5	MC_DQSU_2_P
MC_DQSU_2_N	AR5	MC_DQSU_2_N
MC_DQSU_3_P	AM9	MC_DQSU_3_P
MC_DQSU_3_N	AM10	MC_DQSU_3_N
MC_DML_0	AF1	MC_DML_0
MC_DMU_0	AC4	MC_DMU_0
MC_DML_1	AM2	MC_DML_1
MC_DMU_1	AK7	MC_DMU_1
MC_DML_2	AR6	MC_DML_2
MC_DMU_2	AR3	MC_DMU_2
MC_DML_3	AT10	MC_DML_3
MC_DMU_3	AR9	MC_DMU_3

V_MEMIO

R3R9 40.2 OHM 1% — 2 EMPTY 402

R3R4 40.2 OHM 1% — 2 EMPTY 402

R3R7 40.2 OHM 1% — 2 CH 402

R3R1 0 OHM 5% 402 EMPTY

MC_TEST	AJ11	MC_TEST	
MC_CLK0_DP	AK9	MC_CLK0_DP	OUT 15 16
MC_CLK0_DN	AJ10	MC_CLK0_DN	OUT 15 16
MC_A15	AC8	MC_A<15>	B1 15 16 17
MC_A14	AA7	MC_A<14>	B1 15 16 17
MC_A13	AA10	MC_A<13>	B1 15 16 17
MC_A12	AC6	MC_A<12>	B1 15 16 17
MC_A11	AB4	MC_A<11>	B1 15 16 17
MC_A10	AE10	MC_A<10>	B1 15 16 17
MC_A9	AC7	MC_A<9>	B1 15 16 17
MC_A8	AB9	MC_A<8>	B1 15 16 17
MC_A7	AD9	MC_A<7>	B1 15 16 17
MC_A6	AA8	MC_A<6>	B1 15 16 17
MC_A5	AD5	MC_A<5>	B1 15 16 17
MC_A4	AC10	MC_A<4>	B1 15 16 17
MC_A3	AD7	MC_A<3>	B1 15 16 17
MC_A2	AB5	MC_A<2>	B1 15 16 17
MC_A1	AB7	MC_A<1>	B1 15 16 17
MC_A0	AE7	MC_A<0>	B1 15 16 17
MC_BA0	AE8	MC_BA<0>	B1 15 16 17
MC_BA1	AD4	MC_BA<1>	B1 15 16 17
MC_BA2	AE6	MC_BA<2>	B1 15 16 17
MC_CKE0	AH9	MC_CKE0	OUT 15 16 17
MC_CAS_N	AG10	MC_CAS_N	OUT 15 16 17
MC_RAS_N	AJ7	MC_RAS_N	OUT 15 16 17
MC_CS0_N	AF9	MC_CS0_N	OUT 15 16 17
MC_ODT0	AF4	MC_ODT0	OUT 15 16 17
MC_WE_N	AG7	MC_WE_N	OUT 15 16 17
MC_RESET	AA6	MC_RESET_N	OUT 15 16 17
MC_CKE1	AH7		
MC_CS1_N	AF7		
MC_ODT1	AG8		
MC_VREFDQ	AL7	MC_VREFDQ_MM	OUT 15 16
MC_ZVDDIO	AL10	MC_ZVDDIO	

MICROSOFT CONFIDENTIAL	PROJECT NAME Kingston	PAGE 5/72	CSA PAGE 5/72	FAB G	VER 1.01

SOC: MEMORY PARTITION A & B

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MB_DQ63	V37	MB_DQ63	BI
MB_DQ62	N38	MB_DQ62	BI
MB_DQ61	W38	MB_DQ61	BI
MB_DQ60	P37	MB_DQ60	BI
MB_DQ59	W36	MB_DQ59	BI
MB_DQ58	N35	MB_DQ58	BI
MB_DQ57	V38	MB_DQ57	BI
MB_DQ56	N36	MB_DQ56	BI
MB_DQ55	M37	MB_DQ55	BI
MB_DQ54	Y38	MB_DQ54	BI
MB_DQ53	M38	MB_DQ53	BI
MB_DQ52	Y37	MB_DQ52	BI
MB_DQ51	T38	MB_DQ51	BI
MB_DQ50	W35	MB_DQ50	BI
MB_DQ49	R35	MB_DQ49	BI
MB_DQ48	U36	MB_DQ48	BI
MB_DQ47	R33	MB_DQ47	BI
MB_DQ46	M32	MB_DQ46	BI
MB_DQ45	R31	MB_DQ45	BI
MB_DQ44	N33	MB_DQ44	BI
MB_DQ43	T35	MB_DQ43	BI
MB_DQ42	M34	MB_DQ42	BI
MB_DQ41	P32	MB_DQ41	BI
MB_DQ40	M35	MB_DQ40	BI
MB_DQ39	G35	MB_DQ39	BI
MB_DQ38	L38	MB_DQ38	BI
MB_DQ37	H37	MB_DQ37	BI
MB_DQ36	L35	MB_DQ36	BI
MB_DQ35	H38	MB_DQ35	BI
MB_DQ34	L36	MB_DQ34	BI
MB_DQ33	J35	MB_DQ33	BI
MB_DQ32	K37	MB_DQ32	BI
MB_DQ31	K34	MB_DQ31	BI
MB_DQ30	H34	MB_DQ30	BI
MB_DQ29	L33	MB_DQ29	BI
MB_DQ28	H32	MB_DQ28	BI
MB_DQ27	L31	MB_DQ27	BI
MB_DQ26	H35	MB_DQ26	BI
MB_DQ25	K35	MB_DQ25	BI
MB_DQ24	G31	MB_DQ24	BI
MB_DQ23	C36	MB_DQ23	BI
MB_DQ22	G36	MB_DQ22	BI
MB_DQ21	C38	MB_DQ21	BI
MB_DQ20	G38	MB_DQ20	BI
MB_DQ19	E35	MB_DQ19	BI
MB_DQ18	F38	MB_DQ18	BI
MB_DQ17	D38	MB_DQ17	BI
MB_DQ16	E38	MB_DQ16	BI
MB_DQ15	D34	MB_DQ15	BI
MB_DQ14	C31	MB_DQ14	BI
MB_DQ13	F34	MB_DQ13	BI
MB_DQ12	B32	MB_DQ12	BI
MB_DQ11	F35	MB_DQ11	BI
MB_DQ10	E31	MB_DQ10	BI
MB_DQ9	G33	MB_DQ9	BI
MB_DQ8	D30	MB_DQ8	BI
MB_DQ7	A32	MB_DQ7	BI
MB_DQ6	B36	MB_DQ6	BI
MB_DQ5	D33	MB_DQ5	BI
MB_DQ4	A36	MB_DQ4	BI
MB_DQ3	D35	MB_DQ3	BI
MB_DQ2	A35	MB_DQ2	BI
MB_DQ1	A33	MB_DQ1	BI
MB_DQ0	C35	MB_DQ0	BI
MB_DQSL_0_P	A34	MB_DQSL_0_P	BI
MB_DQSL_0_N	B34	MB_DQSL_0_N	BI
MB_DQSL_1_P	F37	MB_DQSL_1_P	BI
MB_DQSL_1_N	E36	MB_DQSL_1_N	BI
MB_DQSL_2_P	K38	MB_DQSL_2_P	BI
MB_DQSL_2_N	J38	MB_DQSL_2_N	BI
MB_DQSL_3_P	U38	MB_DQSL_3_P	BI
MB_DQSL_3_N	T37	MB_DQSL_3_N	BI
MB_DQSU_0_P	E33	MB_DQSU_0_P	BI
MB_DQSU_0_N	D32	MB_DQSU_0_N	BI
MB_DQSU_1_P	J31	MB_DQSU_1_P	BI
MB_DQSU_1_N	J33	MB_DQSU_1_N	BI
MB_DQSU_2_P	P35	MB_DQSU_2_P	BI
MB_DQSU_2_N	P34	MB_DQSU_2_N	BI
MB_DQSU_3_P	R38	MB_DQSU_3_P	BI
MB_DQSU_3_N	P38	MB_DQSU_3_N	BI
MB_DML_0	C33	MB_DML_0	BI
MB_DMU_0	F32	MB_DMU_0	BI
MB_DML_1	D37	MB_DML_1	BI
MB_DMU_1	K32	MB_DMU_1	BI
MB_DML_2	J36	MB_DML_2	BI
MB_DMU_2	N31	MB_DMU_2	BI
MB_DML_3	R36	MB_DML_3	BI
MB_DMU_3	U35	MB_DMU_3	BI

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V_MEMIO
R3T15 40.2 OHM 1% 2 EMPTY 402
MB_TEST AJ28 MB_TEST
R3T16 40.2 OHM 1% 2 EMPTY 402

MB_CLK0_DP	K30	MB_CLK0_DP	OUT 18 19
MB_CLK0_DN	J29	MB_CLK0_DN	OUT 18 19
MB_A15	V32	MB_A<15>	BI 18 19 20
MB_A14	W29	MB_A<14>	BI 18 19 20
MB_A13	U29	MB_A<13>	BI 18 19 20
MB_A12	W33	MB_A<12>	BI 18 19 20
MB_A11	Y34	MB_A<11>	BI 18 19 20
MB_A10	N29	MB_A<10>	BI 18 19 20
MB_A9	V30	MB_A<9>	BI 18 19 20
MB_A8	W32	MB_A<8>	BI 18 19 20
MB_A7	U32	MB_A<7>	BI 18 19 20
MB_A6	Y32	MB_A<6>	BI 18 19 20
MB_A5	V35	MB_A<5>	BI 18 19 20
MB_A4	R29	MB_A<4>	BI 18 19 20
MB_A3	U31	MB_A<3>	BI 18 19 20
MB_A2	Y35	MB_A<2>	BI 18 19 20
MB_A1	W31	MB_A<1>	BI 18 19 20
MB_A0	T30	MB_A<0>	BI 18 19 20
MB_BA0	T32	MB_BA<0>	BI 18 19 20
MB_BA1	V34	MB_BA<1>	BI 18 19 20
MB_BA2	U33	MB_BA<2>	BI 18 19 20
MB_CKE0	N32	MB_CKE0	OUT 18 19 20
MB_CAS_N	L29	MB_CAS_N	OUT 18 19 20
MB_RAS_N	M30	MB_RAS_N	OUT 18 19 20
MB_CS0_N	R32	MB_CS0_N	OUT 18 19 20
MB_ODT0	T34	MB_ODT0	OUT 18 19 20
MB_WE_N	P30	MB_WE_N	OUT 18 19 20
MB_RESET	Y30	MB_RESET_N	OUT 18 19 20
MB_CKE1	H30		
MB_CS1_N	F30		
MB_ODT1	G29		

V_MEMIO
R2T1 40.2 OHM 1% CH 402
R2T2 MB_VREFDQ 1 2 MB_VREFDQ_MM OUT 18 19 0 OHM 5% 402 EMPTY
MB_VREFDQ G32
MB_ZVDDIO J32 MB_ZVDDIO

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MA_DQ63	AM28	MA_DQ63	BI
MA_DQ62	AN31	MA_DQ62	BI
MA_DQ61	AP28	MA_DQ61	BI
MA_DQ60	AM30	MA_DQ60	BI
MA_DQ59	AR28	MA_DQ59	BI
MA_DQ58	AM31	MA_DQ58	BI
MA_DQ57	AN29	MA_DQ57	BI
MA_DQ56	AR32	MA_DQ56	BI
MA_DQ55	AV32	MA_DQ55	BI
MA_DQ54	AU28	MA_DQ54	BI
MA_DQ53	AR31	MA_DQ53	BI
MA_DQ52	AV28	MA_DQ52	BI
MA_DQ51	AT31	MA_DQ51	BI
MA_DQ50	AV29	MA_DQ50	BI
MA_DQ49	AR29	MA_DQ49	BI
MA_DQ48	AT29	MA_DQ48	BI
MA_DQ47	AN33	MA_DQ47	BI
MA_DQ46	AT37	MA_DQ46	BI
MA_DQ45	AM32	MA_DQ45	BI
MA_DQ44	AR36	MA_DQ44	BI
MA_DQ43	AP32	MA_DQ43	BI
MA_DQ42	AP37	MA_DQ42	BI
MA_DQ41	AR34	MA_DQ41	BI
MA_DQ40	AT38	MA_DQ40	BI
MA_DQ39	AU36	MA_DQ39	BI
MA_DQ38	AV33	MA_DQ38	BI
MA_DQ37	AV36	MA_DQ37	BI
MA_DQ36	AU32	MA_DQ36	BI
MA_DQ35	AR35	MA_DQ35	BI
MA_DQ34	AT33	MA_DQ34	BI
MA_DQ33	AV35	MA_DQ33	BI
MA_DQ32	AR33	MA_DQ32	BI
MA_DQ31	AL33	MA_DQ31	BI
MA_DQ30	AJ31	MA_DQ30	BI
MA_DQ29	AM34	MA_DQ29	BI
MA_DQ28	AJ33	MA_DQ28	BI
MA_DQ27	AM35	MA_DQ27	BI
MA_DQ26	AH35	MA_DQ26	BI
MA_DQ25	AL31	MA_DQ25	BI
MA_DQ24	AH34	MA_DQ24	BI
MA_DQ23	AJ35	MA_DQ23	BI
MA_DQ22	AP38	MA_DQ22	BI
MA_DQ21	AK38	MA_DQ21	BI
MA_DQ20	AR38	MA_DQ20	BI
MA_DQ19	AK37	MA_DQ19	BI
MA_DQ18	AM37	MA_DQ18	BI
MA_DQ17	AM38	MA_DQ17	BI
MA_DQ16	AN38	MA_DQ16	BI
MA_DQ15	AG36	MA_DQ15	BI
MA_DQ14	AB37	MA_DQ14	BI
MA_DQ13	AG35	MA_DQ13	BI
MA_DQ12	AC38	MA_DQ12	BI
MA_DQ11	AH38	MA_DQ11	BI
MA_DQ10	AB38	MA_DQ10	BI
MA_DQ9	AG38	MA_DQ9	BI
MA_DQ8	AA35	MA_DQ8	BI
MA_DQ7	AA38	MA_DQ7	BI
MA_DQ6	AJ38	MA_DQ6	BI
MA_DQ5	AA36	MA_DQ5	BI
MA_DQ4	AJ36	MA_DQ4	BI
MA_DQ3	AE38	MA_DQ3	BI
MA_DQ2	AH37	MA_DQ2	BI
MA_DQ1	AD37	MA_DQ1	BI
MA_DQ0	AF38	MA_DQ0	BI
MA_DQSL_0_P	AE35	MA_DQSL_0_P	BI
MA_DQSL_0_N	AE36	MA_DQSL_0_N	BI
MA_DQSL_1_P	AL35	MA_DQSL_1_P	BI
MA_DQSL_1_N	AL36	MA_DQSL_1_N	BI
MA_DQSL_2_P	AV34	MA_DQSL_2_P	BI
MA_DQSL_2_N	AU34	MA_DQSL_2_N	BI
MA_DQSL_3_P	AU30	MA_DQSL_3_P	BI
MA_DQSL_3_N	AV30	MA_DQSL_3_N	BI
MA_DQSU_0_P	AC35	MA_DQSU_0_P	BI
MA_DQSU_0_N	AC36	MA_DQSU_0_N	BI
MA_DQSU_1_P	AK34	MA_DQSU_1_P	BI
MA_DQSU_1_N	AK35	MA_DQSU_1_N	BI
MA_DQSU_2_P	AN35	MA_DQSU_2_P	BI
MA_DQSU_2_N	AN36	MA_DQSU_2_N	BI
MA_DQSU_3_P	AR30	MA_DQSU_3_P	BI
MA_DQSU_3_N	AP30	MA_DQSU_3_N	BI
MA_DML_0	AD38	MA_DML_0	BI
MA_DMU_0	AF37	MA_DMU_0	BI
MA_DML_1	AL38	MA_DML_1	BI
MA_DMU_1	AK32	MA_DMU_1	BI
MA_DML_2	AT35	MA_DML_2	BI
MA_DMU_2	AP34	MA_DMU_2	BI
MA_DML_3	AV31	MA_DML_3	BI
MA_DMU_3	AM29	MA_DMU_3	BI

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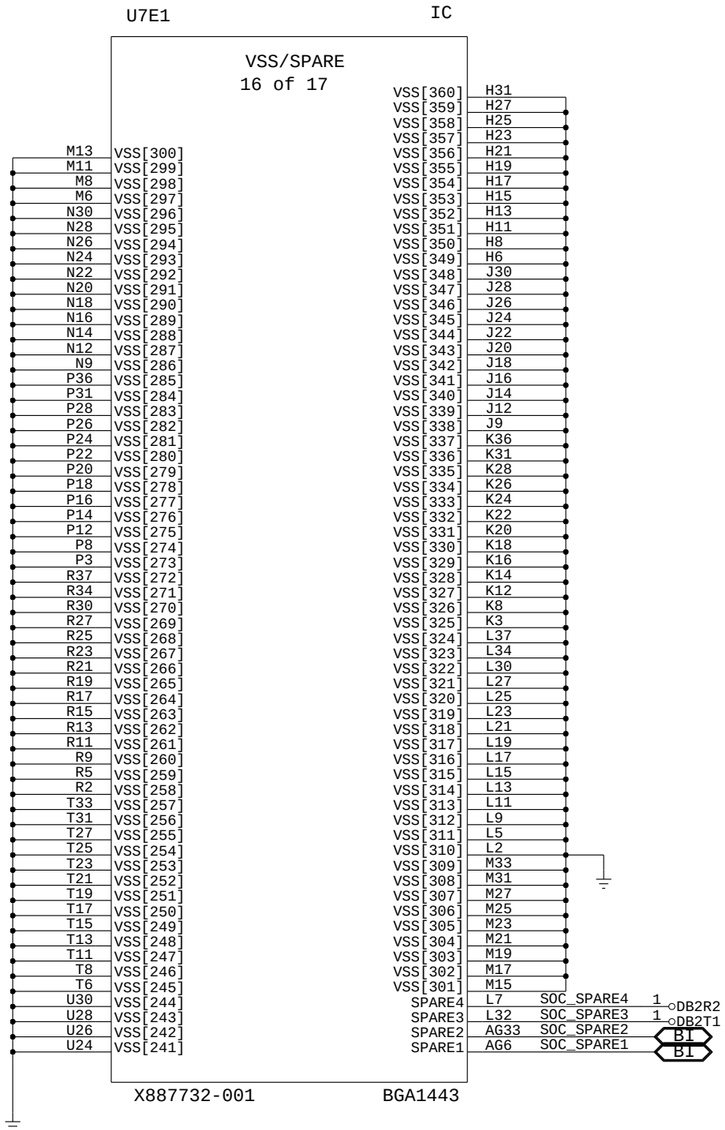
V_MEMIO
R3T14 40.2 OHM 1% 2 EMPTY 402
MA_TEST AK28 MA_TEST
R3T13 40.2 OHM 1% 2 EMPTY 402

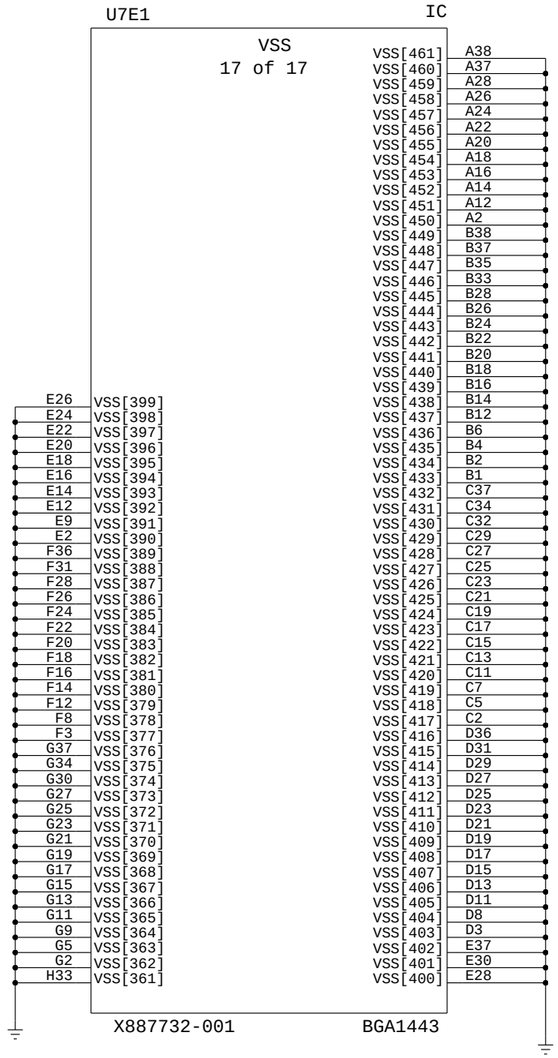
MA_CLK0_DP	AK30	MA_CLK0_DP	OUT 21 22
MA_CLK0_DN	AJ29	MA_CLK0_DN	OUT 21 22
MA_A15	AC31	MA_A<15>	BI 21 22 23
MA_A14	AA32	MA_A<14>	BI 21 22 23
MA_A13	AA29	MA_A<13>	BI 21 22 23
MA_A12	AC33	MA_A<12>	BI 21 22 23
MA_A11	AB35	MA_A<11>	BI 21 22 23
MA_A10	AE29	MA_A<10>	BI 21 22 23
MA_A9	AC32	MA_A<9>	BI 21 22 23
MA_A8	AB30	MA_A<8>	BI 21 22 23
MA_A7	AD30	MA_A<7>	BI 21 22 23
MA_A6	AA31	MA_A<6>	BI 21 22 23
MA_A5	AD34	MA_A<5>	BI 21 22 23
MA_A4	AC29	MA_A<4>	BI 21 22 23
MA_A3	AD32	MA_A<3>	BI 21 22 23
MA_A2	AB34	MA_A<2>	BI 21 22 23
MA_A1	AB32	MA_A<1>	BI 21 22 23
MA_A0	AE32	MA_A<0>	BI 21 22 23
MA_BA0	AE31	MA_BA<0>	BI 21 22 23
MA_BA1	AD35	MA_BA<1>	BI 21 22 23
MA_BA2	AE33	MA_BA<2>	BI 21 22 23
MA_CKE0	AH30	MA_CKE0	OUT 21 22 23
MA_CAS_N	AG29	MA_CAS_N	OUT 21 22 23
MA_RAS_N	AJ32	MA_RAS_N	OUT 21 22 23
MA_CS0_N	AF30	MA_CS0_N	OUT 21 22 23
MA_ODT0	AF35	MA_ODT0	OUT 21 22 23
MA_WE_N	AG32	MA_WE_N	OUT 21 22 23
MA_RESET	AA33	MA_RESET_N	OUT 21 22 23
MA_CKE1	AH32		
MA_CS1_N	AF32		
MA_ODT1	AG31		

V_MEMIO
R3T11 40.2 OHM 1% 2 CH 402
R3T12 MA_VREFDQ 1 2 MA_VREFDQ_MM OUT 21 22 0 OHM 5% 402 EMPTY
MA_VREFDQ AL32
MA_ZVDDIO AL29 MA_ZVDDIO

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	6/72	6/72	G	1.01

SOC: VSS,SPARE





MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	7/72	7/72	G	1.01

SOC:DEBUG,SB SIGNALS

V_SOC1P8
R4T1 10 KOHM 1% 2 EMPTY 402
R3T17 10 KOHM 1% 2 EMPTY 402
A0 BYPASS FOR ARLENE A0/A1 BRING UP
NO SECURITY RISK FOR B0

U7E1 IC
JTAG/DEBUG
4 of 17

RTC_RETAIL R4T2 0 OHM 5% 402 CH
RTC_DEBUG R4T3 20 MOHM 5% 402 CH
RTC_DEBUG Y4T1 SM XTAL 32 KHZ
RTC_DEBUG C4T1 15 PF 5% 50 V NPO 402
RTC_DEBUG C4T3 15 PF 5% 50 V NPO 402

	ATE_TSTCLK_EN	AK11	ATE_TSTCLK_EN	
		AL24	DLY_PSP_RESET	
	A0_BYPASS	AK16	A0_BYPASS	
26 OUT	SOC_THERMTRIP_N	AN15	THERMTRIP*	
29 IN X32K_X2_R	X32K_X2	AM23	X32K_X2	
	X32K_X1	AN24	X32K_X1	
	GPIO_3P3_3	AK19	NEW_GPIO13	
	GPIO_3P3_2	AN25	NEW_GPIO12	
	GPIO_3P3_1	AK21	NEW_GPIO11	
68	VREG_BURN_EN	AK20	NEW_GPIO10	
	GPIO_1P8_3	AL21	NEW_GPIO9	
	GPIO_1P8_2	AL20	NEW_GPIO8	
	GPIO_1P8_1	AM19	NEW_GPIO7	
	GPIO_1P8_0	AM17	NEW_GPIO6	
	GPIO_POSTOUT_5	AM26	NEW_GPIO5	
	GPIO_POSTOUT_4	AR25	NEW_GPIO4	
	GPIO_POSTOUT_3	AP27	NEW_GPIO3	
	GPIO_POSTOUT_2	AP26	NEW_GPIO2	
	GPIO_POSTOUT_1	AR26	NEW_GPIO1	
	GPIO_POSTOUT_0	AT27	NEW_GPIO0	
67 IN	DBREQ_L	AK26	DBREQ_N	
67 OUT	DBRDY	AL26	DBRDY	
67 IN	TRST_L	AN27	TRST_N	
67 26 IN	TMS	AK24	TMS	
OUT	TDO	AK25	TDO	
67 26 IN	TDI	AK22	TDI	
67 26 IN	TCK	AL23	TCK	
67 26 IN	SOC_RST_N	AR17	RESET_N	
67 44 26 IN	SOC_PWR_OK	AR16	PWROK	
	SOC_TEST32_P	AL18	TEST32_P	
	SOC_TEST32_N	AK18	TEST32_N	
	SOC_TEST30_P	AK17	TEST30_P	
	SOC_TEST30_N	AL17	TEST30_N	
	SOC_TEST28_P	AL15	TEST28_P	
	SOC_TEST28_N	AK15	TEST28_N	
67 OUT	TEST19	AJ21	TEST19	
67 OUT	TEST18	AL27	TEST18	
	TEST17	AL11	TEST17	
	TEST16	AK13	TEST16	
	TEST15	AL14	TEST15	
	TEST14	AM16	TEST14	
	TEST10	AK12	TEST10	
	TEST9	AM14	TEST9	
		AJ22	TEST6	
	SOC_TEST5	AK14	TEST5	
	SOC_TEST4	AL13	TEST4	

DEBUG R5T24 1 KOHM 5% CH 402
DEBUG R5T25 1 KOHM 5% CH 402
DEBUG R3R12 0 OHM 5% EMPTY 402

V_SOC1P8
R3T8 10 KOHM 5% 2 EMPTY 402
R3T7 10 KOHM 1% 2 CH 402

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TMON_CAL1	AM22	TMON_CAL1 1	FTP	FT3T2
TMON_CAL0	AM20	TMON_CAL0 1	FTP	FT3T1
PSEN	AG13			
ALERT_N	AN16	SP_SMC_INT_N	OUT 26	
SVT	E10	SVT	IN 44	
SVD	C9	SVD	BI 44	
SVC	F10	SVC	OUT 44	
SID	AP18	SID		
SIC	AP17	SIC		
RTCCLK	AK27	RTCCLK	BI	
M_VREF	AH11	M_VREF	IN 8	
VDDCR_MEM_SENSE	AE19	SOC_MEMCORE_S	BI	
VDD_MEM0_SENSE	AF5	SOC_MEMIO_S0	BI	
VDD_MEM1_SENSE	AF34	SOC_MEMIO_S1	DB2R4 1 BI	
VSS_GFX_SENSE	B8	SOC_GFXCPU_S_RTN	OUT 44	
VDD_GFX_SENSE	B9	SOC_GFXCORE_S	OUT 45	
VDD_NB_SENSE	AC25	SOC_NBCORE_S	DB2R3 1 BI	
			DB2R1	
VDD_CORE_SENSE	A8	SOC_CPUCORE_S	OUT 44	
VDD_095_SENSE	AG17			

V_SOCPHY

V_1P8STBY
DEBUG R3T4 475 OHM 1% 2 EMPTY 402
R3T2 1% 243 OHM CH 402
SB_SOC_DATA BI 26
DEBUG C3T7 47 PF 5% 50 V EMPTY 402
V_1P8STBY
DEBUG R3T3 475 OHM 1% 2 EMPTY 402
R3T1 1% 243 OHM CH 402
SB_SOC_CLK IN 26
DEBUG C3T4 47 PF 5% 50 V EMPTY 402

DB2R6 1

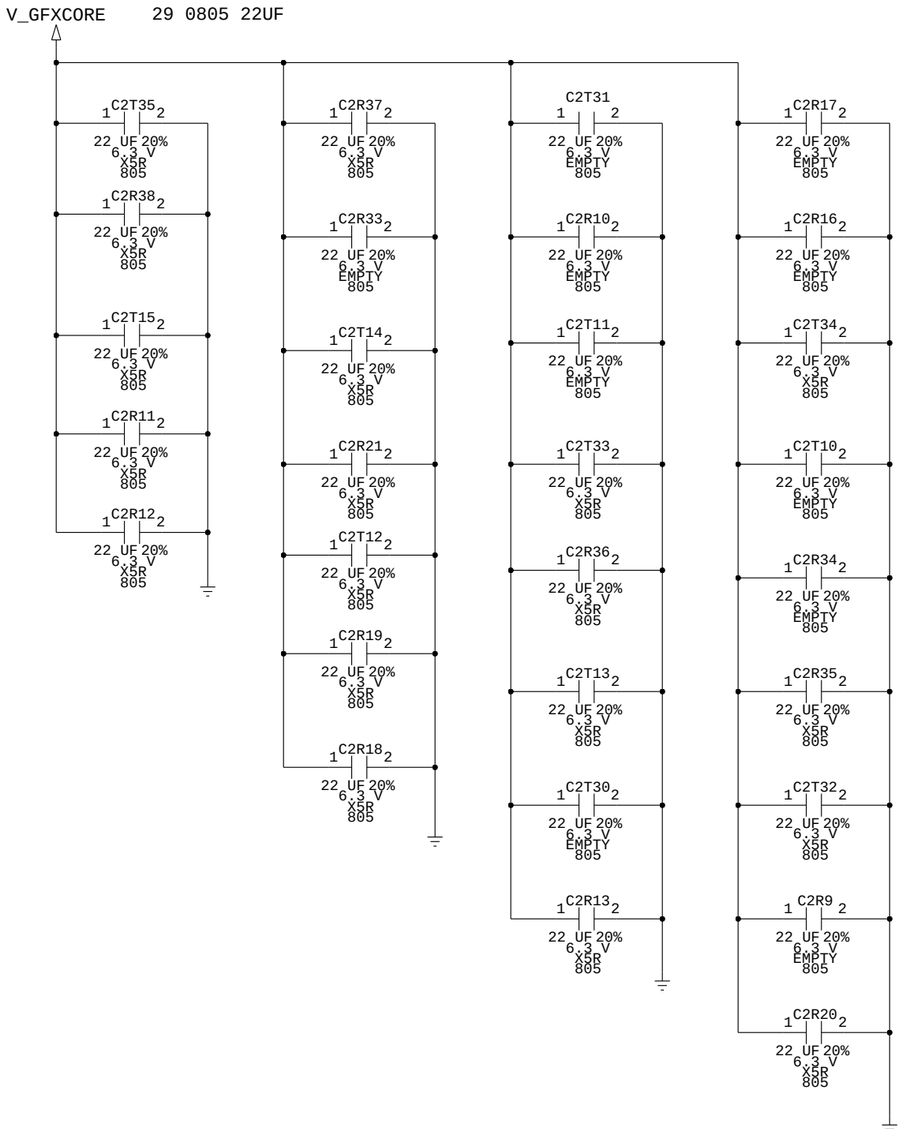
V_CPUCORE
DB2T2 1
V_GFXCORE
DB2R5 1
SENSE CONNECTION FOR VRTB

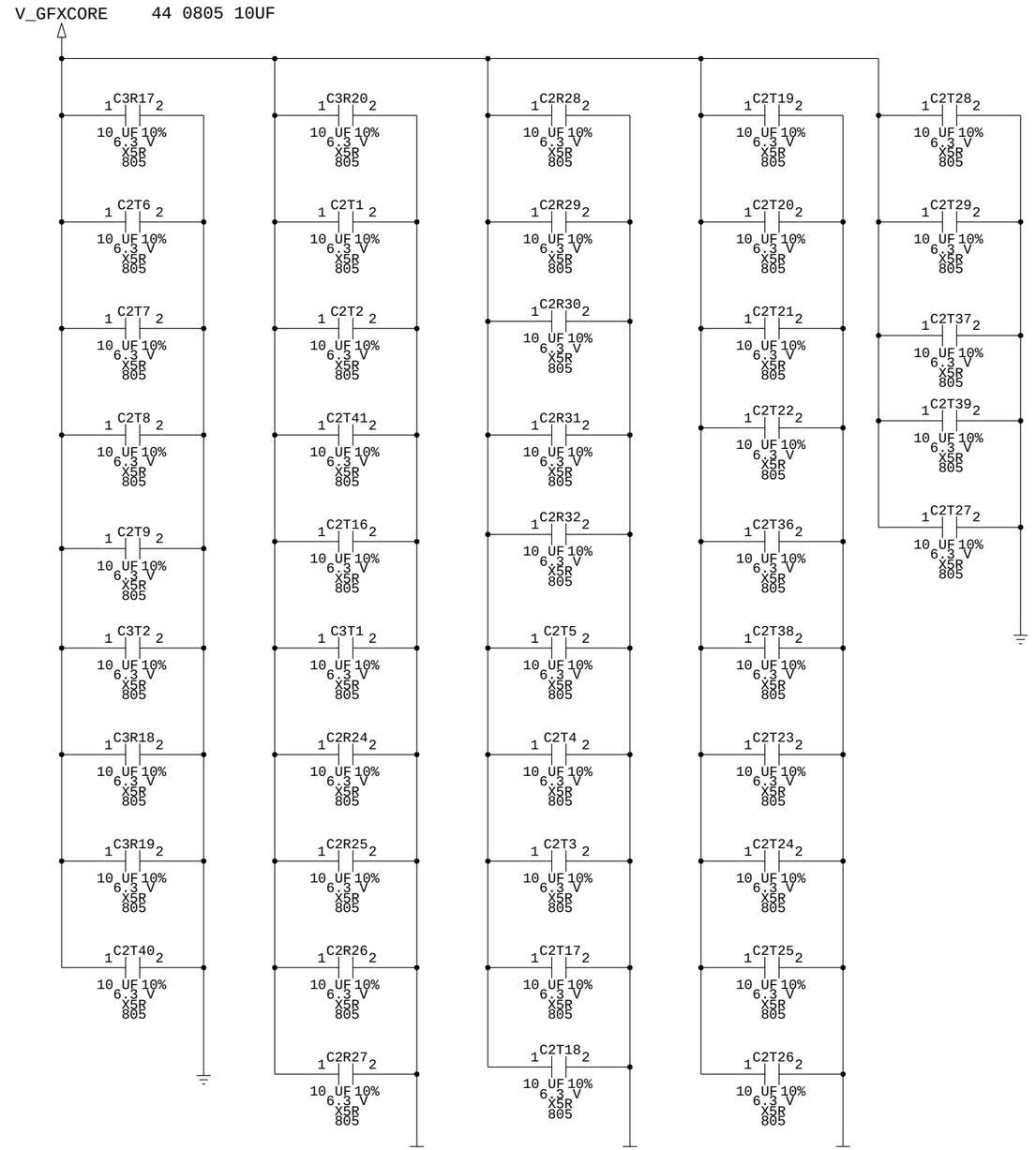
V_MEMIO
R3R2 1 KOHM 1% 2 CH 402
R7D1 M_VREF_W 0 OHM 5% 402 EMPTY
M_VREF OUT 8
R3R3 1 KOHM 1% 2 CH 402
C3R21 1 UF 10% 6.3 V X5R 402

NET M_VREF_W
LEFT FOR 8L
MEMORY MARGINING

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	8/72	8/72	G	1.01

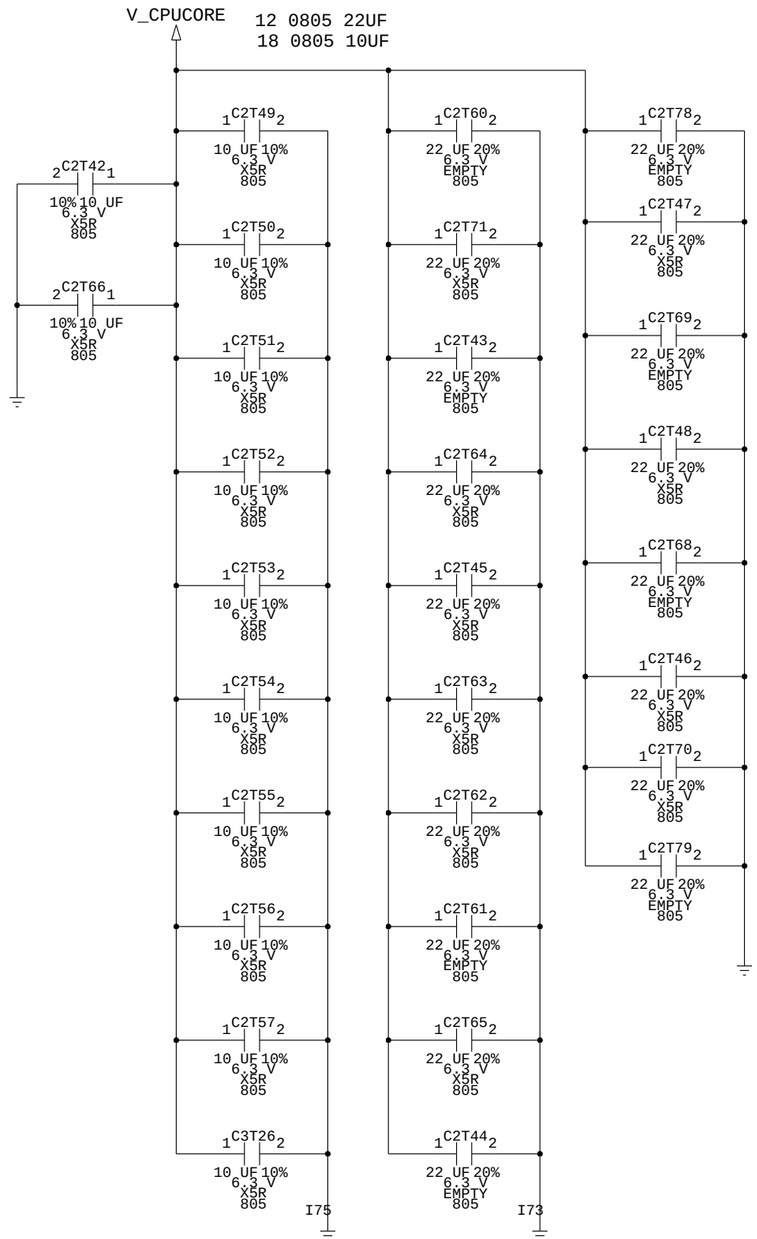
SOC: DECOUPLING

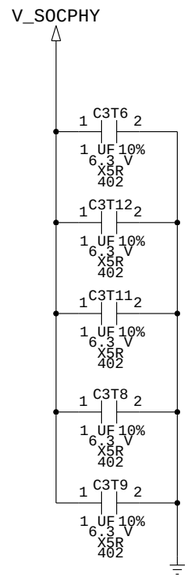


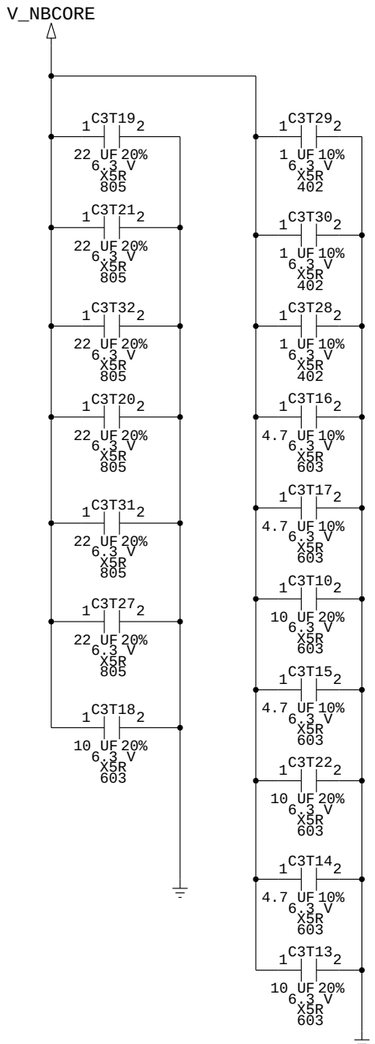


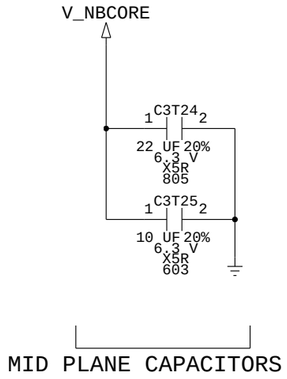
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	9/72	9/72	G	1.01

SOC: DECOUPLING



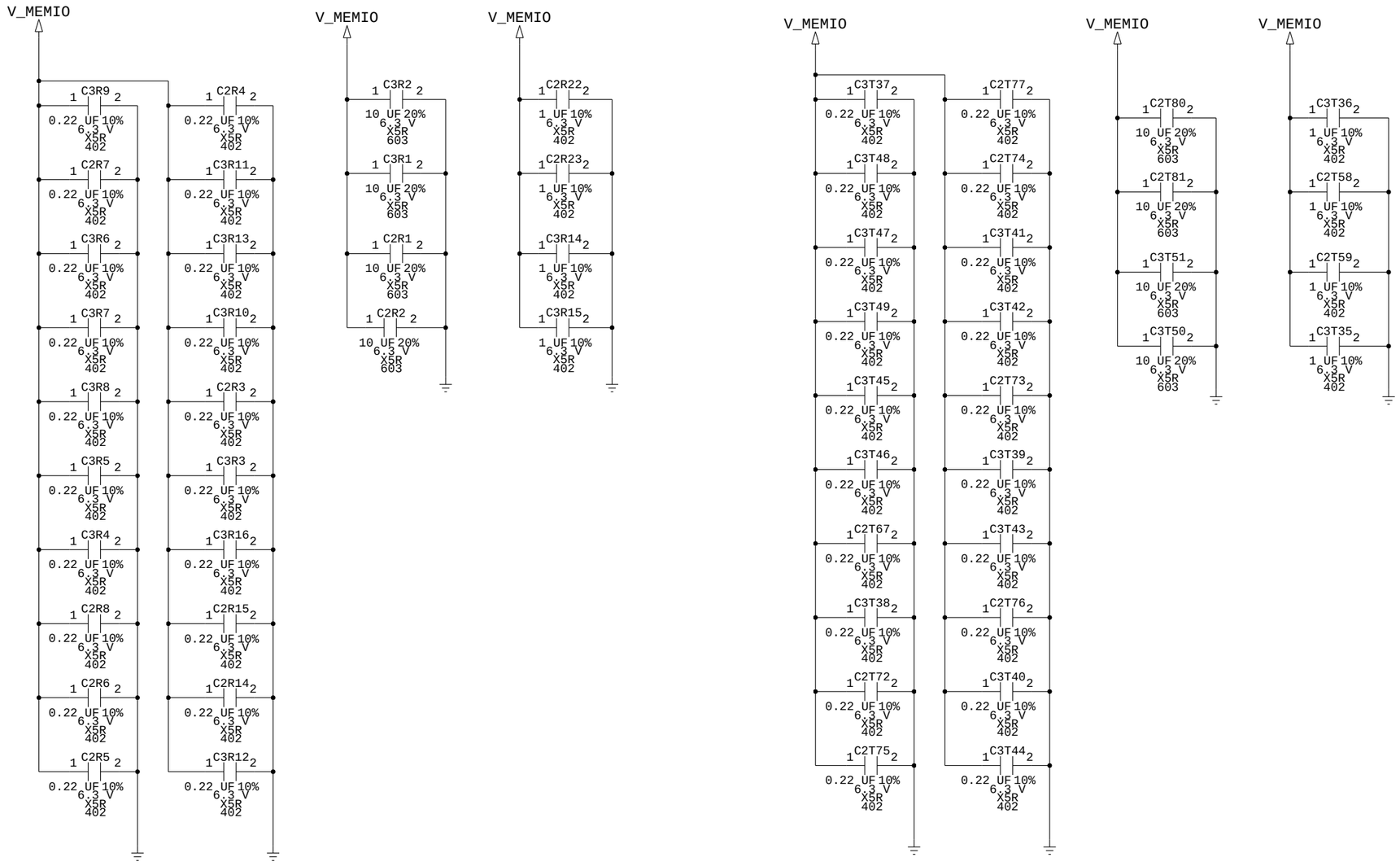






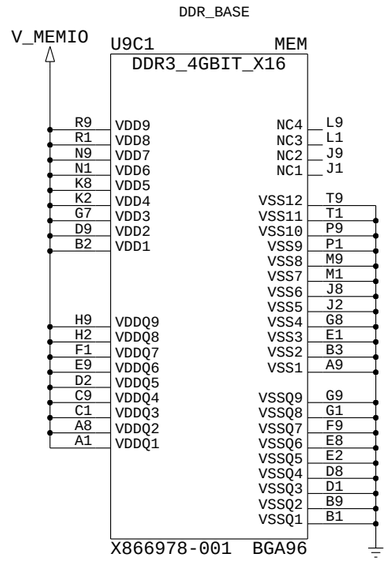
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	10/72	10/72	G	1.01

SOC: DECOUPLING

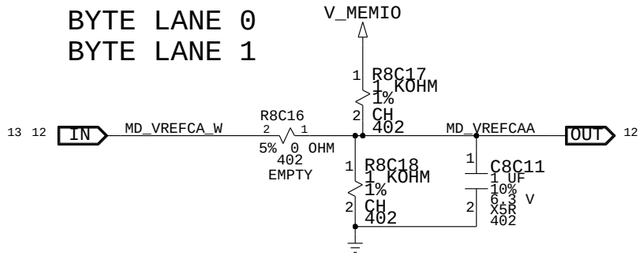


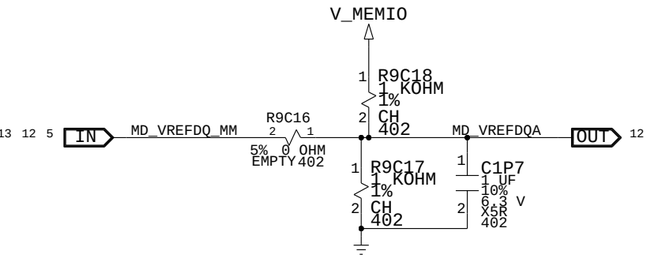
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	11/72	11/72	G	1.01

MEMORY: CHANNEL D

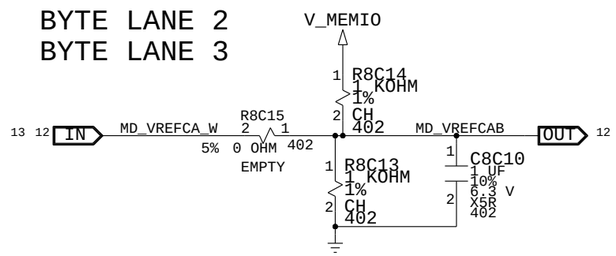


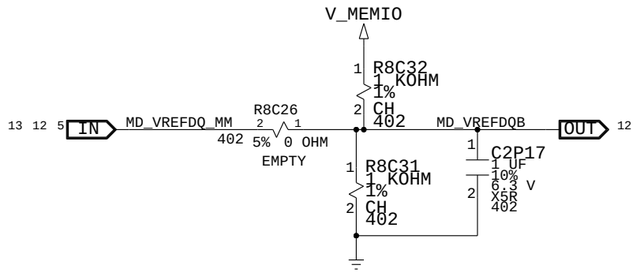
BYTE LANE 0
BYTE LANE 1

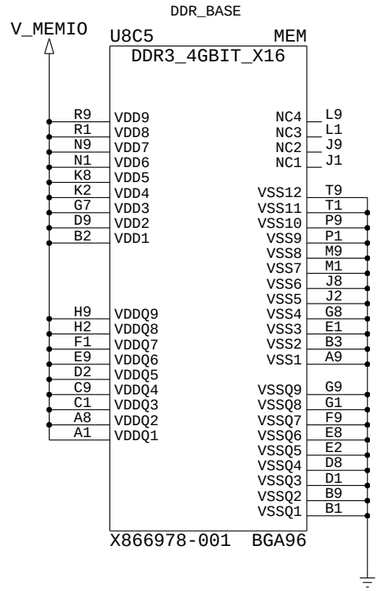


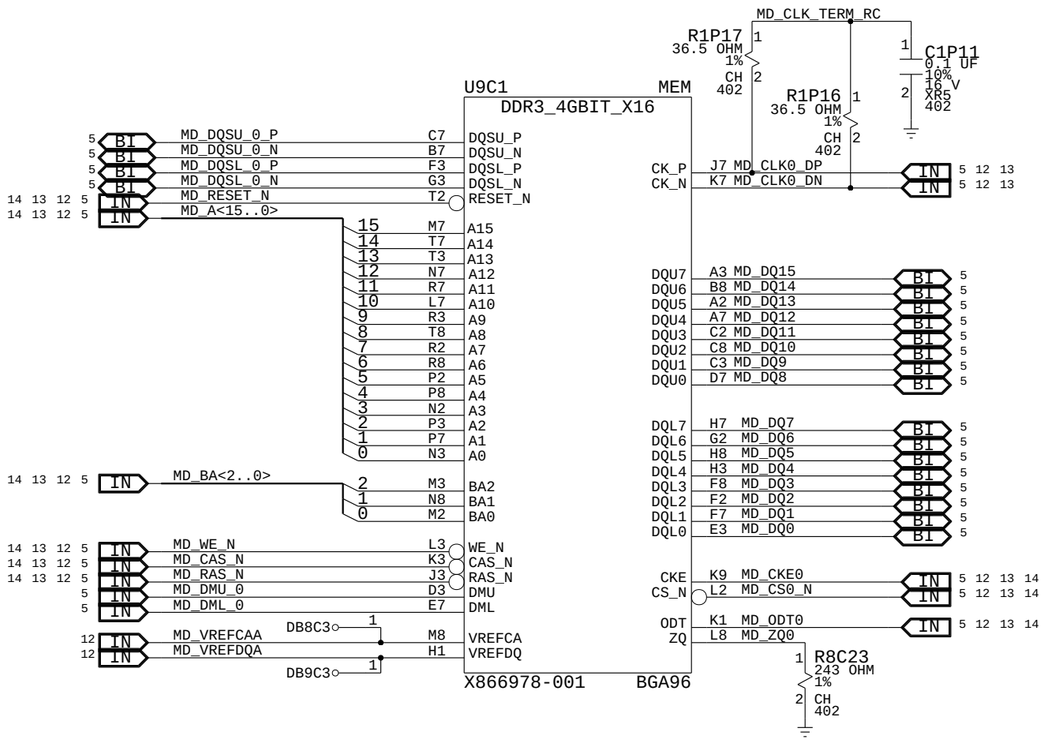


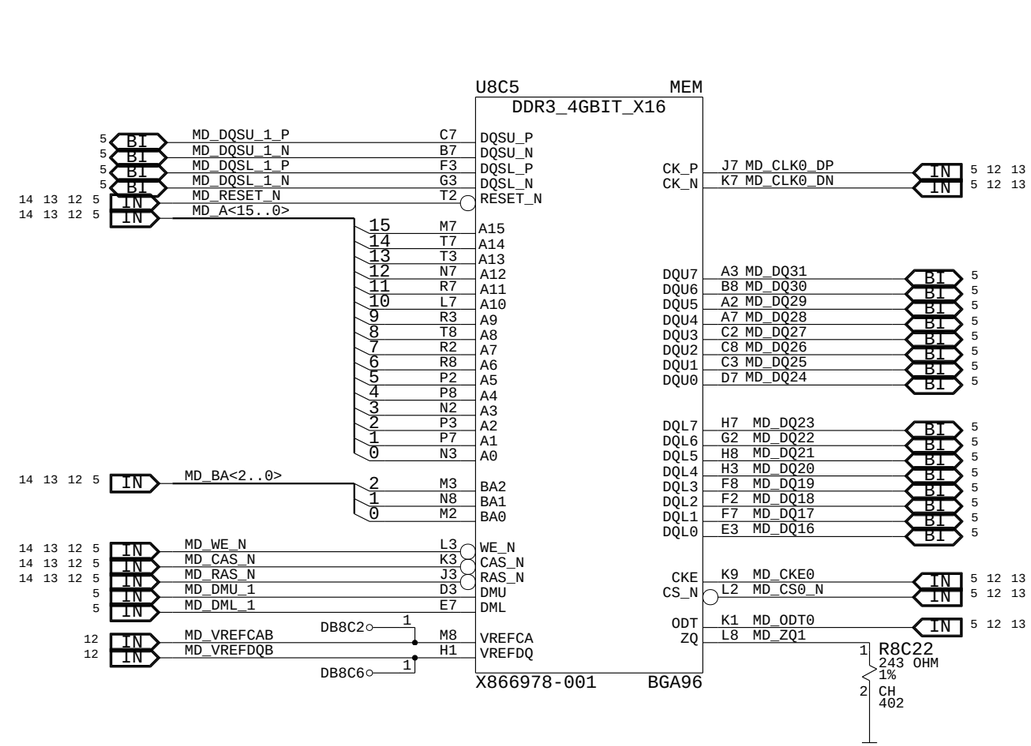
BYTE LANE 2
BYTE LANE 3









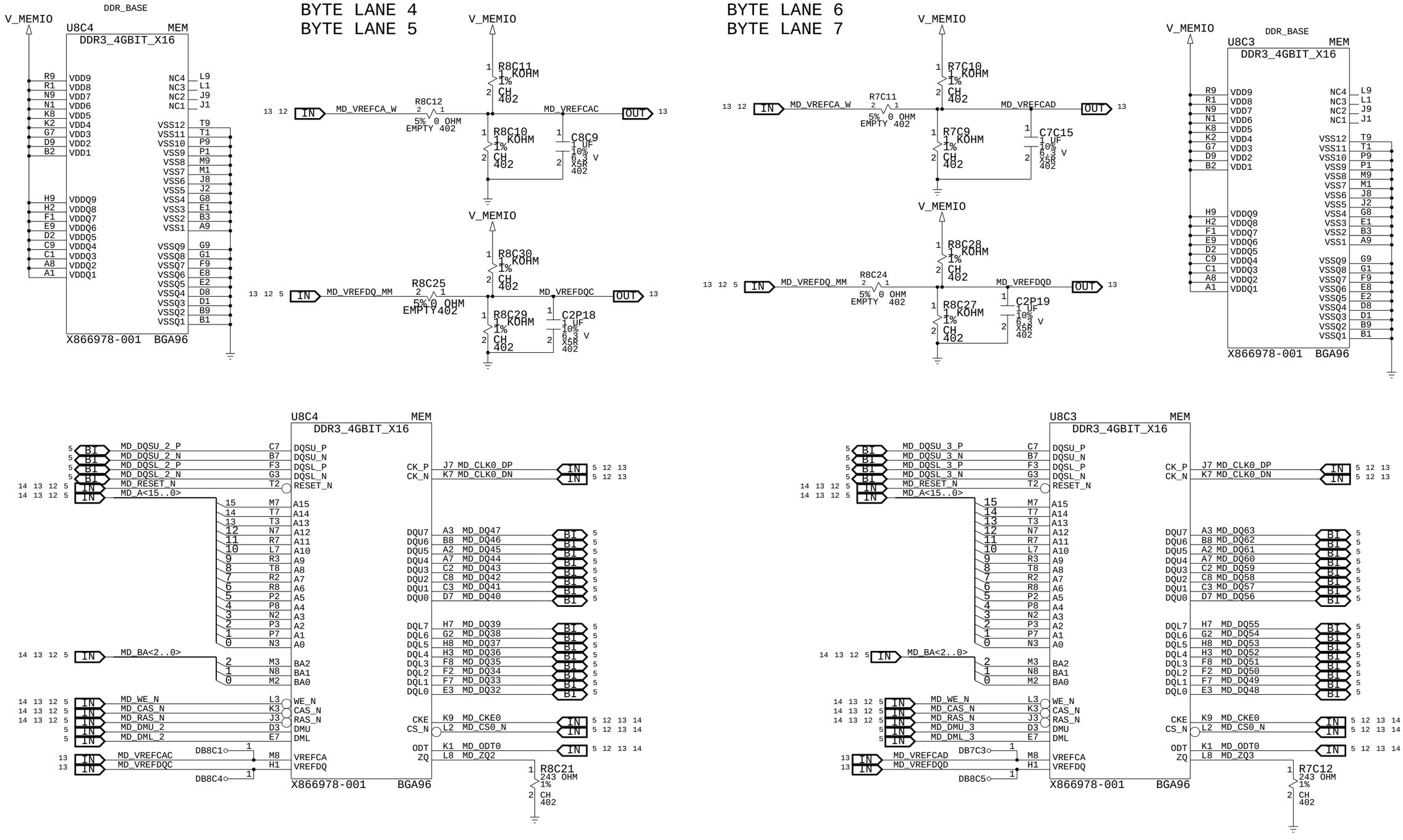


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	12/72	12/72	G	1.01

MEMORY: CHANNEL D

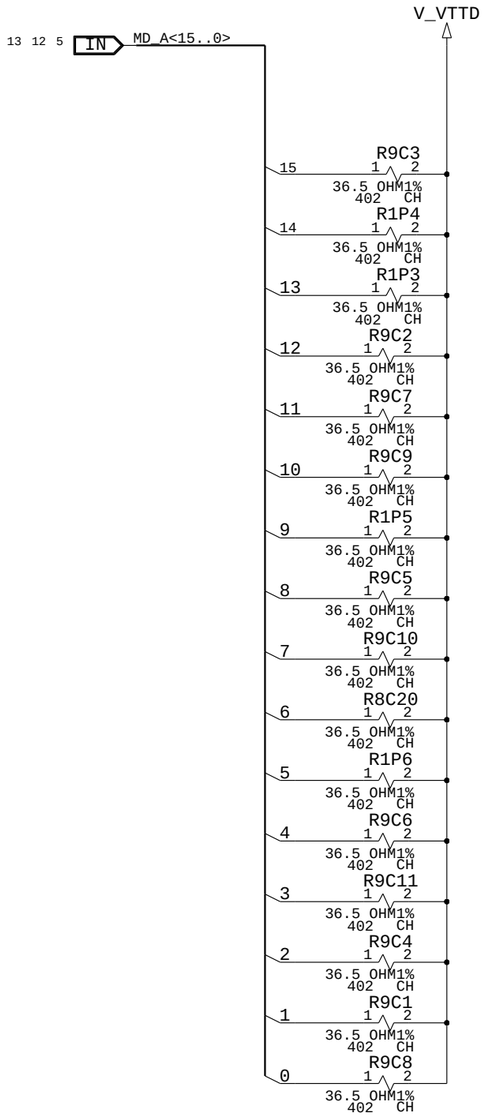
BYTE LANE 4
BYTE LANE 5

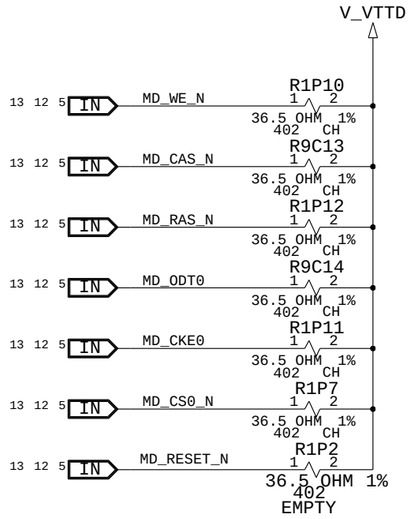
BYTE LANE 6
BYTE LANE 7

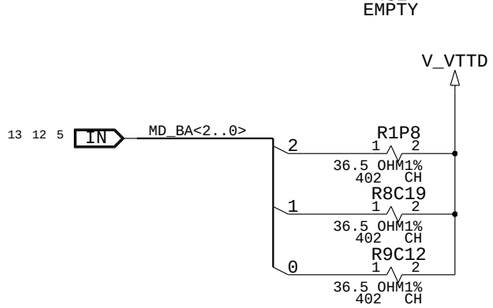


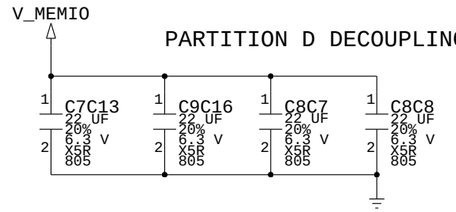
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	13/72	13/72	G	1.01

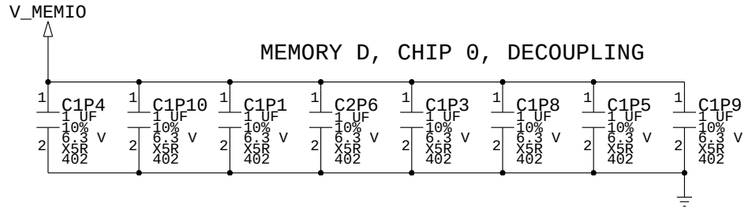
MEMORY: CHANNEL D,DECOUPLING & TERMINATION

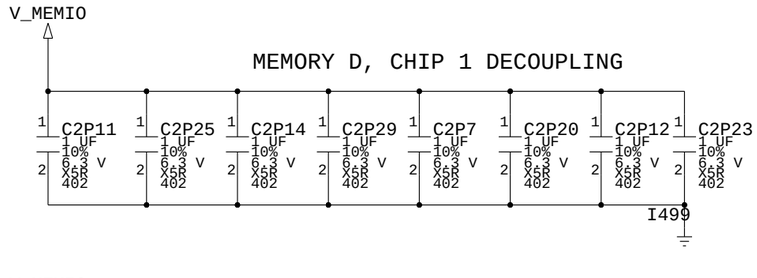


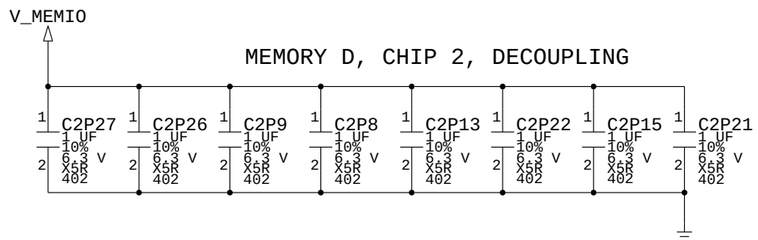


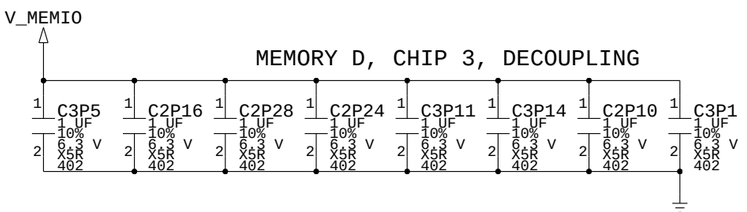












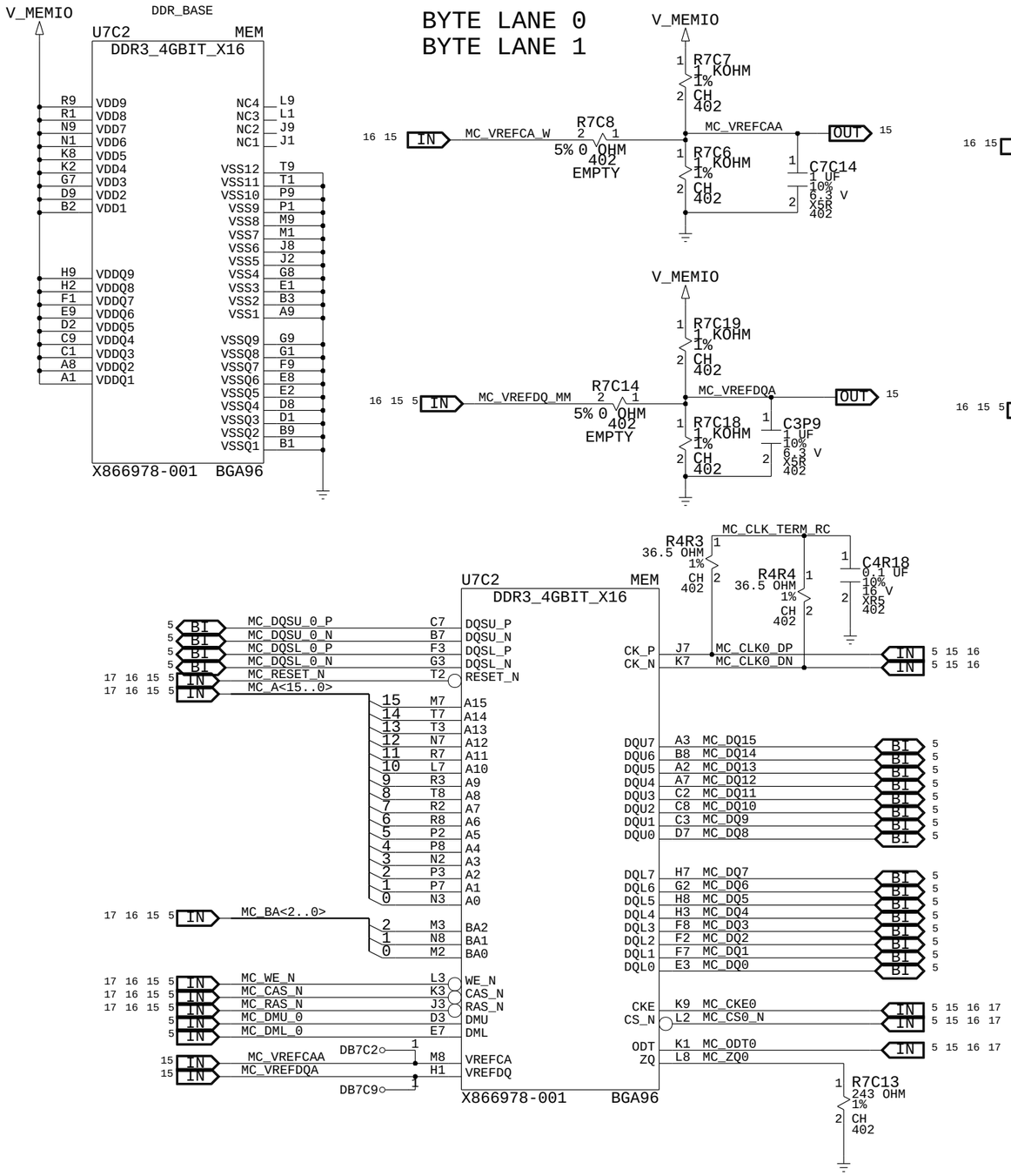
MS_PART#	MATL	REF_DES DESCR.	BOM PROPERTY
X903915-001	IC	MEM,SM,4GB,DDR3,E-DIE,20NM,2133MHZ,96FBGA,K4W4G1646E-BC1A,TYP,SAMSUNG U9C1,U8C5,U8C4,U8C3,U7C2,U7C1,U6D2,U6D3	DDR_SAMSUNG
X903915-001	IC	MEM,SM,4GB,DDR3,E-DIE,20NM,2133MHZ,96FBGA,K4W4G1646E-BC1A,TYP,SAMSUNG U9F5,U8F3,U8F2,U8F1,U7F2,U7F1,U6E5,U6E3	DDR_SAMSUNG
X903814-001	IC	MEM,SM,4GB,DDR3,POLARIS,25NM,2133MHZ,FBGA96,H5TQ4G63CFR-TEC,TYP,HYNIX U9C1,U8C5,U8C4,U8C3,U7C2,U7C1,U6D2,U6D3	DDR_HYNIX_25NM
X903814-001	IC	MEM,SM,4GB,DDR3,POLARIS,25NM,2133MHZ,FBGA96,H5TQ4G63CFR-TEC,TYP,HYNIX U9F5,U8F3,U8F2,U8F1,U7F2,U7F1,U6E5,U6E3	DDR_HYNIX_25NM
X945095-001	IC	MEM,SM,4GB,256X16,DDR3,96 FBGA,NT5CB256M16DP-FL,TYP U9C1,U8C5,U8C4,U8C3,U7C2,U7C1,U6D2,U6D3	DDR_NANYA_30NM
X945095-001	IC	MEM,SM,4GB,256X16,DDR3,96 FBGA,NT5CB256M16DP-FL,TYP U9F5,U8F3,U8F2,U8F1,U7F2,U7F1,U6E5,U6E3	DDR_NANYA_30NM

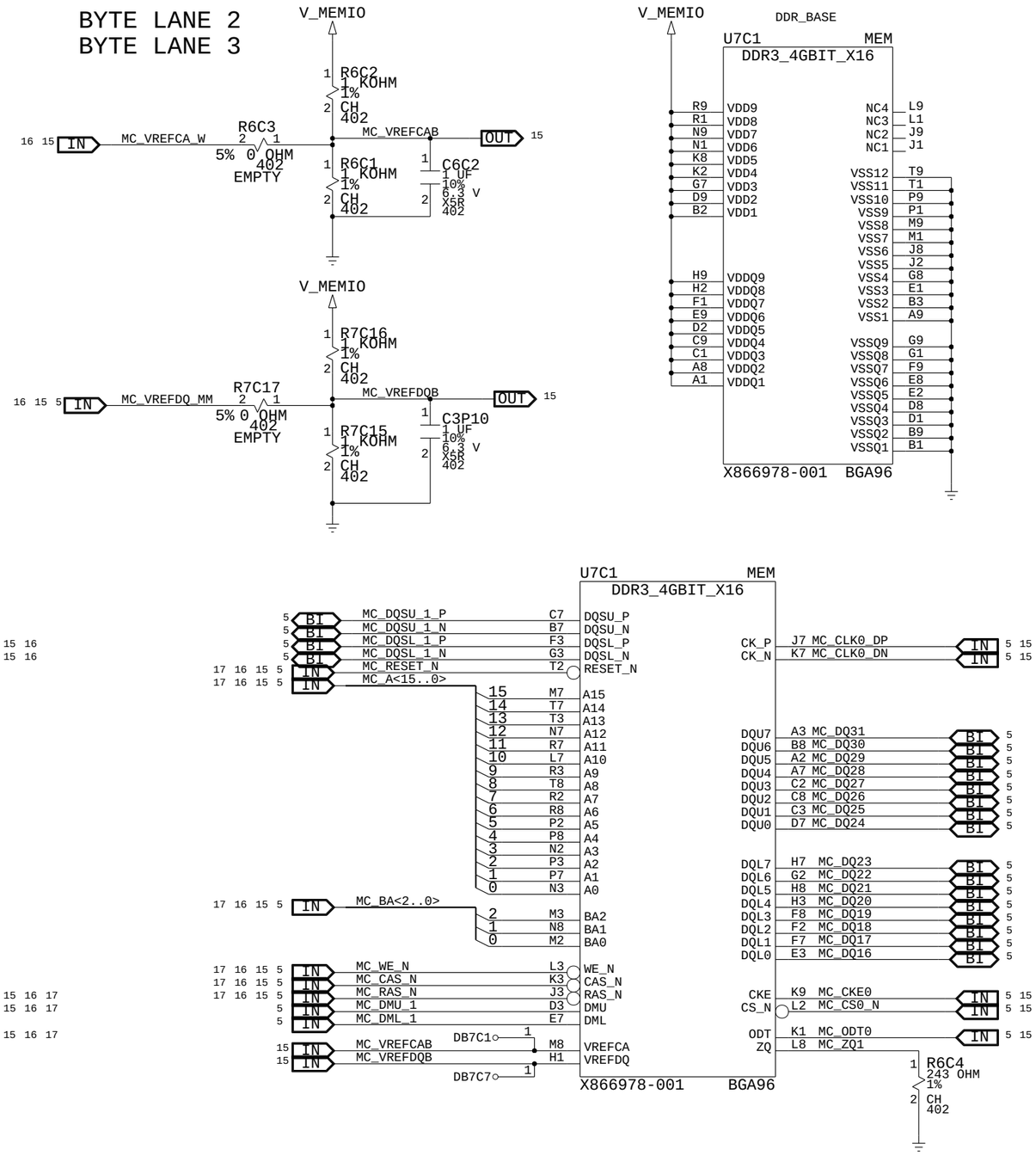
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	14/72	14/72	G	1.01

MEMORY: CHANNEL C

BYTE LANE 0
BYTE LANE 1

BYTE LANE 2
BYTE LANE 3



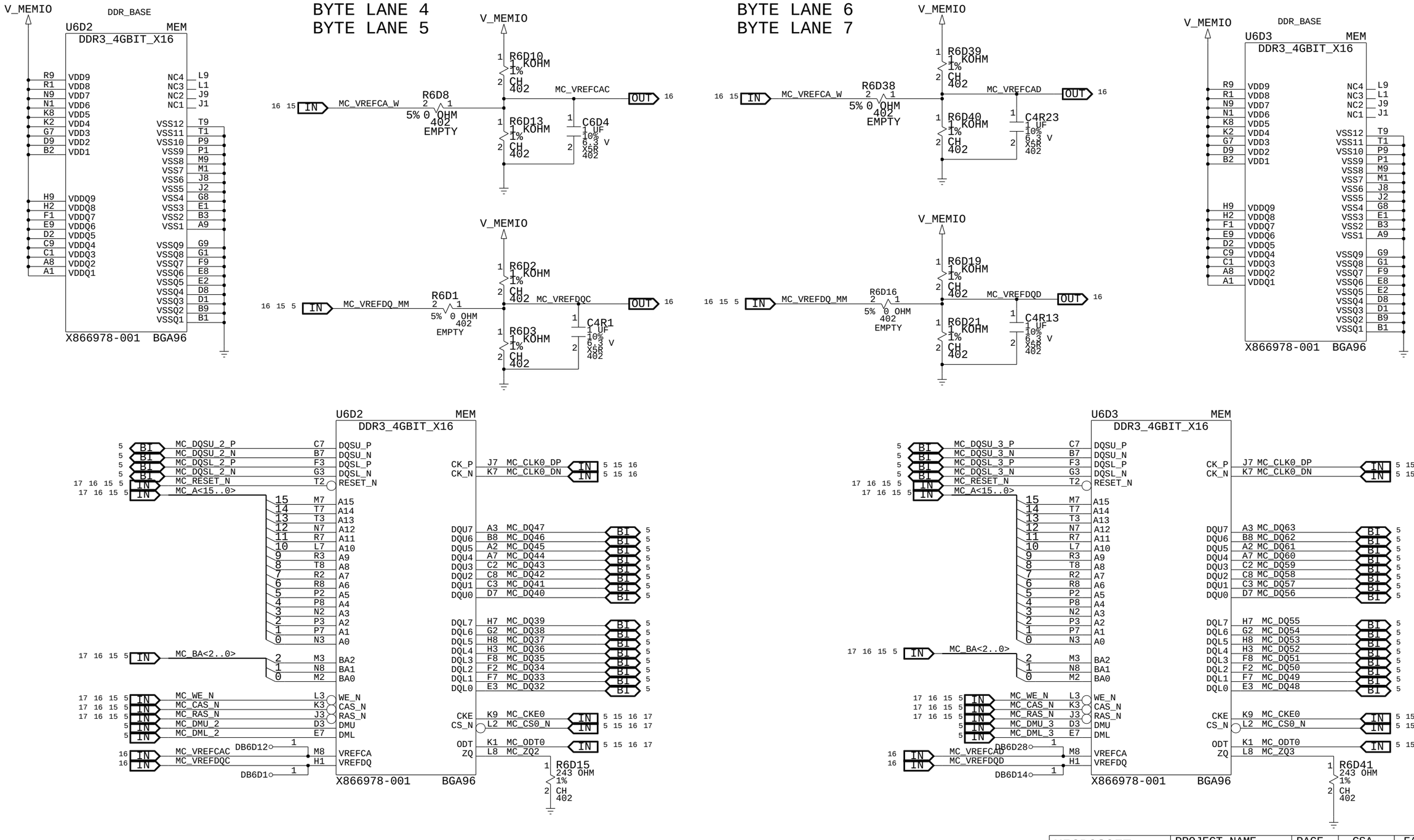


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	15/72	15/72	G	1.01

MEMORY: CHANNEL C

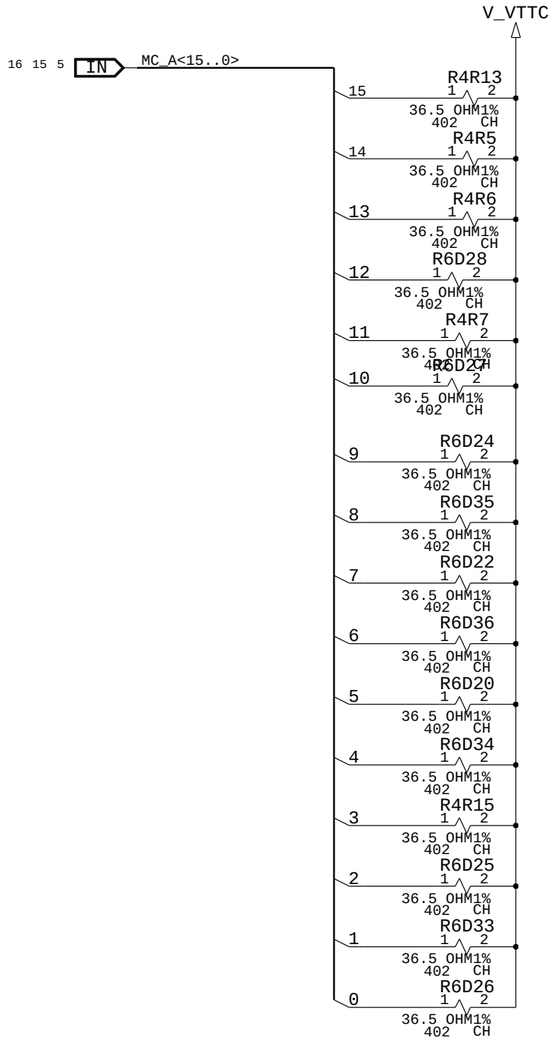
BYTE LANE 4
BYTE LANE 5

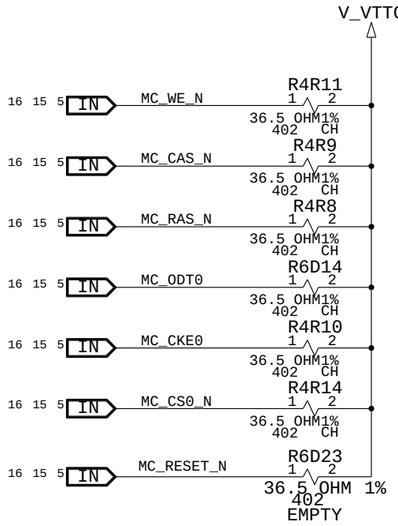
BYTE LANE 6
BYTE LANE 7

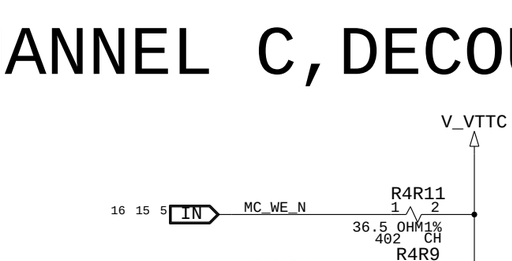


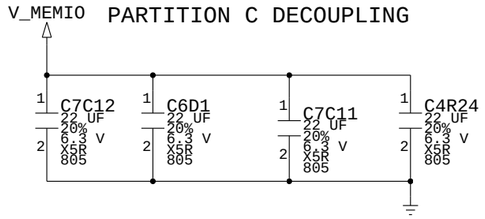
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	16/72	16/72	G	1.01

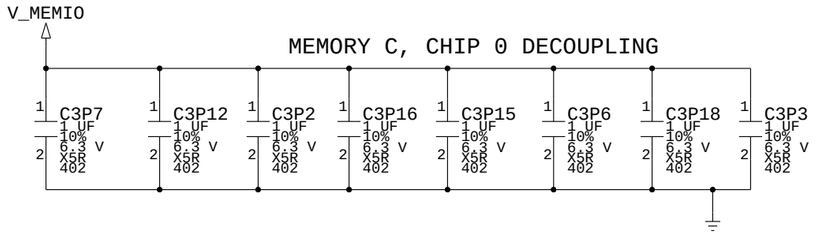
MEMORY: CHANNEL C,DECOUPLING & TERMINATION

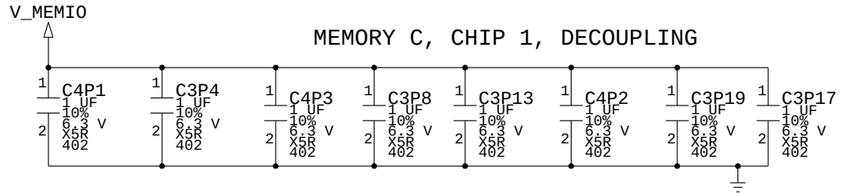


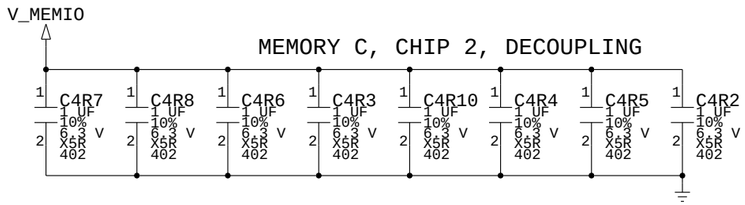


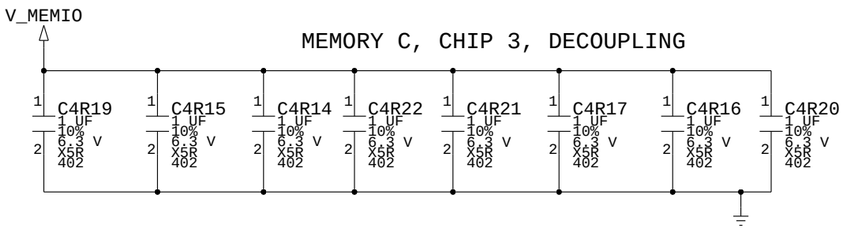










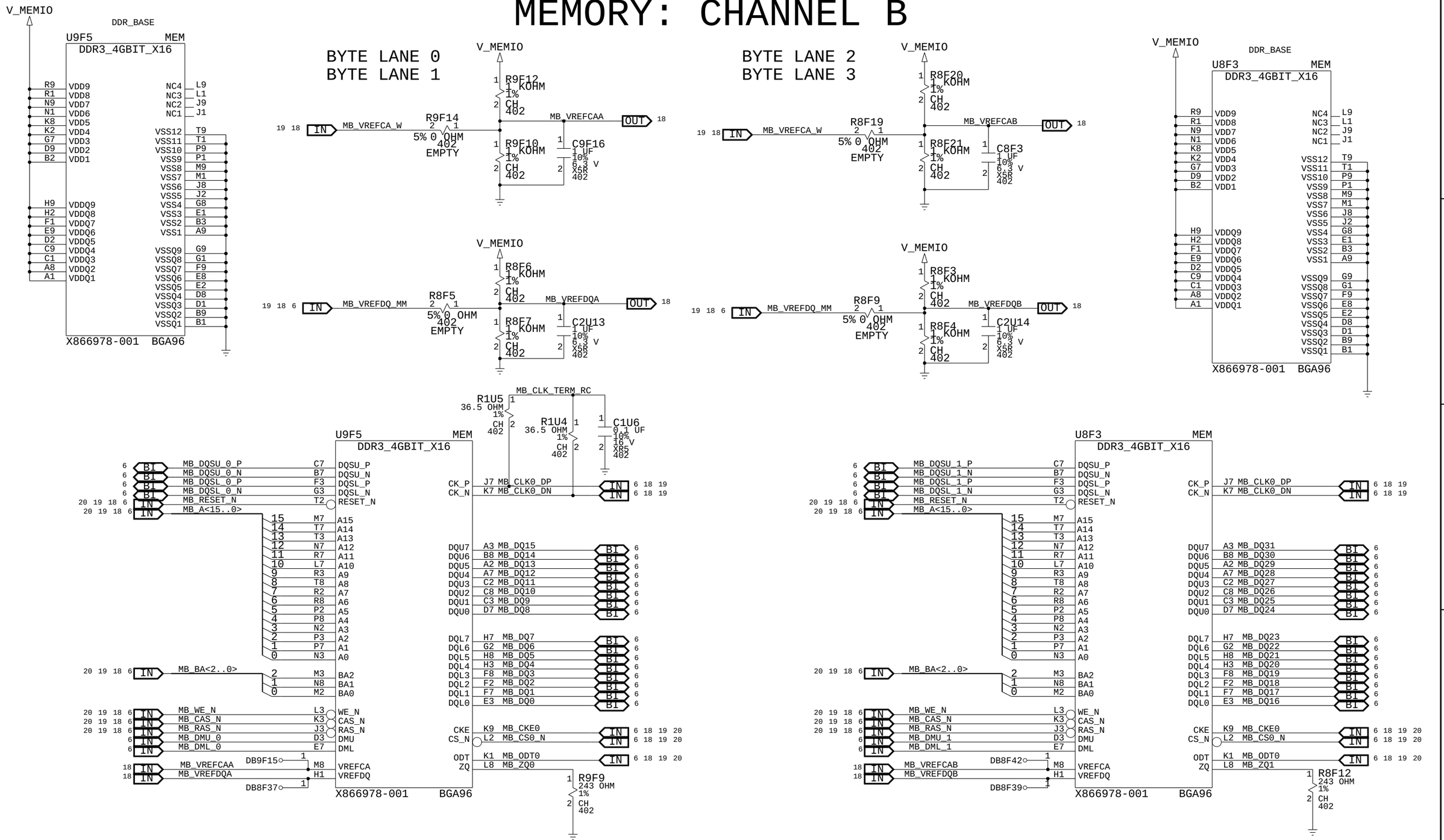


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	17/72	17/72	G	1.01

MEMORY: CHANNEL B

BYTE LANE 0
BYTE LANE 1

BYTE LANE 2
BYTE LANE 3

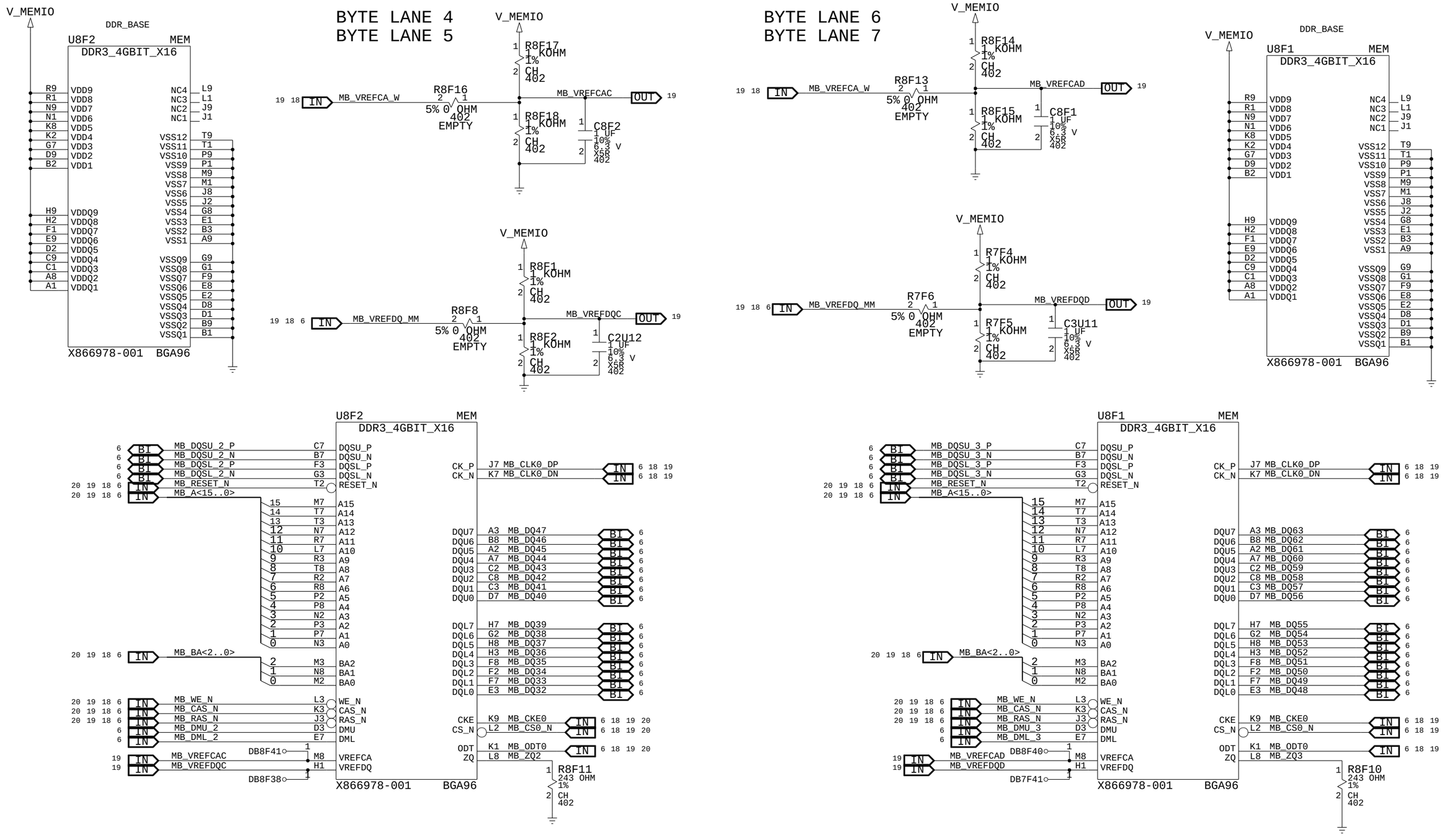


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	18/72	18/72	G	1.01

MEMORY: CHANNEL B

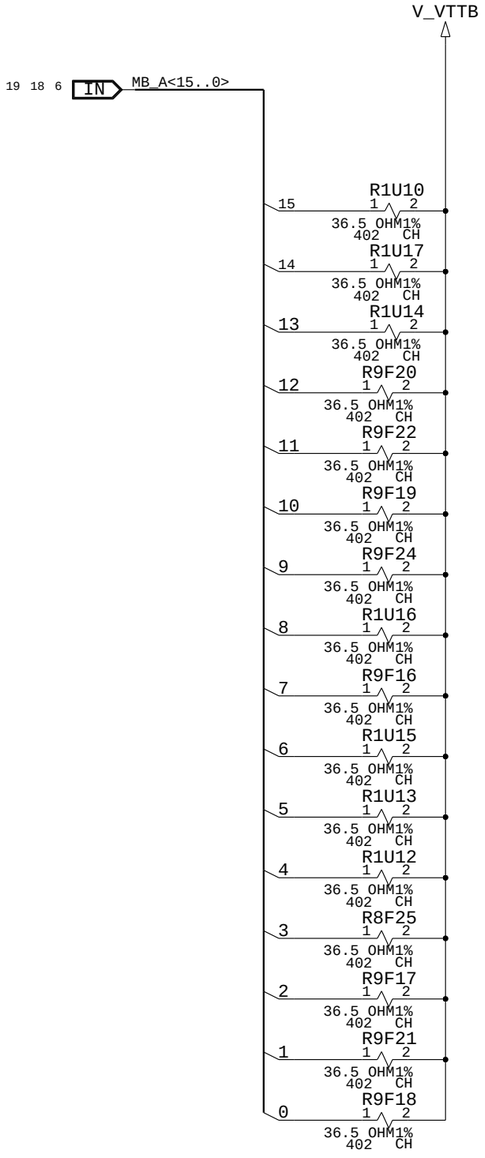
BYTE LANE 4
BYTE LANE 5

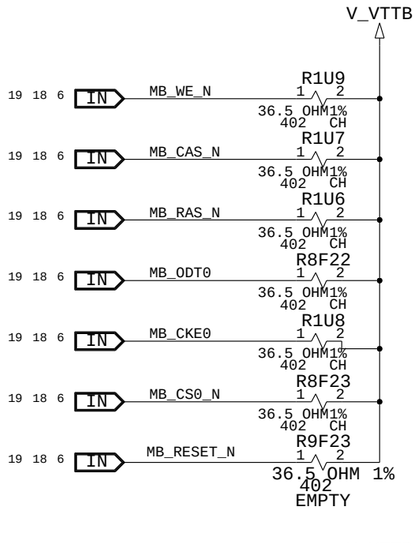
BYTE LANE 6
BYTE LANE 7

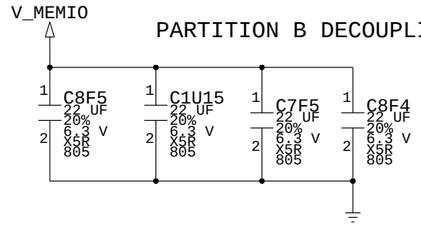


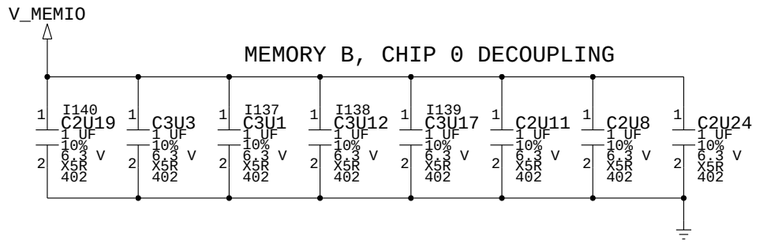
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	19/72	19/72	G	1.01

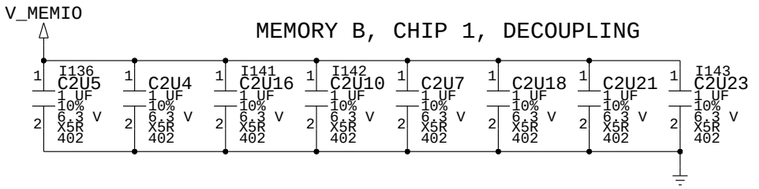
MEMORY: CHANNEL B,DECOUPLING & TERMINATION

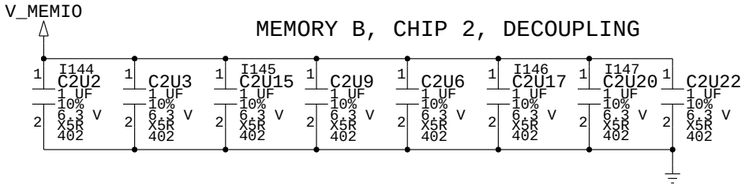


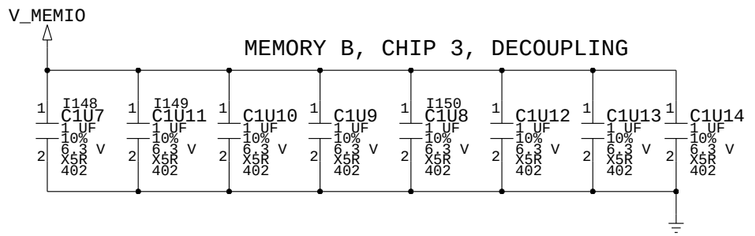








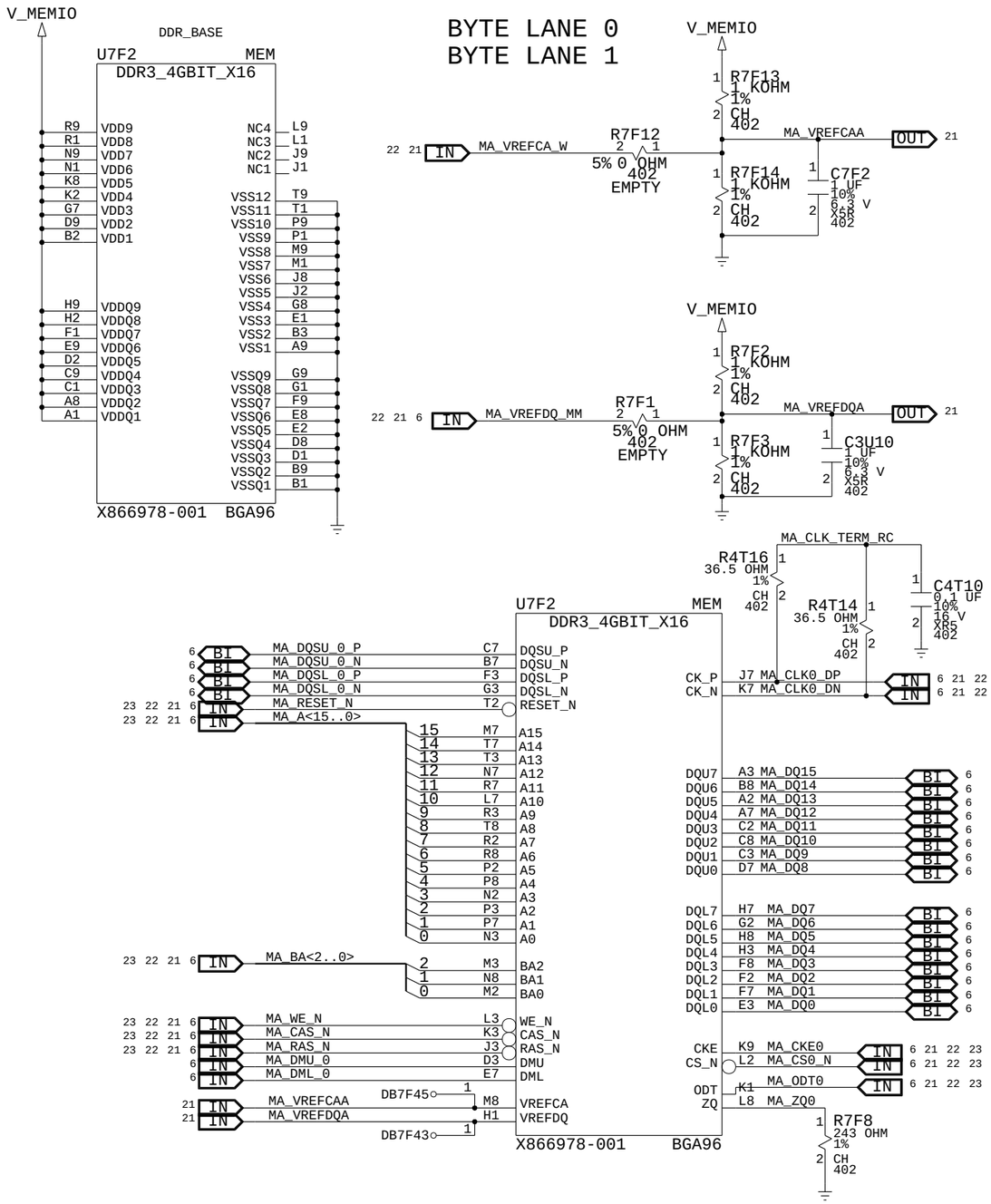




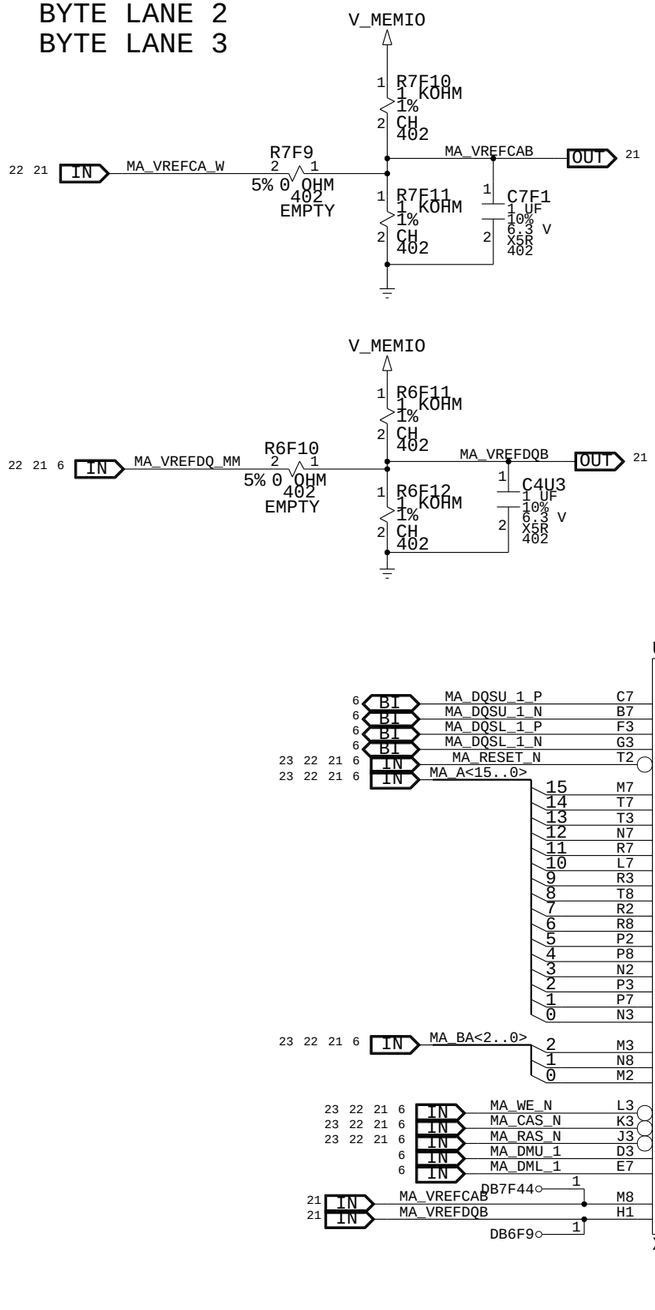
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	20/72	20/72	G	1.01

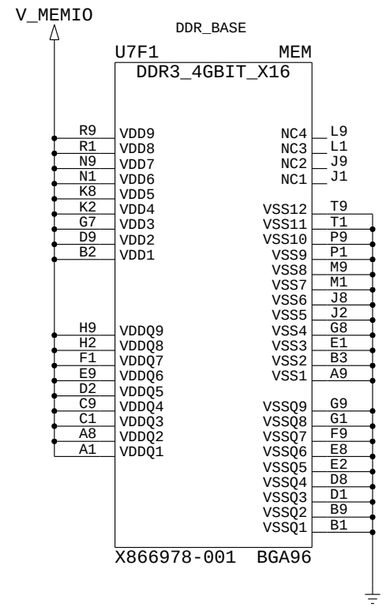
MEMORY: CHANNEL A

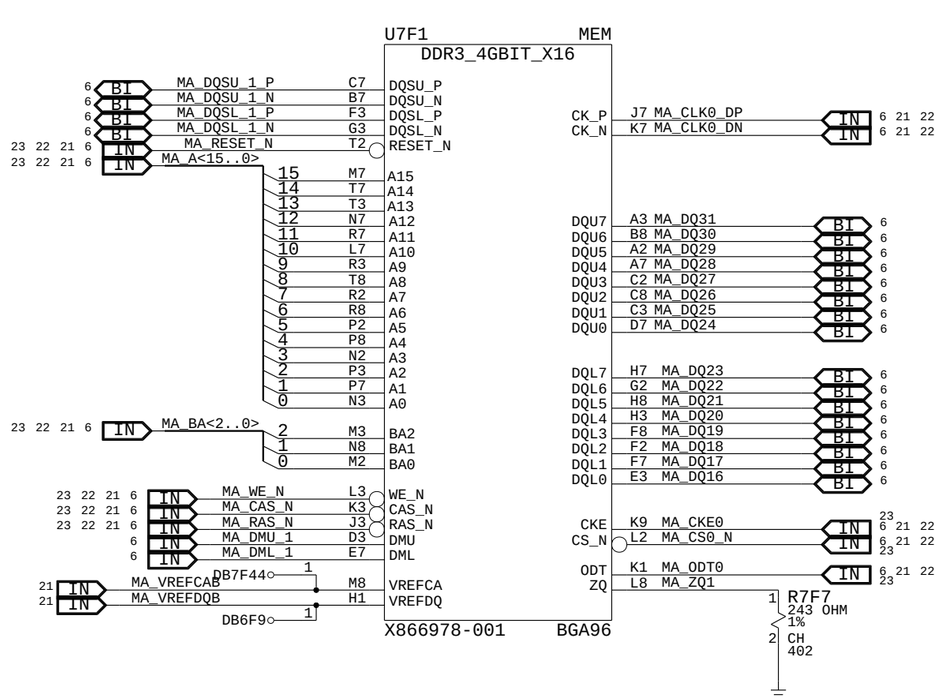
BYTE LANE 0
BYTE LANE 1



BYTE LANE 2
BYTE LANE 3



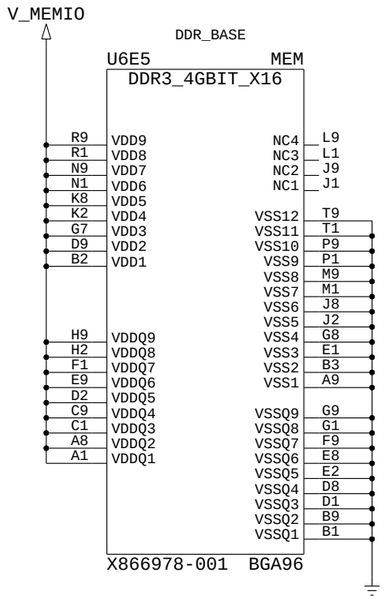


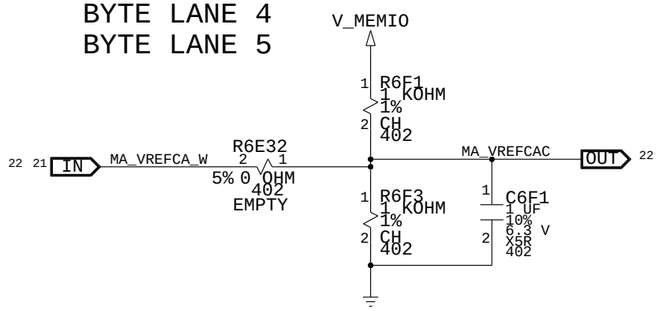


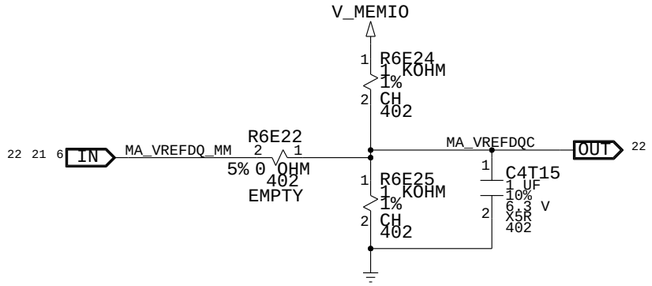
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	21/72	21/72	G	1.01

MEMORY: CHANNEL A

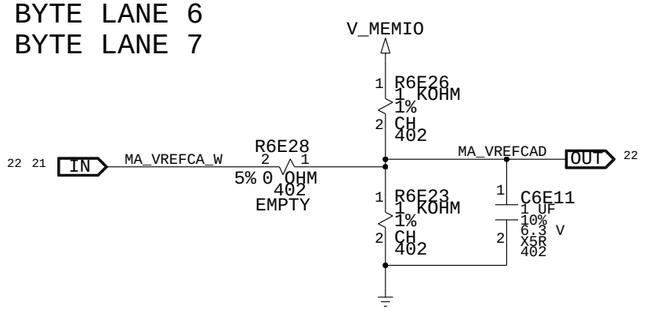
BYTE LANE 4
BYTE LANE 5

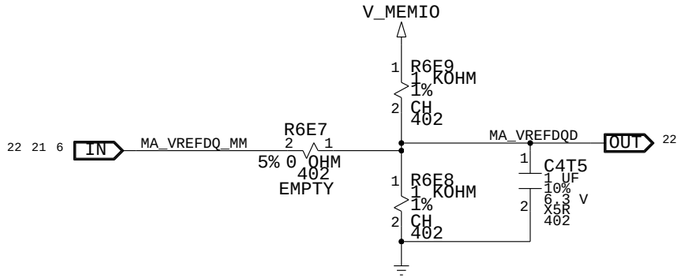


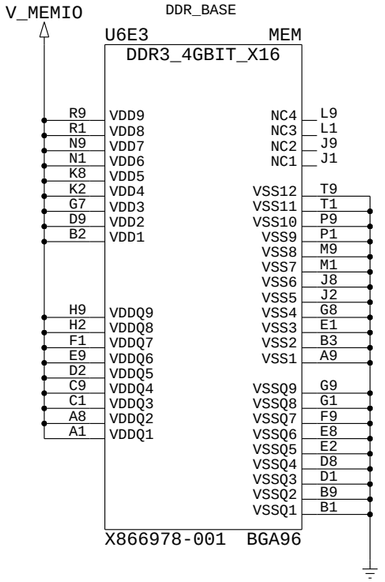


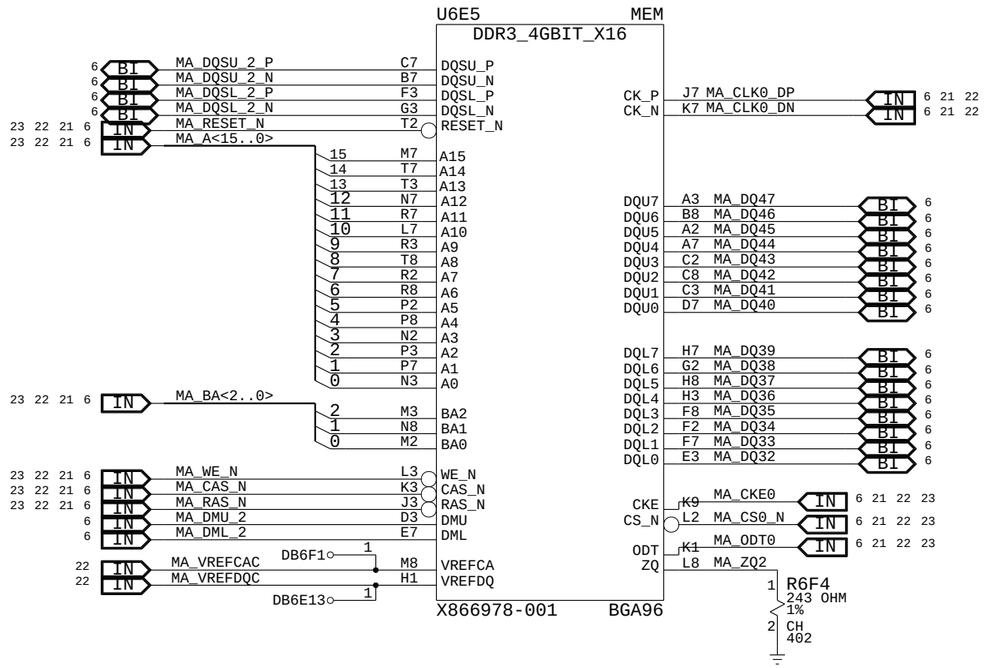


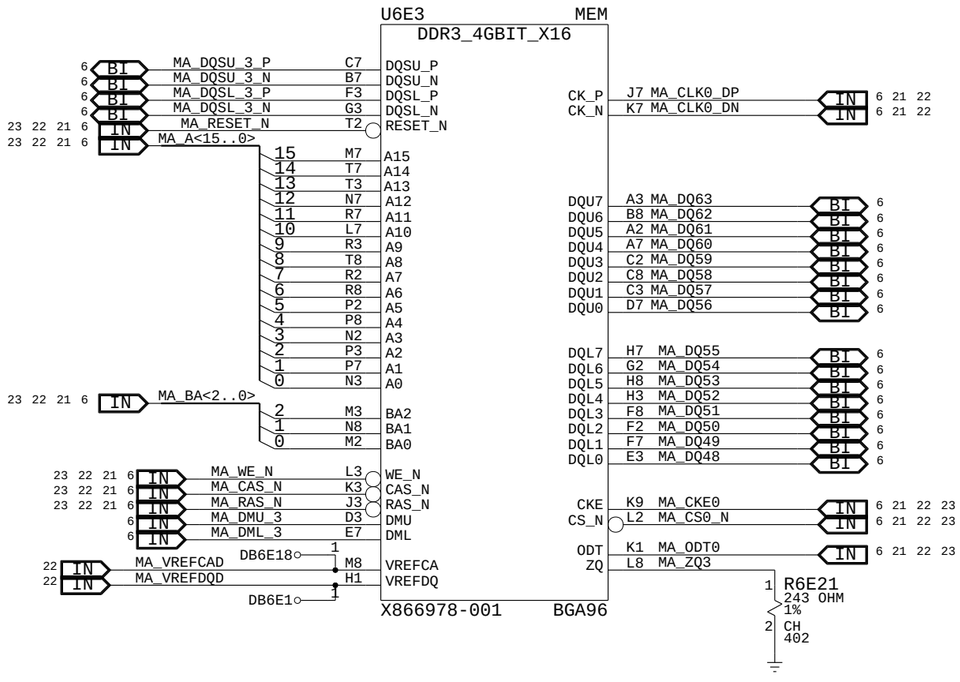
BYTE LANE 6
BYTE LANE 7





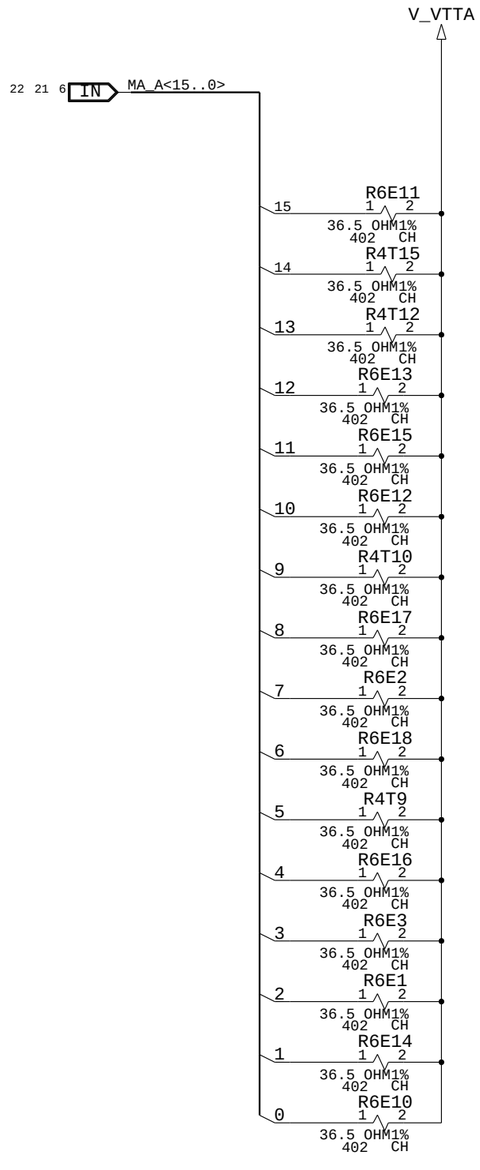


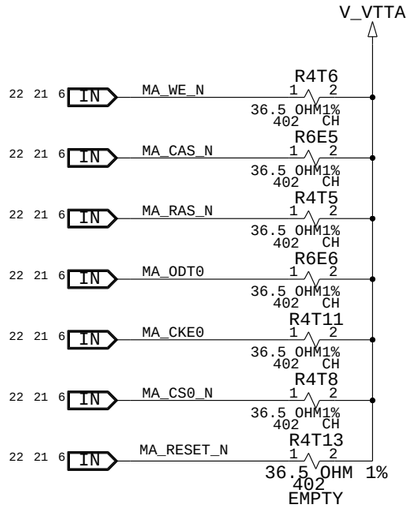


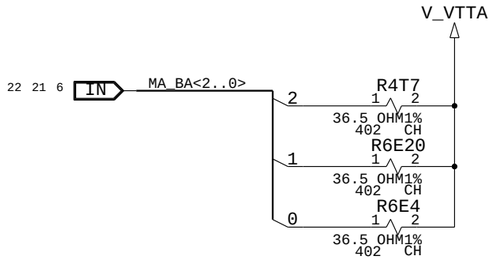


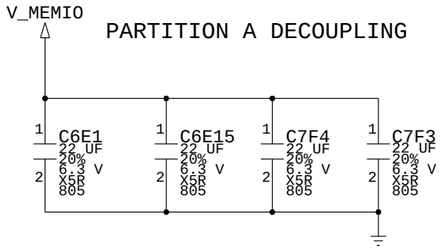
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	22/72	22/72	G	1.01

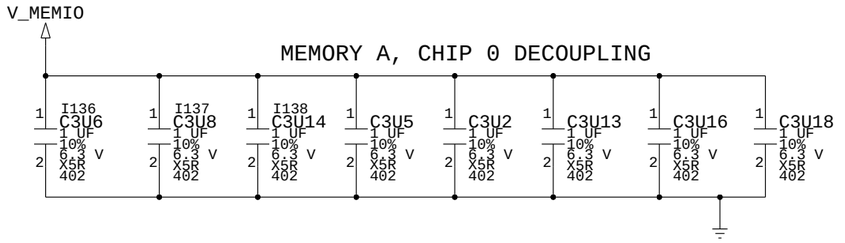
MEMORY: CHANNEL A,DECOUPLING & TERMINATION

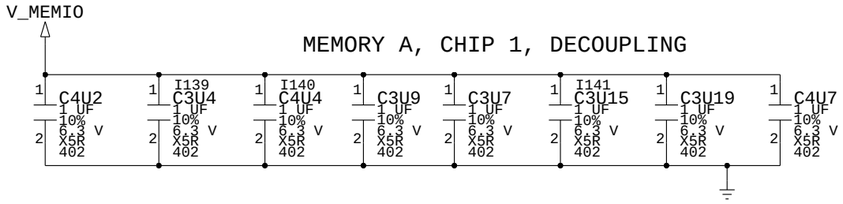


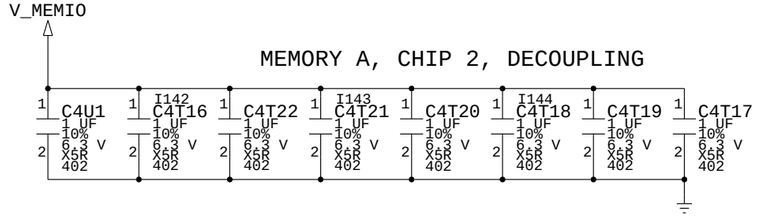


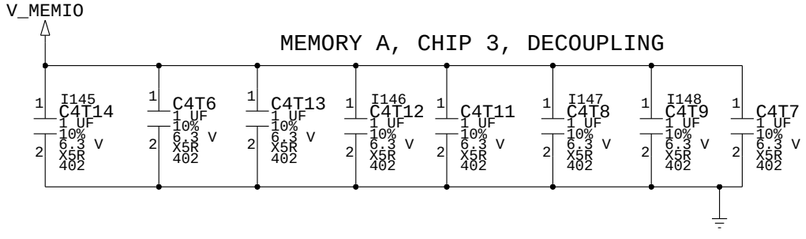












MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	23/72	23/72	G	1.01

KIC: USB

U4D2 KIC IC
6 of 11

USBD_D0_DP	AF8			USB2_WIFI_DP	BI	36
USBD_D0_DN	AE8			USB2_WIFI_DN	BI	36
USBC_D0_DP	AC4			USB2_AUX_DP	BI	35
USBC_D0_DN	AC3			USB2_AUX_DN	BI	35
USBC_D1_RXP	Y1	USB3_AUX_RP	IN 35			
USBC_D1_RXN	Y2	USB3_AUX_RN	IN 35			
USBC_D1_TXP	W3	USB3_AUX_TP_C	C6B12 0.1 UF 10% 16 V XR5 402	USB3_AUX_TP	OUT	35
USBC_D1_TXN	W4	USB3_AUX_TN_C	C6B11 0.1 UF 10% 16 V XR5 402	USB3_AUX_TN	OUT	35
USBB_D2_RXP	V1	USB3_BP_P1_RP	IN 35			
USBB_D2_RXN	V2	USB3_BP_P1_RN	IN 35			
USBB_D2_TXP	U3	USB3_BP_P1_TP_C	C6B8 0.1 UF 10% 16 V XR5 402	USB3_BP_P1_TP	OUT	35
USBB_D2_TXN	U4	USB3_BP_P1_TN_C	C6B7 0.1 UF 10% 16 V XR5 402	USB3_BP_P1_TN	OUT	35
USBB_D0_DP	AE4			USB2_BP_P1_DP	BI	35
USBB_D0_DN	AE3			USB2_BP_P1_DN	BI	35
USBB_D1_DP	AD2			USB2_BP_P0_DP	1 DB4D19	
USBB_D1_DN	AD1			USB2_BP_P0_DN	1 DB4D20	
USBB_D3_RXP	T1	USB3_BP_P0_RP	DB4D17 1			
USBB_D3_RXN	T2	USB3_BP_P0_RN	DB4D18 1			
USBB_D3_TXP	R3			USB3_BP_P0_TP	1 DB4D21	
USBB_D3_TXN	R4			USB3_BP_P0_TN	1 DB4D22	
USBA_D2_RXP	AA8	USB3_FP_P0_RP	IN 35			
USBA_D2_RXN	AA7	USB3_FP_P0_RN	IN 35			
USBA_D2_TXP	AB10	USB3_FP_P0_TP_C	C4F14 0.1 UF 10% 16 V XR5 402	USB3_FP_P0_TP	OUT	35
USBA_D2_TXN	AB9	USB3_FP_P0_TN_C	C4F15 0.1 UF 10% 16 V XR5 402	USB3_FP_P0_TN	OUT	35
USBA_D0_DP	AC7			USB2_FP_P0_DP	BI	35
USBA_D0_DN	AC8			USB2_FP_P0_DN	BI	35
USBA_D1_DP	AF6			USB2_ACC_DP	BI	41
USBA_D1_DN	AF5			USB2_ACC_DN	BI	41
USBA_D3_RXP	AB1	USB3_ACC_RP	DB4C10 1			
USBA_D3_RXN	AB2	USB3_ACC_RN	DB4C11 1			
USBA_D3_TXP	AA3			USB3_ACC_TP	1 DB4C12	
USBA_D3_TXN	AA4			USB3_ACC_TN	1 DB4C13	
USBA_D1_TXRTUNE	AB11		R6T1 174 OHM 1% CH 402			

USBA_D2_AMOUT	AA5	USBA_D2_AMOUNT 1	DB6R2
USBA_D1_AMOUT	AE6	USBA_D1_AMOUNT 1	DB4D7
USBA_D0_AMOUT	AC11	USBA_D0_AMOUNT 1	DB4E3
USBB_D2_AMOUT	R5	USBB_D2_AMOUNT 1	DB5R2
USBB_D1_AMOUT	AD5	USBB_D1_AMOUNT 1	DB4D12
USBB_D0_AMOUT	AF4	USBB_D0_AMOUNT 1	DB4D13
USBC_D1_AMOUT	W5	USBC_D1_AMOUNT 1	DB6R1
USBC_D0_AMOUT	AC5	USBC_D0_AMOUNT 1	DB6R3
USBD_D0_AMOUT	AE9	USBD_D0_AMOUNT 1	DB4E2

WIFI

REAR 0

REAR 1

NOT USED

FRONT

FRONT PANEL ACC

NOT USED

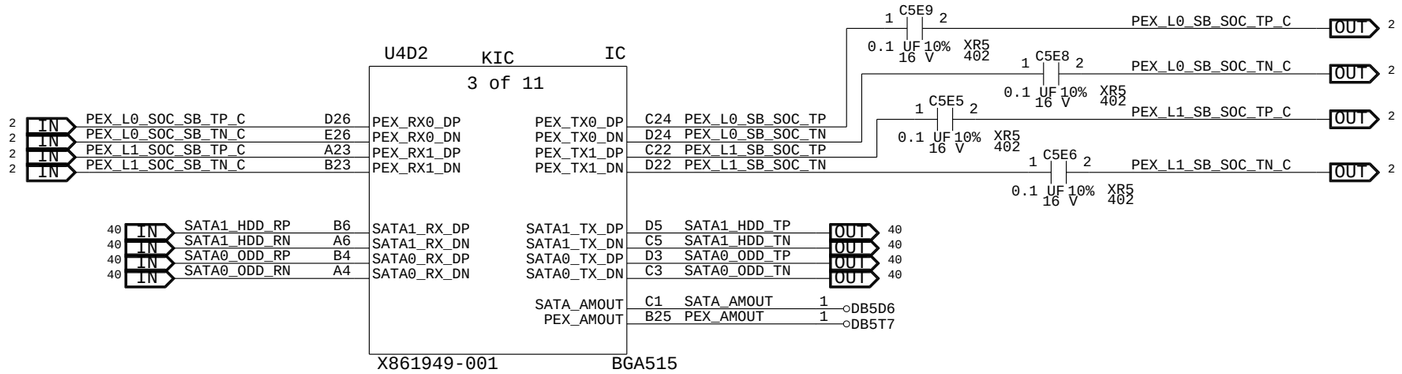
NOT USED

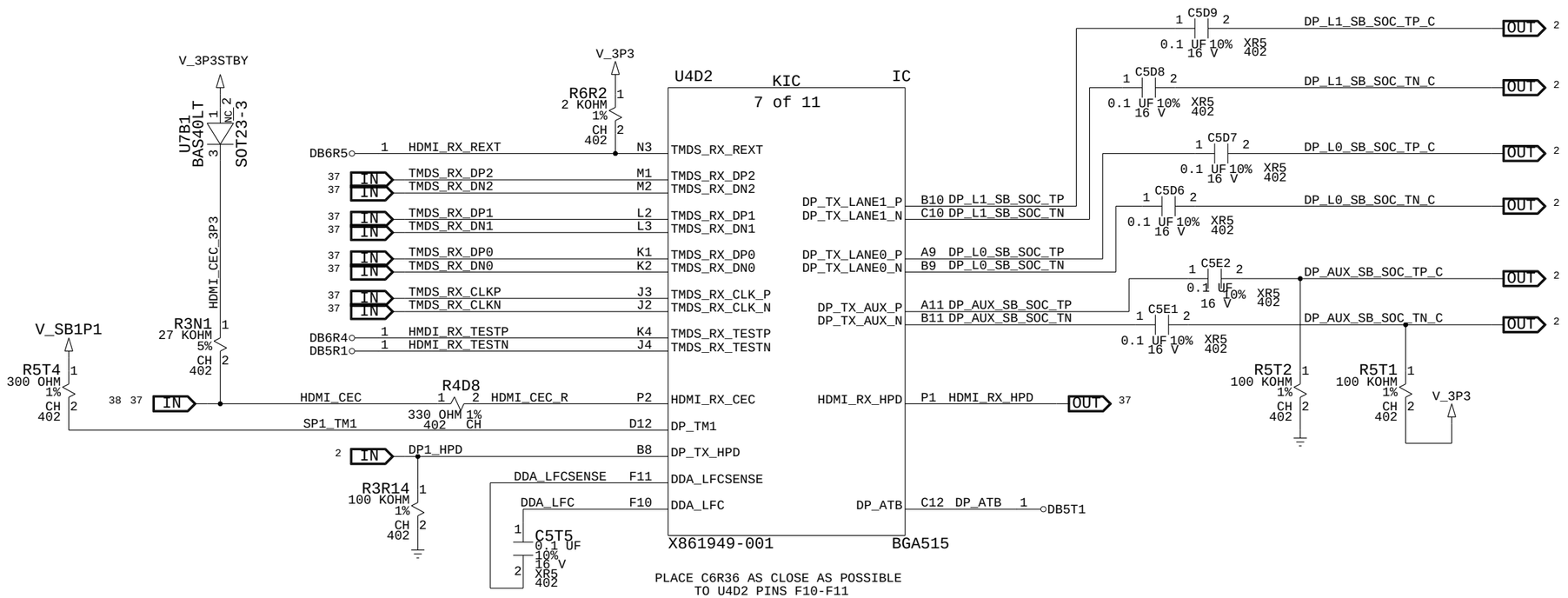
X861949-001

BGA515

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	24/72	24/72	G	1.01

KIC: PCIEX,SATA,VIDEO





MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	25/72	25/72	G	1.01

KIC: SMC

U4D2 KIC IC — 4 of 11

Pin	Signal
AF23	SMC_RST_N_IN
V23	SMC_P4_GPIO7
V22	SMC_P4_GPIO6
T23	SMC_P4_GPIO5
U23	SMC_P4_GPIO4
U22	SMC_P4_GPIO3
U21	SMC_P4_GPIO2
T21	SMC_P4_GPIO1
R23	SMC_P4_GPIO0
L21	SMC_P3_GPIO7
M22	SMC_P3_GPIO6
N22	SMC_P3_GPIO5
N21	SMC_P3_GPIO4
P21	SMC_P3_GPIO3
R21	SMC_P3_GPIO2
R22	SMC_P3_GPIO1
P23	SMC_P3_GPIO0
V25	SMC_UART1_RXD
V26	SMC_UART1_CTS
AE21	SMBUS3_SDA
AD21	SMBUS3_SCK
AE20	SMBUS2_SDA
AF20	SMBUS2_SCK
H6	SMBUS1_SDA
G6	SMBUS1_SCK
G5	SMBUS0_SDA
F5	SMBUS0_SCK
AC14	SECURITY_BYPASS
N26	SMC_IR_IN
T24	SMC_TXD
V24	ODD_3P3V_STBY_EN_N
G2	VREG_CPUGFX_PWRGD
F2	COLD_RESET_N
G1	HDD_PWR_EN
F1	SOC_THERMTRIP_R
G3	ODD_PWR_EN
H5	ODD_OPEN
J5	SP_SMC_INT_R
J6	ODD_WAKE_N
J21	TDI_R
H21	SOC_PWR_OK_R
J22	SOC_RST_N_R
K22	TDO_R
J23	FAN_PWM
H25	IR_MOD_TIMER_IN
J25	IR_BLAST_SMC_OUT
H26	LED_NEXUS
M23	HDMI_V_5P0_EN
M24	IR_MOD_TIMER_OUT
L23	DP0_HPD_SB
K24	WIFI_RESET_N
L25	ENET_LPMODE_N
M25	HDMI_RX_DDC_5V_E_R
L26	SMC_DBG_LED0_SWO
K26	VREG_1P1_ACTIVE_EN

X861949-001 BGA515

3.3V LEVEL SHIFTER

U5E2 XSTR

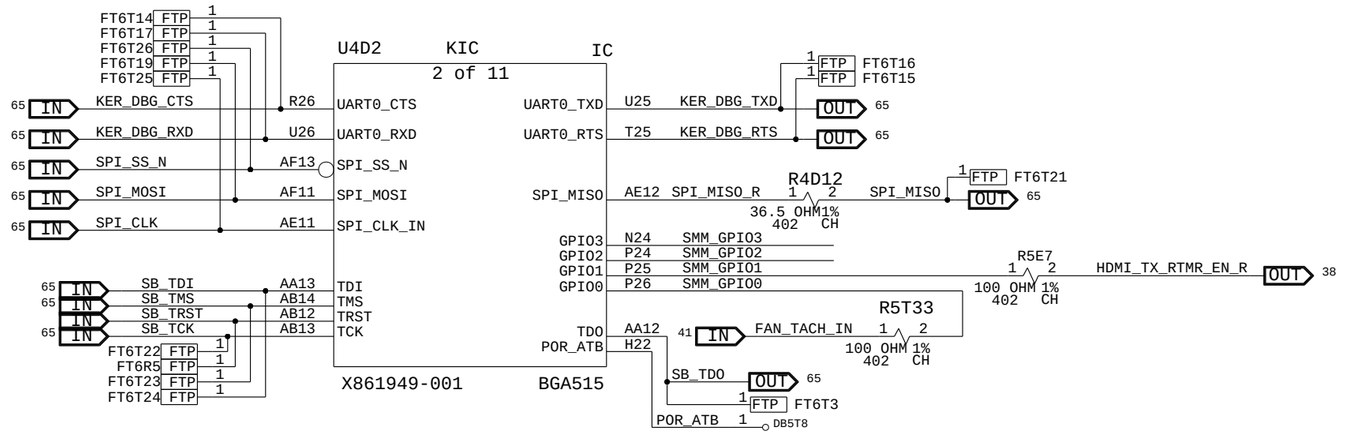
USED BY SMC IR BLASTER SOLUTION

SMC_IR_BLASTER_GPIO'S

SMC_P1_GPIO1 > IR_SMC_BLAST_OUT (TIMER 0)
SMC_P0_GPIO6 > IR_MOD_TIMER_OUT (TIMER 1)
SMC_P4_GPIO2 > IR_MOD_TIMER_IN (TIMER 2)

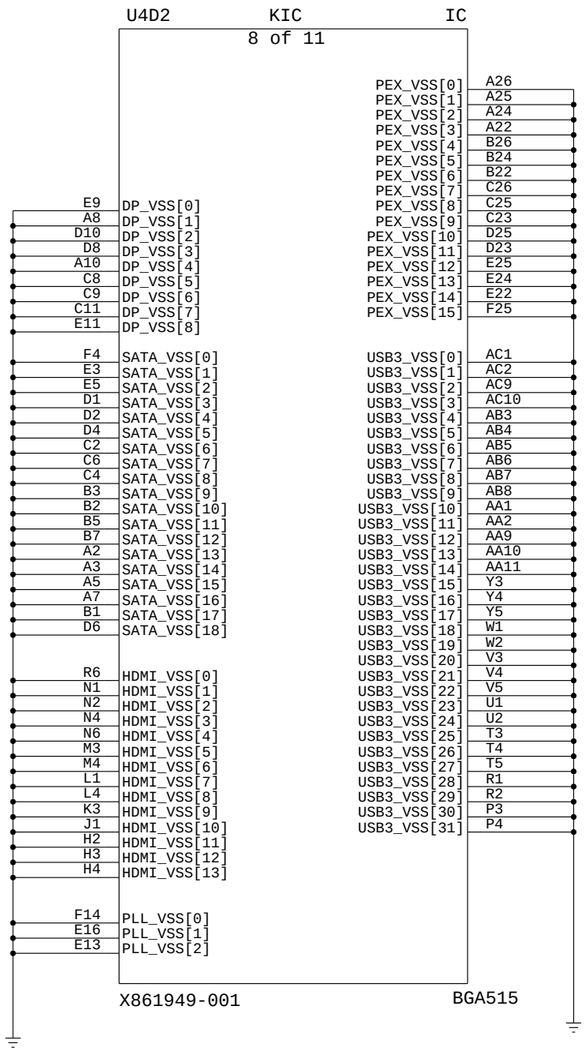
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	26/72	26/72	G	1.01

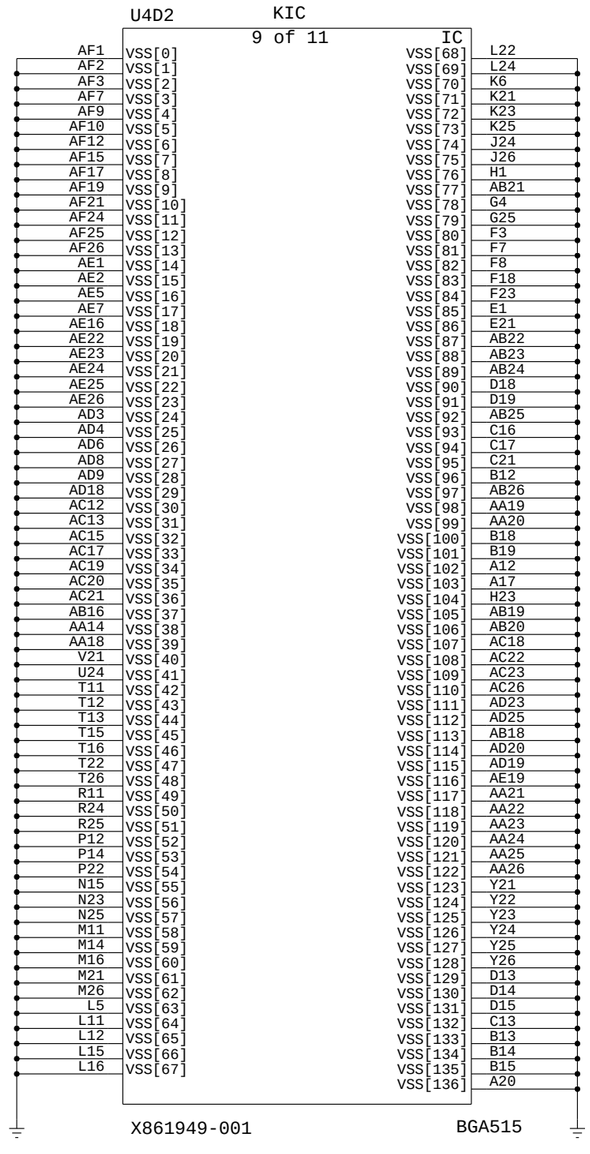
KIC: FACET



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	27/72	27/72	G	1.01

KIC: POWER





MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	28/72	28/72	G	1.01

KIC: CLOCKS,STRAPPING,POR

SEE PAGE 32 FOR ADDITIONAL INFORMATION
100 MHZ CLOCK ENABLE AND ROUTING

32 IN PEX_REF_CLK1_REQN_R

R4B3 0 OHM 5% 402 CH

V_3P3

R4C26 100 KOHM 5% EMPTY 402

R4B10 100 KOHM 5% EMPTY 402

R5T14 100 KOHM 1% CH 402

R5T15 100 KOHM 5% EMPTY 402

V_1P8STBY

KIC_BASE

U4D2 KIC 1 of 11 IC

	G22	V_VDD18_DET	
PEX_REF_CLK1_REQN	C18	PEX_REF_CLK1_REQN	
PEX_REF_CLK2_REQN	C19	PEX_REF_CLK2_REQN	
	AD22	VPGM	
DB5T2 BKUP_SB_SYNC_CLKP	E18	100M_SYNC_CLK_BYPASS_DP	
DB5T3 BKUP_SB_SYNC_CLKN	E19	100M_SYNC_CLK_BYPASS_DN	
SMC_XTAL_OUT	A18	XTAL_CLK_OUT	
SMC_XTAL_IN	A19	XTAL_CLK_IN	
	AD24	TEST	
SMC_BYPASS_CLK_IN	D21	SMC_BYPASS_CLK_IN	
BKUP_SB_PEX_REF_CLKP	E20	PCIE_BYPASS_CLK_DP	
BKUP_SB_PEX_REF_CLKN	D20	PCIE_BYPASS_CLK_DN	
DB5T4 250M_CLK_BYPASS_DP	F20	250M_CLK_BYPASS_DP	
DB5T6 250M_CLK_BYPASS_DN	F21	250M_CLK_BYPASS_DN	
DB5T10 BKUP_SB_AV_CLKP	D16	AV_CLK_BYPASS_DP	
DB5T9 BKUP_SB_AV_CLKN	D17	AV_CLK_BYPASS_DN	

X861949-001

R5E1 1 KOHM 1% 402 CH

R5E2 249 KOHM 1% 402 CH

R5E5 49.9 OHM 1% EMPTY 402

R5T17 10 KOHM 1% 402 CH

Y5D1 25 MHZ SM XTAL

C5E3 30 PF 5% 50 V NPO 402

C5E4 30 PF 5% 50 V NPO 402

DEBUG
CONN J5F3 MINI RF
GND1 GND2 IO1
X864758-001 SM

DEBUG
CONN J5E2 MINI RF
GND1 GND2 IO1
X864758-001 SM

ULTRA MINIATURE COAX
TO SMA CABLE ASSEMBLY
DIGIKEY PN: 931-1098-ND

V_12P0_GATED

C5T27 1 UF 10% 16 V X5R 603

R5T26 11.3 KOHM 1% CH 402

R5T26/V_EXT_DET IS NOW CONNECTED TO V_12P0_GATED

R5T18 1.13 KOHM 1% CH 402

V_EXT_DET	G21	V_EXT_DET	
V_EXT_OK	F22	V12P0_PWRGD	OUT 53
SMC_RST_N_OUT	AF22	SMC_RST_N_OUT_R	
I27			DB5E5
CLK_MONITOR	E17	SB_CLK_MONITOR	
RTC_CLK_OUT	A21	RTC_CLK_OUT	
EMMC_CLK	AC16	MMC_CLK	OUT 33
ENET_REF_CLK	C20	ENET_REF_CLK	
SOC_NS_SYNC_CLK_DP	B16	SOC_SYNC_NS_100M_CLKP	OUT 2
SOC_NS_SYNC_CLK_DN	B17	SOC_SYNC_NS_100M_CLKN	OUT 2
SOC_NS_AV_CLK_DP	A13	SOC_AV_NS_100M_CLKP	OUT 2
SOC_NS_AV_CLK_DN	A14	SOC_AV_NS_100M_CLKN	OUT 2
PEX_REF_CLK2_DP	B20	SOC_NB_PEX_SS_100M_CLKP	OUT 2
PEX_REF_CLK2_DN	B21	SOC_NB_PEX_SS_100M_CLKN	OUT 2
PEX_REF_CLK1_DP	C14	ENET_PEX_SS_100M_CLKP	OUT 32
PEX_REF_CLK1_DN	C15	ENET_PEX_SS_100M_CLKN	OUT 32
PEX_REF_CLK0_DP	A15	SOC_PEX_SS_100M_CLKP	OUT 2
PEX_REF_CLK0_DN	A16	SOC_PEX_SS_100M_CLKN	OUT 2

BGA515

DB5E6
KEEP DB55E5 AND DB5E6
CLOSE TO EACH OTHER IN
LAYOUT

R4E30 1 KOHM 1% 402 CH

SMC_RST_N OUT 26 65

C4E3 270 PF 10% 50 V X7R 402

RTC_RETAIL
R5E4 0 OHM 5% 402 CH

X32K_X2_R OUT 8

R5T16 100 OHM 1% 402 CH PEX_XTAL2 OUT 32

R5E6 69.8 KOHM 1% EMPTY 402

MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X861949-005	IC	U4D2	IC,KRAKEN SB,BGA515	KIC_RETAIL
X861949-003	IC	U4D2	IC,KRAKEN SB,BGA515	KIC_DEV

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	29/72	29/72	G	1.01

KIC: POWER

FET PREVENTS LEAKAGE THROUGH KRAKEN
FROM STANDBY RAILS TO 3P3 RAIL

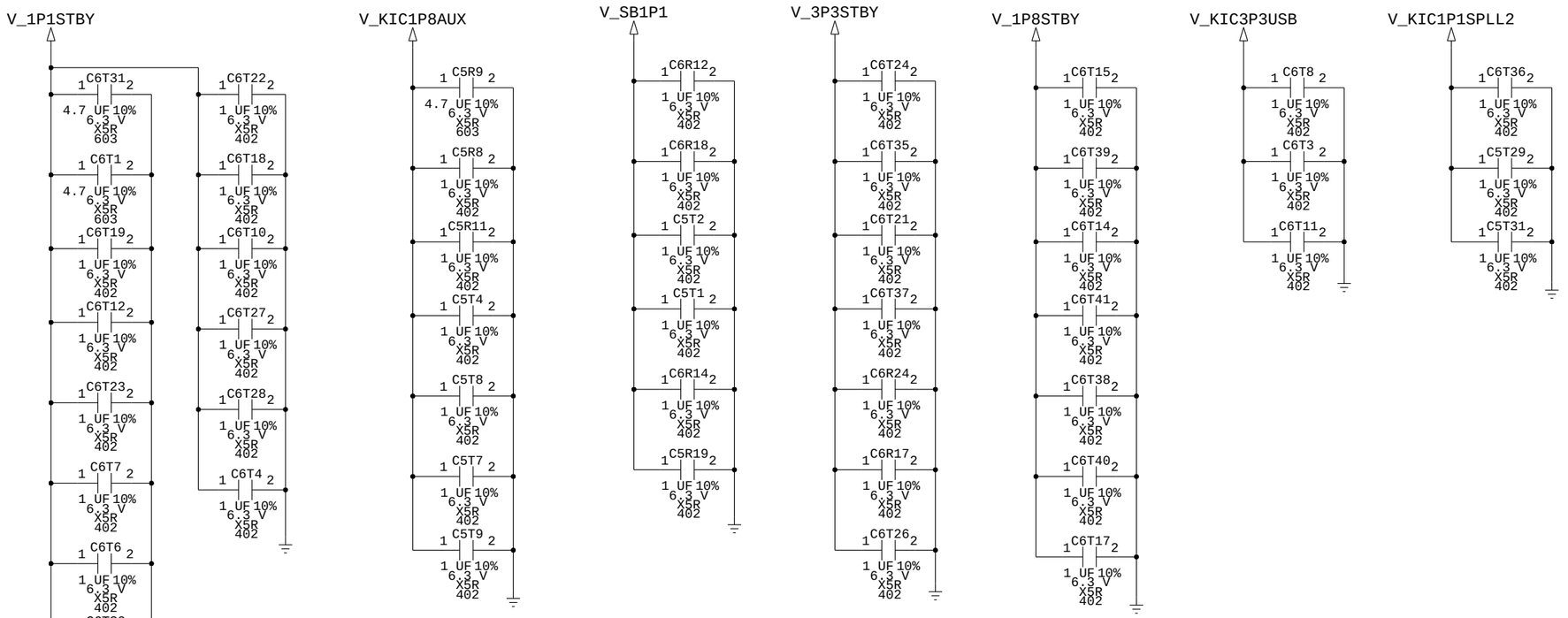
U4D2 KIC IC
10 of 11

11 of 11

X861949-001 BGA515

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	30/72	30/72	G	1.01

KIC: DECOUPLING



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	31/72	31/72	G	1.01

ETHERNET CONTROLLER

RED = 1 GBPS LINK
GREEN = 10 MBPS LINK
NO INDICATION FOR 100 MBPS LINK

DEBUG D4C1 LED RED-GREEN SM

V_3P3_ENET

R4C1 10 KOHM 1% CH 402

R4C3 0 OHM 5% 402 CH

R4B9 10 KOHM 1% EMPTY 402

FT6P3 FTP

ENET_LPMODE_N

U6N1 RTL8111HM IC

MDIP3 9 MDIN3 10 MDIP2 6 MDIN2 7 MDIP1 4 MDIN1 5 MDIP0 1 MDIN0 2

DEBUG R4C6 511 OHM 1% 402 CH

DEBUG R4C7 511 OHM 1% CH 402

ENET_LED2_R
ENET_LED0_R
ENET_LED2 25 LED2
ENET_GPO 26 LED1/GPO
ENET_LED0 27 LED0

R6N2 0 OHM 5% 402 CH

PEX_XTAL2

LAYOUT: PLACE R6N2 AND R6P1
PIN 1 ON TOP OF EACH OTHER

R6P1 1.2 KOHM 1% 402 EMPTY

R6N1 2.49 KOHM 1% CH 402

DB4B1

ENET_RSET 31 RSET

XTAL2_R
XTAL1_R
ENET_XTAL2 29 CKXTAL2
ENET_XTAL1 28 CKXTAL1

Y6P1 25 MHZ SM EMPTY

R6P2 0 OHM 5% 402 EMPTY

C6N1 27 PF 5% 25 V EMPTY 402

C6P1 27 PF 5% 25 V EMPTY 402

ENET_ISOLATE_N 20 ISOLATE_N

PEX_SOC_ENET_TP_C 13 HSIP
PEX_SOC_ENET_TN_C 14 HSIN

ENET_PEX_SS_100M_CLKP 15 REFCLK_P
ENET_PEX_SS_100M_CLKN 16 REFCLK_N
PERST_N 19

X896524-002 QFN33 MPAD 33

V_3P3_ENET

R4C5 10 KOHM 1% CH 402

FT6P2 FTP

R4C8 10 KOHM 1% EMPTY 402

WIFI_PEX_SS_100M_CLKP
WIFI_PEX_SS_100M_CLKN

R4C23 0 OHM 5% 402 EMPTY

R4C24 0 OHM 5% 402 EMPTY

V_3P3_ENET

R4C4 10 KOHM 1% EMPTY 402

ENET_PCIE_RST_N

FT6P6 FTP

R4C2 10 KOHM 1% CH 402

V_3P3_ENET

PLACE 0.1UF CLOSE TO PINS 11, 23 AND 32
PLACE 4.7UF CLOSE TO PINS 11 AND 32

C4B9 0.1 UF 10% 16 V XR5 402
C4C3 0.1 UF 10% 16 V XR5 402
C4B7 0.1 UF 10% 16 V XR5 402
C4B4 4.7 UF 10% 6.3 V X5R 603
C4B12 4.7 UF 10% 6.3 V X5R 603

V_1P0_ENET

PLACE 0.1UF CLOSE TO PINS 3, 8, 22 AND 30

C4C2 0.1 UF 10% 16 V XR5 402
C4C1 0.1 UF 10% 16 V XR5 402
C4B10 0.1 UF 10% 16 V XR5 402
C4B11 1 UF 10% 16 V XR5 402
C4C4 1 UF 10% 6.3 V X5R 402
C4B3 4.7 UF 10% 6.3 V X5R 603

AVDD33_2 32
AVDD33_1 11
DVDD10 22
AVDD10_3 30
AVDD10_2 8
AVDD10_1 3
VDDREG 23

PLACE 0.1UF CLOSE TO PIN 24

C4C6 0.1 UF 10% 16 V XR5 402

L6P1 4.7 UH X821308-001 EMPTY

FT6P5

DB4C1

R4C10 0 OHM 5% 805 CH

REGOUT 24 ENET_REGOUT

LANWAKE_N 21 ENET_WAKE_N

CLKREQ_N 12 PEX_REF_CLK1_REQN_R

V_3P3_ENET

R4C9 10 KOHM 1% CH 402

V_3P3_ENET

R4B8 10 KOHM 1% CH 402

OUT 29

HSOP 17
HSON 18

PEX_ENET_SOC_TP

C6P2 0.1 UF 10% 16 V XR5 402

PEX_ENET_SOC_TP_C OUT 2

PEX_ENET_SOC_TN

C6P3 0.1 UF 10% 16 V XR5 402

PEX_ENET_SOC_TN_C OUT 2

SOFT START
NOM. VOLTAGE: 3.3V
MAX POWER: 570MW

V_3P3

Q4B3 FET SOT23 X889118-002

V_3P3_ENET

C4B6 1 UF 10% 6.3 V X5R 402

C4B8 0.22 UF 10% 10 V X7R 402

C4B2 10 UF 20% 6.3 V EMPTY 603

C4B5 10 UF 20% 6.3 V X5R 603

R4B5 1 KOHM 1% EMPTY 402

V_ENET_IR_OUT_G

R4B6 100 OHM 1% 402 CH

V_ENET_IR_OUT_CGD

R4B7 20 KOHM 1% CH 402

NOTES:

1. THE RTL8111HM IS CONFIGURED FROM REALTEK WITH EFUSE "ENABLE LAN DISABLE" MODE SET, WHICH PROVIDES DEFAULT SUPPORT FOR SIGNAL ENET_LPMODE_N. THIS FEATURE CAN BE OVERRIDDEN BY SOFTWARE
2. TO SUPPORT A PCIE WIFI INTERFACE (J3C1), THE FOLLOWING BOM CHANGES ARE NEEDED:
   - A: POPULATE R4C23 AND R4C24, WHICH WILL SEND THE 100 MHZ CLOCK TO THE WIFI INTERFACE. IN ADDITION, IT IS CRITICAL THAT EACH LEG OF THE CLOCK BE TERMINATED AT THE FAR END INTO 50 OHMS.
   - B: ON PAGE 2, POPULATE AC COUPLING CAPS C6P6 AND C6P7
   - C: FOR PRELIMINARY DEVELOPMENT TESTING, NO OTHER BOM CHANGES ARE REQUIRED. THE ETHERNET CONTROLLER WILL MANAGE THE ENABLING OF THE 100 MHZ CLOCK.
   - D: FOR PRODUCTION RETAIL, ON PAGE 29 REMOVE R4B3 AND POPULATE R4C26. THIS CHANGE WILL REQUIRE MODIFIED SMC FIRMWARE TO EXPLICITLY ENABLE THE 100 MHZ CLOCK.

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	32/72	32/72	G	1.01

EMMC

MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X934261-001	IC	U3D1	MEM,SM,8GB,EMMC,15NM,THGBMFG6C1LBAILL	EMMC_TOSHIBA
X904693-001	IC	U3D1	MEM,SM,8GB,EMMC,FLASH,16NM,BGA153,KLM8G1GEND-B031,SAMSUNG	EMMC_SAMSUNG
X939723-001	IC	U3D1	MEM,SM,8GB,64GBX1,H26M41204HPR	EMMC_HYNIX_5P0
X800617-001	CH	R4E9,R4E13,R4E11,R4E17,R4E15,R4E21,R4E19,R4E23	RES-FXD,SM,49.9 OHM 1%,1/16W,0402	EMMC_TOSHIBA,EMMC_SAMSUNG
X809936-001	CH	R4E9,R4E13,R4E11,R4E17,R4E15,R4E21,R4E19,R4E23	RES-FXD,SM,100 OHM 1%,1/16W,0402	EMMC_HYNIX_5P0

V_1P8STBY

R4E4 R4E8 R4E12 R4E10 R4E16 R4E14 R4E20 R4E18 R4E22 — 402 CH — 10 KOHM 1%

EMMC_BASE R4E9 — 0 OHM 5% 402 CH
EMMC_BASE R4E13 — 0 OHM 5% 402 CH
EMMC_BASE R4E11 — 0 OHM 5% 402 CH
EMMC_BASE R4E17 — 0 OHM 5% 402 CH
EMMC_BASE R4E15 — 0 OHM 5% 402 CH
EMMC_BASE R4E21 — 0 OHM 5% 402 CH
EMMC_BASE R4E19 — 0 OHM 5% 402 CH
EMMC_BASE R4E23 — 0 OHM 5% 402 CH

33 BI MMC_DATA<7>
33 BI MMC_DATA<6>
33 BI MMC_DATA<5>
33 BI MMC_DATA<4>
33 BI MMC_DATA<3>
33 BI MMC_DATA<2>
33 BI MMC_DATA<1>
33 BI MMC_DATA<0>

R4E7 — 0 OHM 5% 402 CH — MMC_CMD
R4E5 — 0 OHM 5% 402 CH — 29 BI MMC_CLK
R4E1 — 1 KOHM 1% 402 CH — FT6T4 FTP 1 — 33 BI MMC_RST_N
R4E2 — 10 KOHM 1% — EMPTY 402 — FT6T2 FTP 1

V_1P8STBY
R4E6 — 10 KOHM 1% — CH 402

R7T2 — 10 KOHM 5% 402 EMPTY — MMC_DS_5P0

EMMC_BASE
U3D1 IC
EMMC_153PIN
1 of 3
X894231-002 BGA153

MMC_DATA_R<7>	B6	DATA7
MMC_DATA_R<6>	B5	DATA6
MMC_DATA_R<5>	B4	DATA5
MMC_DATA_R<4>	B3	DATA4
MMC_DATA_R<3>	B2	DATA3
MMC_DATA_R<2>	A5	DATA2
MMC_DATA_R<1>	A4	DATA1
MMC_DATA_R<0>	A3	DATA0
MMC_DS_5P0	H5	DATA_STROBE
MMC_CMD_R	M5	CMD
MMC_CLK_R	M6	CLK
MMC_RST_N_R	K5	RSTN

VDD4 P5, VDD3 P3, VDD2 N4, VDD1 M4, VDD0 C6
VDDF3 K9, VDDF2 F5, VDDF1 E6, VDDF0 J10
VDDI C2 VDDI_ROUTE
VSS10 J5 MMC_VSS10_5P0, VSS9 A6, VSS8 P4, VSS7 N2, VSS6 C4 MMC_VSS9_5P0, VSS5 N5, VSS4 G5, VSS3 P6, VSS2 E7, VSS1 K8, VSS0 H10

V_1P8STBY
C7T1 0.1 UF 10% 16 V XR5 402
C4E1 0.1 UF 10% 16 V XR5 402
C6T13 2.2 UF 20% 6.3 V X5R 402
C7T5 1 UF 10% 6.3 V X5R 402
C6T16 0.1 UF 10% 16 V XR5 402
C6T25 0.1 UF 10% 16 V XR5 402
C6T20 2.2 UF 20% 6.3 V X5R 402

V_3P3STBY
C7T2 0.1 UF 10% 16 V XR5 402
C7T3 0.1 UF 10% 16 V XR5 402
C7T4 2.2 UF 20% 6.3 V X5R 402

R7T1 — 0 OHM 5% 402 EMPTY — MMC_VSS10_5P0
R6T2 — 0 OHM 5% 402 EMPTY — MMC_VSS9_5P0
C4E2 1 UF 10% 6.3 V X5R 402

MMC_X NETS ARE DUMMY NETS TO IMPROVE PAD ADHESION

U4D2 KIC IC
5 of 12
I446
X861949-001 BGA515

MMC_DATA7	AE18	7	
MMC_DATA6	AD15	6	
MMC_DATA5	AE15	5	
MMC_DATA4	AD16	4	
MMC_DATA3	AF16	3	
MMC_DATA2	AF18	2	
MMC_DATA1	AD17	1	
MMC_DATA0	AE17	0	
MMC_CMD	AF14	MMC_CMD	OUT 33
MMC_RESET_N	AE14	MMC_RST_N	OUT 33

MMC_DATA<7..0> BI 33

U3D1 IC
EMMC_153PIN
2 of 3
X894231-002 BGA153

MMC_X7	A1	NC0	NC35	E1
D4	A2	NC1	NC36	E2
	A8	NC2	NC37	E3
	A9	NC3	NC38	E12
	A10	NC4	NC39	E13
	A11	NC5	NC40	E14
	A12	NC6	NC41	F1
MMC_X6	A13	NC7	NC42	F2
MMC_X4	A14	NC8	NC43	F3
MMC_X8	B1	NC9	NC44	F12
	B7	NC10	NC45	F13
	B8	NC11	NC46	F14
	B9	NC12	NC47	G1
	B10	NC13	NC48	G2
	B11	NC14	NC49	G12
	B12	NC15	NC50	G13
	B13	NC16	NC51	G14
MMC_X5	B14	NC17	NC52	H1
D3	C1	NC18	NC53	H2
D5	C3	NC19	NC54	H3
D7	C7	NC20	NC55	H12
	C8	NC21	NC56	H13
	C9	NC22	NC57	H14
	C10	NC23	NC58	J1
	C11	NC24	NC59	J2
	C12	NC25	NC60	J3
	C13	NC26	NC61	J12
	C14	NC27	NC62	J13
	D1	NC28	NC63	J14
	D2	NC29	NC64	K1
	D3	NC30	NC65	K2
	D4	NC31	NC66	K3
	D12	NC32	NC67	K12
	D13	NC33	NC68	K13
	D14	NC34	NC69	K14

DB3E7 DB3E6 DB3E5 DB3E4 DB3E3 DB3E2

VDDI_ROUTE

U3D1 IC
EMMC_153PIN
3 of 3
X894231-002 BGA153

	A7	RFU1	NC80	M8
D6	C5	RFU2	NC81	M9
	E5	RFU3	NC82	M10
	E8	RFU4	NC83	M11
	E9	RFU5	NC84	M12
	E10	RFU6	NC85	M13
	F10	RFU7	NC86	M14
	G3	RFU8	NC87	N1 MMC_X11
	G10	RFU9	NC88	N3
			NC89	N6
	K6	RFU12	NC90	N7
	K7	RFU13	NC91	N8
	K10	RFU14	NC92	N9
	P7	RFU15	NC93	N10
	P10	RFU16	NC94	N11
	L1	NC70	NC95	N12
	L2	NC71	NC96	N13
	L3	NC72	NC97	N14 MMC_X3
	L12	NC73	NC98	P1 MMC_X10
	L13	NC74	NC99	P2 MMC_X12
	L14	NC75	NC100	P8
	M1	NC76	NC101	P9
	M2	NC77	NC102	P11
	M3	NC78	NC103	P12
	M7	NC79	NC104	P13 MMC_X2
			NC105	P14 MMC_X1

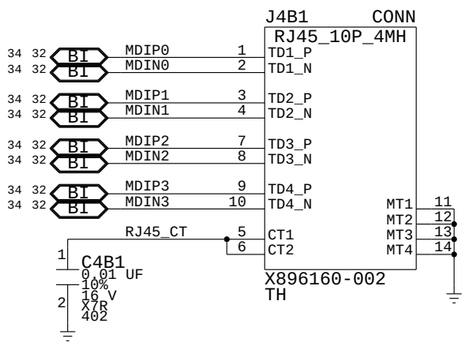
DUMMY_EMMC_P10
1 DB3E1

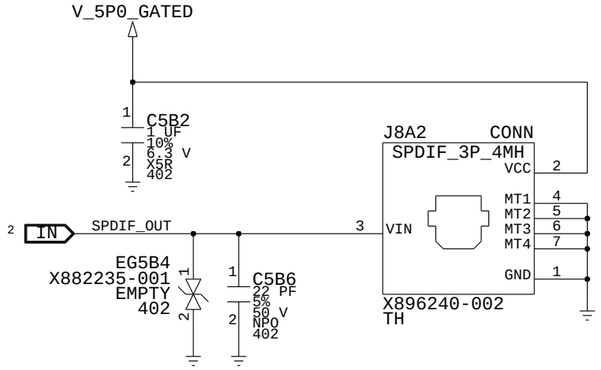
SIGNALS ROUTED TO NC PINS FOR ESCAPE

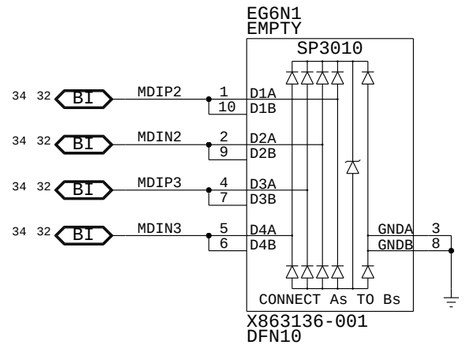
D7	PIN B6 ROUTED TO PIN C7, C8
D6	PIN B5 ROUTED TO PIN C5
D5	PIN B4 ROUTED TO PIN C3, D3
D4	PIN B3 ROUTED TO PIN A2
D3	PIN B2 ROUTED TO PIN C1
VDDI	PIN C2 ROUTED TO PIN D1
GND	PIN N2 ROUTED TO PIN M1

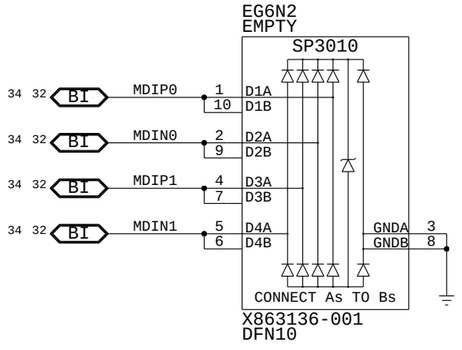
MICROSOFT CONFIDENTIAL	PROJECT NAME: Kingston	PAGE: 33/72	CSA PAGE: 33/72	FAB: G	VER: 1.01

CONN: RJ45,TOSLINK



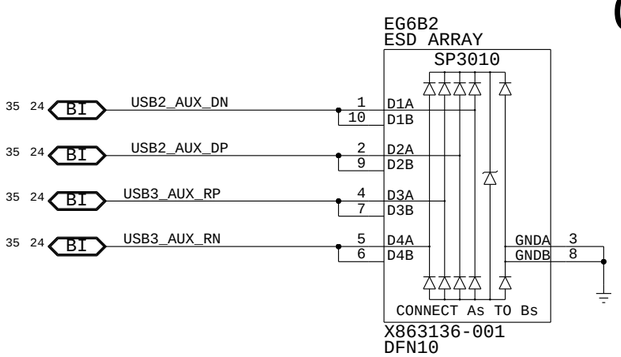


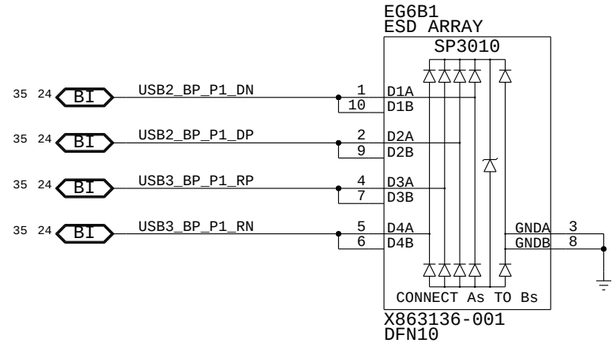


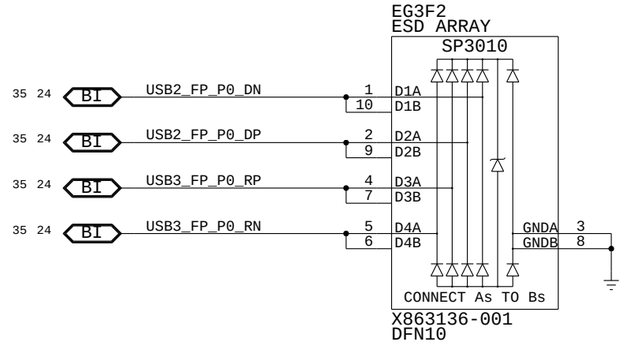


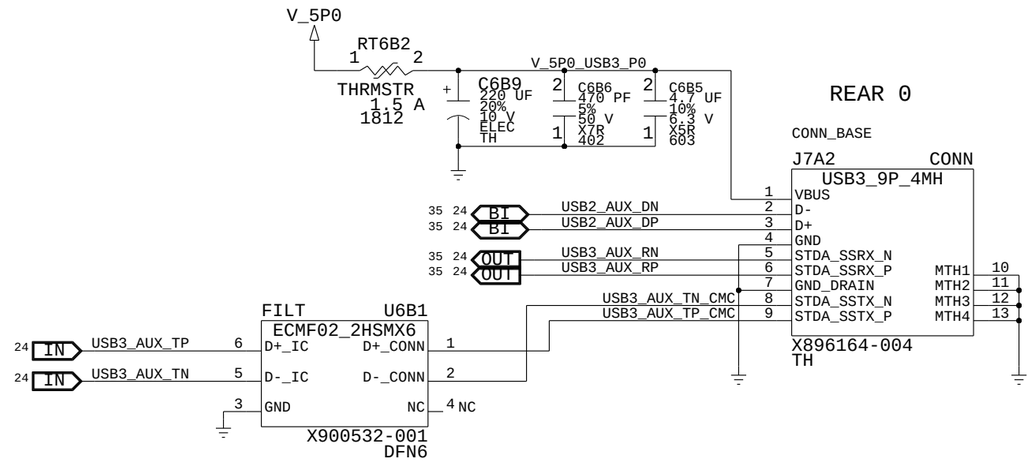
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	34/72	34/72	G	1.01

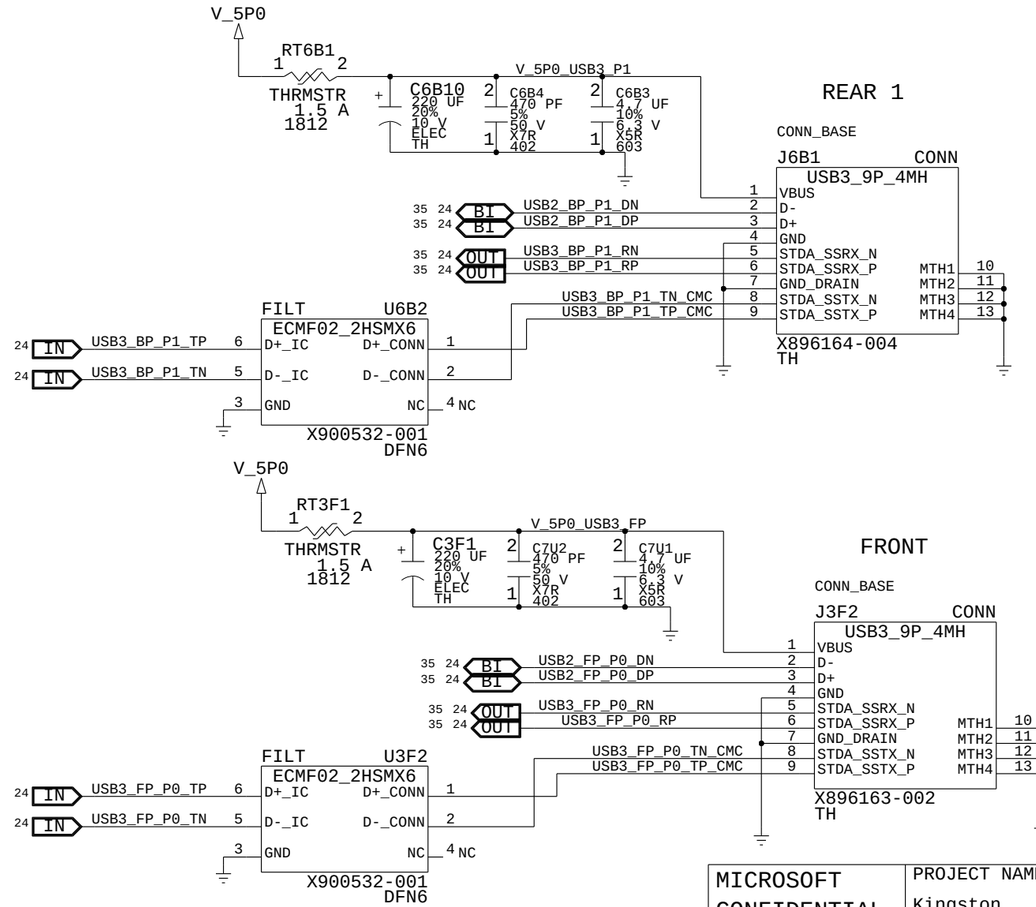
CONN: USB (FRONT & REAR)









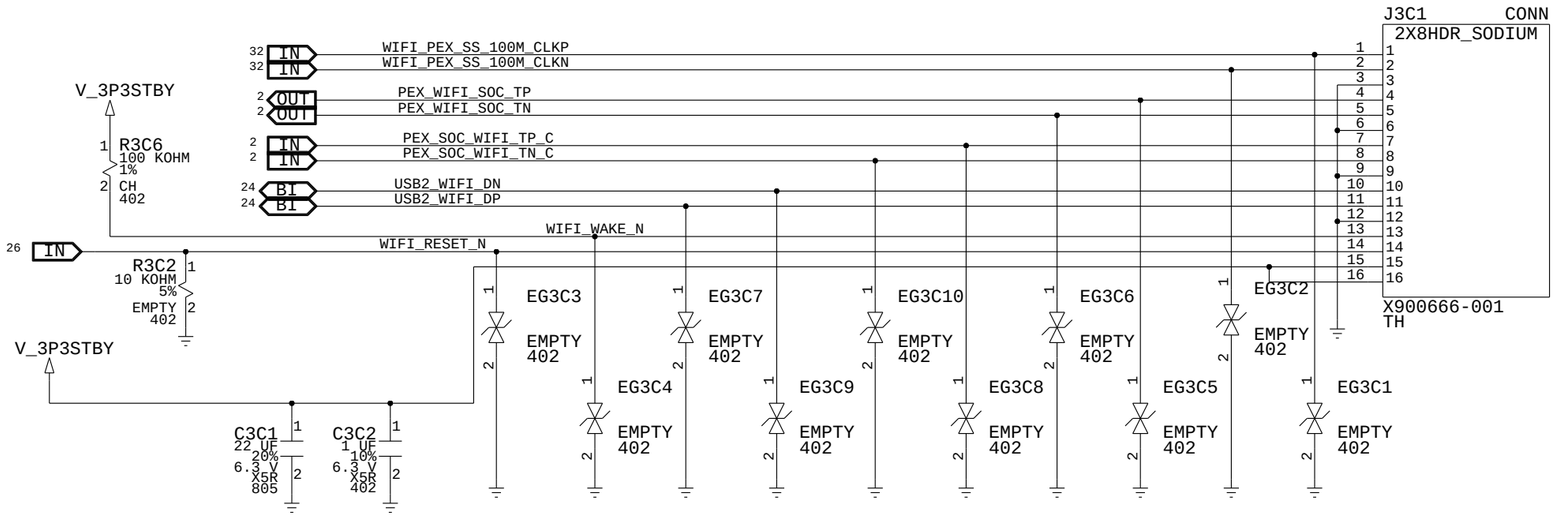


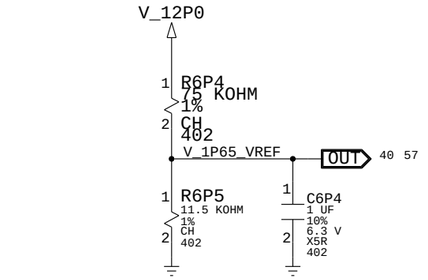
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X896164-004	CONN	J7A2,J6B1	CONN-USB,TH,009,FEMALE,USB3-A,BLACK NI, KGS	USB_REAR_FOXC_BLACK
X896164-003	CONN	J7A2,J6B1	CONN-USB,SM,009,FEMALE,USB3-A,PLAIN NI, KGS	USB_REAR_FOXC_PLAIN

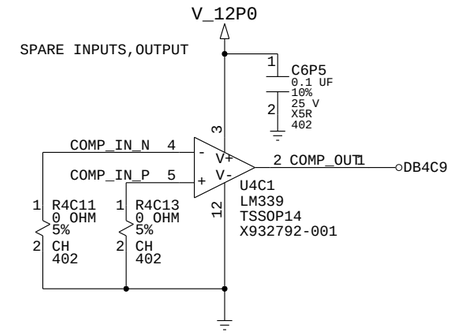
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X896163-004	CONN	J3F2	CONN-USB,TH,009,FEMALE,FRONT,USB3-A,BLACK NICKEL,KGS	USB_FRONT_FOXC
X907318-002	CONN	J3F2	CONN-USB,TH,009,FEMALE,FRONT,USB3-A,KGS,(QUAL-AMPHENOL)	USB_FRONT_AMPH

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	35/72	35/72	G	1.01

CONN: WIFI

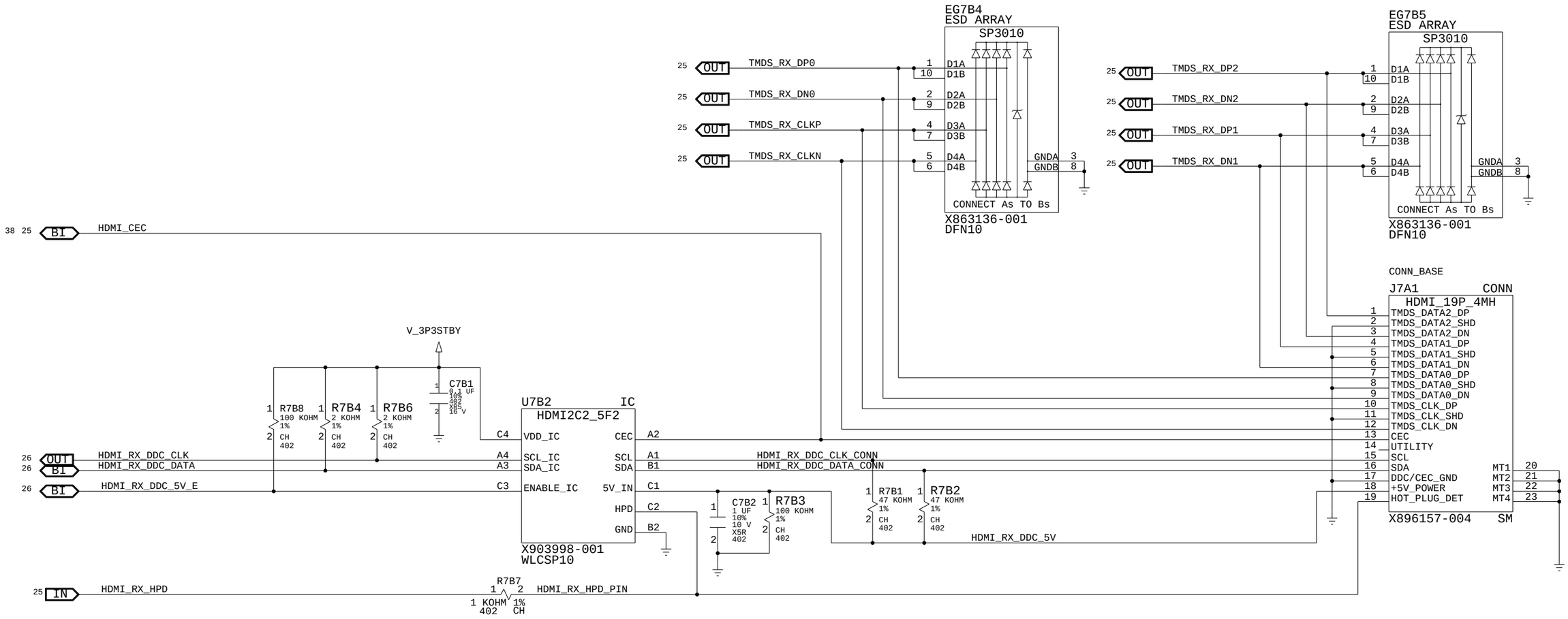






MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	36/72	36/72	G	1.01

CONN: HDMI IN



MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X896157-006	CONN	J7A1,J8A1	CONN-HDMI,SM.019,FEMALE,W/EMI FLANGES,BLACK NI,KGS	HDMI_FOXC_BLACK
X896157-005	CONN	J7A1,J8A1	CONN-HDMI,SM,019,FEMALE,W/ EMI FLANGES,PLAIN NI, KGS	HDMI_FOXC_PLAIN

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	37/72	37/72	G	1.01

CONN: HDMI OUT

DP159 CONTROL SLAVE ADDRESS			
SELECTED	A1/A0	7-BIT ADDRESS	8-BIT ADDRESS
-->	00	0X5E	BC/BD
	01	0X5D	BA/BB
	10	0X5C	B8/B9
	11	0X5B	B6/B7

V_3P3STBY
C2P31 0.1 UF 10% 402 X5R 16 V
C3N1 0.1 UF 10% 402 X5R 16 V
C2P30 10 UF 20% 603 X5R 6.3 V

V_1P1STBY
C8C13 0.1 UF 10% 402 X5R 16 V
C8C12 0.01 UF 10% 402 X7R 16 V
C7B4 10 UF 20% 603 X5R 6.3 V
C7B3 22 UF 20% 805 X5R 6.3 V
C8C14 0.1 UF 10% 402 X5R 16 V
C7B7 0.1 UF 10% 402 X5R 16 V
C7B5 0.01 UF 10% 402 X7R 16 V

U8B2 IC
SN75DP159
3 OF 3
POWER
VDD_1 12
VDD_2 19
VDD_3 20
VDD_4 31
VDD_5 40
VCC_1 11
VCC_2 37
GND_1 15
GND_2 35
MPAD 41
X944186-001
QFN41

65 49 44 41 26 IN SMBUS_CLK — R8C2 0 OHM 5% CH 402 — HDMI_TX_RTMR_SCL_CTL
65 49 44 41 26 BI SMBUS_DATA — R8C33 0 OHM 5% CH 402 — HDMI_TX_RTMR_SDA_CTL
27 IN HDMI_TX_RTMR_EN_R — R3N2 0 OHM 5% CH 402 — HDMI_TX_RTMR_EN
FT3N1 FTP

HDMI_TX_RTMR_EN — R3N3 10 KOHM 1% CH 402 — C7B6 0.1 UF 10% 6.3 V EMPTY 402

U8B2 IC
SN75DP159
1 OF 3
CONTROL
SCL_CTL 13
SDA_CTL 14
OE 36
I2C_EN/PIN 8 DP159_I2C_EN
PRE_SEL 16 DP159_PRE_SEL
EQ_SEL/A0 17 DP159_EQ_SEL_A0
HDMI_SEL_N/A1 23 DP159_HDMI_SEL_N_A1
SLEW_CTL 34 DP159_SLEW_CTL
VSADJ 18
X944186-001
QFN41

V_3P3STBY
R3P1 60.4 KOHM 1% CH 402
R8C36 60.4 KOHM 1% EMPTY 402
R8C37 60.4 KOHM 1% EMPTY 402
R2N4 60.4 KOHM 1% EMPTY 402
R7B13 60.4 KOHM 1% EMPTY 402
R3P2 60.4 KOHM 1% EMPTY 402
R8C35 60.4 KOHM 1% EMPTY 402
R8B13 60.4 KOHM 1% CH 402
R2N3 60.4 KOHM 1% CH 402
R7B14 60.4 KOHM 1% CH 402
DP159_VSADJ — R8B12 4.64 KOHM 1% CH 402

EG8B6
ESD ARRAY
SP3010
D1A 1
D1B 10
D2A 2
D2B 9
D3A 4
D3B 7
D4A 5
D4B 6
GNDA 3
GNDB 8
CONNECT As TO Bs
X863136-001
DFN10

EG8B7
ESD ARRAY
SP3010
D1A 1
D1B 10
D2A 2
D2B 9
D3A 4
D3B 7
D4A 5
D4B 6
GNDA 3
GNDB 8
CONNECT As TO Bs
X863136-001
DFN10

CONN_BASE
J8A1 CONN
HDMI_19P_4MH
TMDS_DATA2_DP 1
TMDS_DATA2_SHD 2
TMDS_DATA2_DN 3
TMDS_DATA1_DP 4
TMDS_DATA1_SHD 5
TMDS_DATA1_DN 6
TMDS_DATA0_DP 7
TMDS_DATA0_SHD 8
TMDS_DATA0_DN 9
TMDS_CLK_DP 10
TMDS_CLK_SHD 11
TMDS_CLK_DN 12
CEC 13
UTILITY 14
SCL 15
SDA 16
DDC/CEC_GND 17
+5V_POWER 18
HOT_PLUG_DET 19
MT1 20
MT2 21
MT3 22
MT4 23
X896157-004 SM

U8B2 IC
SN75DP159
2 OF 3
HDMI
IN_D2_P 1 — OUT_D2_P 30
IN_D2_N 2 — OUT_D2_N 29
IN_D1_P 4 — OUT_D1_P 27
IN_D1_N 5 — OUT_D1_N 26
IN_D0_P 6 — OUT_D0_P 25
IN_D0_N 7 — OUT_D0_N 24
IN_CLK_P 9 — OUT_CLK_P 22
IN_CLK_N 10 — OUT_CLK_N 21
SCL_SRC 38 — SCL_SNK 32
SDA_SRC 39 — SDA_SNK 33
HPD_SRC 3 — HPD_SNK 28
X944186-001
QFN41

2 IN TMDS_TX_DP2 — C7C3 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DP2_C
2 IN TMDS_TX_DN2 — C7C4 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DN2_C
2 IN TMDS_TX_DP1 — C7C5 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DP1_C
2 IN TMDS_TX_DN1 — C7C6 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DN1_C
2 IN TMDS_TX_DP0 — C7C7 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DP0_C
2 IN TMDS_TX_DN0 — C7C8 0.1 UF 10% 16 V XR5 402 — TMDS_TX_DN0_C
2 IN TMDS_TX_CLKP — C7C9 0.1 UF 10% 16 V XR5 402 — TMDS_TX_CLKP_C
2 IN TMDS_TX_CLKN — C7C10 0.1 UF 10% 16 V XR5 402 — TMDS_TX_CLKN_C

TMDS_TX_DP2_CONN 1
TMDS_TX_DN2_CONN 3
TMDS_TX_DP1_CONN 4
TMDS_TX_DN1_CONN 6
TMDS_TX_DP0_CONN 7
TMDS_TX_DN0_CONN 9
TMDS_TX_CLKP_CONN 10
TMDS_TX_CLKN_CONN 12
HDMI_CEC 13
HDMI_TX_SCL_CONN 15
HDMI_TX_SDA_CONN 16
V_5P0_HDMI_OUT 18
HDMI_TX_HPD_CONN 19

V_3P3STBY
R7B11 10 KOHM 1% CH 402
R7B12 10 KOHM 1% CH 402

HDMI_TX_HPD_RTMR
HDMI_TX_SDA_RTMR
HDMI_TX_SCL_RTMR

R7B9 0 OHM 5% CH 402 — HDMI_TX_SCL_R
R7B10 0 OHM 5% CH 402 — HDMI_TX_SDA_R
HDMI_TX_HPD_R

V_3P3STBY
R3N4 2 KOHM 1% CH 402
2 OUT DP0_HPD — R3P3 0 OHM 5% CH 402

V_3P3STBY
R3N5 2 KOHM 1% CH 402
2 IN SOC_DDC_CLK — R2N5 0 OHM 5% EMPTY 402
2 BI SOC_DDC_DATA — R2N6 0 OHM 5% EMPTY 402

26 IN SB_DDC_CLK — R8B8 0 OHM 5% CH 402
26 BI SB_DDC_DATA — R8B6 0 OHM 5% CH 402

V_3P3STBY — C8B3 0.1 UF 10% 16 V X5R 402
V_5P0 — C8B2 0.1 UF 10% 16 V X5R 402

R8B7 0 OHM 5% CH 402
R8B5 0 OHM 5% CH 402

HDMI LOAD SWITCH
NOM.VOLTAGE: 5V
PEAK CURRENT: 0.2A

U8B1 IC
HDMI2C4_5F2
VDD_5V C3
VDD_IC B4
SCL_IC A4 — HDMI_TX_SCL_RTMR_R
SDA_IC A3 — HDMI_TX_SDA_RTMR_R
ENABLE_IC B3
HPD_IC C4 — HDMI_TX_HPD_RTMR_R
CEC B2
SCL A2
SDA A1
5V_OUT C2
HPD C1 — HDMI_TX_HPD_BUF
GND B1
X934148-001
WLCSP12

26 OUT DP0_HPD_SB — R2N2 0 OHM 5% CH 402 — R8B4 0 OHM 5% CH 402 — HDMI_TX_HPD_RTMR_R

26 IN HDMI_V_5P0_EN — R8B11 10 KOHM 1% CH 402
FT2N1 FTP

37 25 BI
HDMI_TX_SCL_BUF
HDMI_TX_SDA_BUF
HDMI_TX_HPD_BUF

R8B9 0 OHM 5% CH 402
R8B2 0 OHM 5% CH 402
R8B3 0 OHM 5% CH 402
R8B1 2 KOHM 1% CH 402
R8B10 2 KOHM 1% CH 402
C8B1 1 UF 10% 10 V X5R 402
R2N1 1 KOHM 1% 402 CH 1/16W

DATA P LINE DELAY ADDED TO BOARD TO COMPENSATE FOR DP159 INTRAPAIR SKEW

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	38/72	38/72	G	1.01

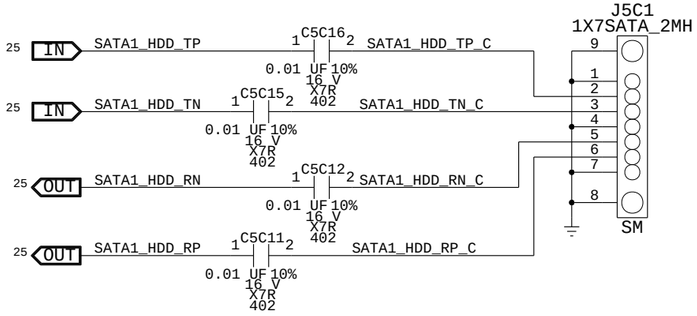
CONN: HDMI SUPPORT

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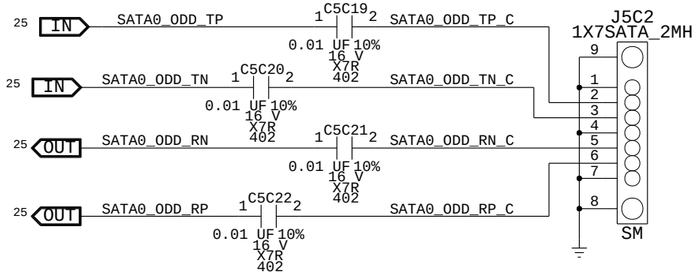
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	39/72	39/72	G	1.01

CONN: ODD & HDD

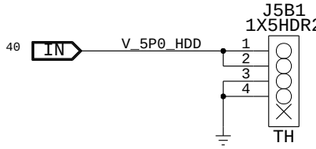
HDD SATA

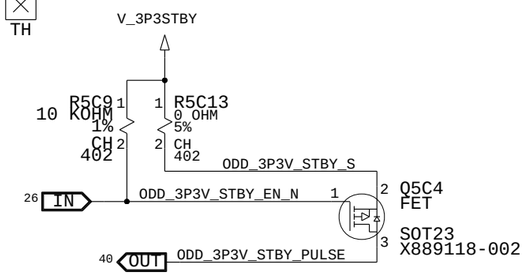


ODD SATA

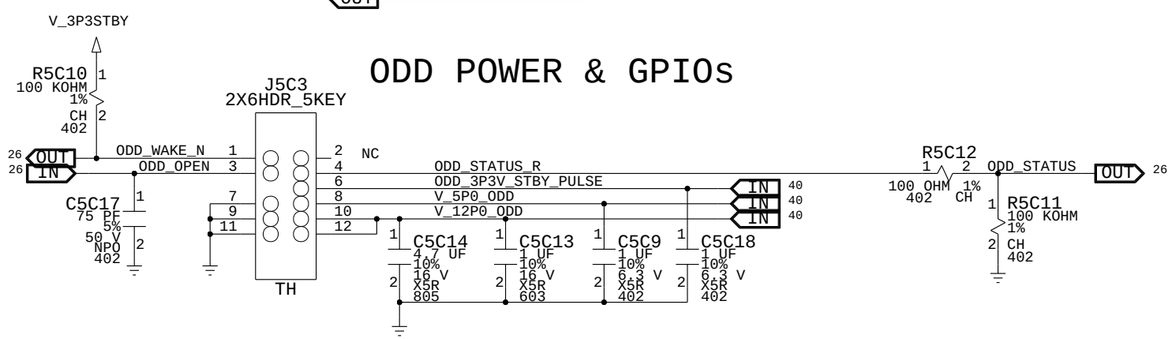


HDD POWER



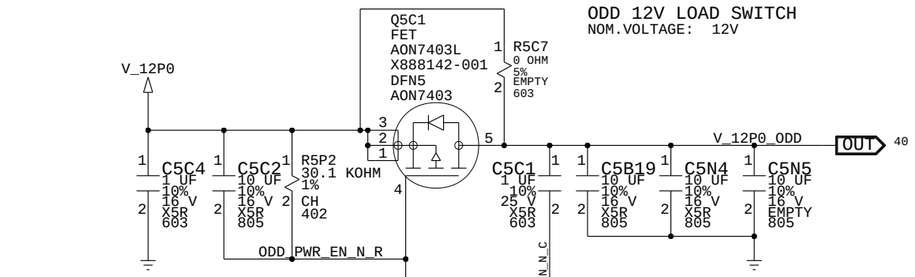


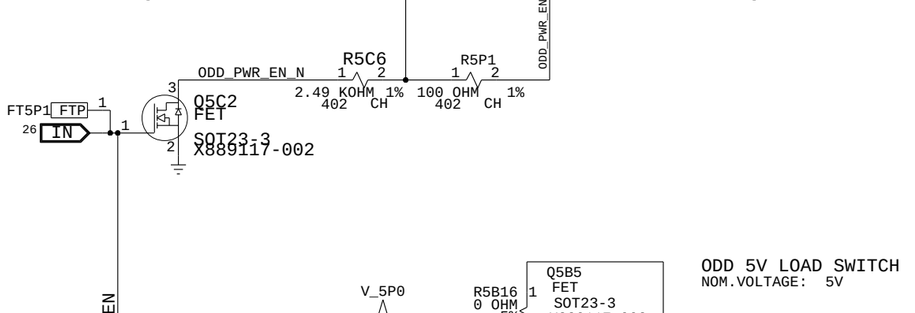
ODD POWER & GPIOs



ODD 12V LOAD SWITCH

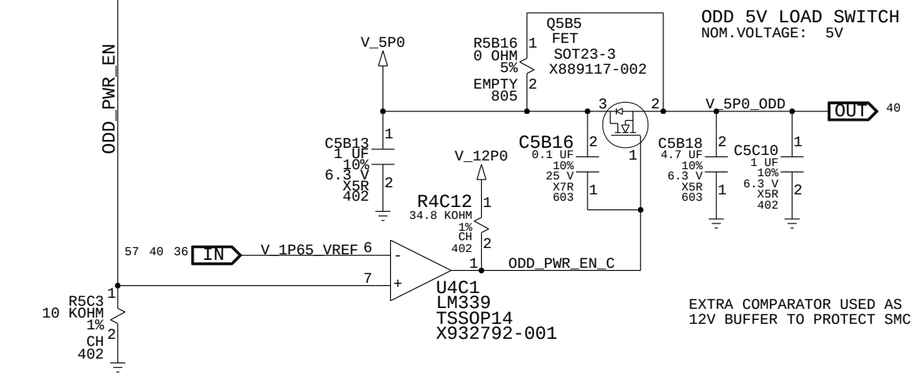
NOM.VOLTAGE: 12V





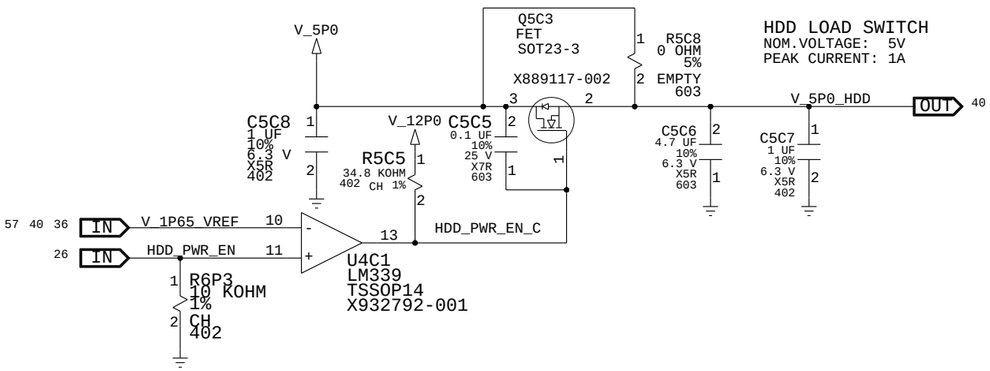
ODD 5V LOAD SWITCH

NOM.VOLTAGE: 5V



HDD LOAD SWITCH

NOM.VOLTAGE: 5V
PEAK CURRENT: 1A



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	40/72	40/72	G	1.01

CONN: FRONT PANEL,FAN, AUDIO

R6F15 10 KOHM 1% 402 CH — PWRSW_N → PWRSW_N_R
R6F16 10 KOHM 1% 402 CH — EJECTSW_N → EJECTSW_N_R
R6G1 2.49 KOHM 1% 402 CH — BINDSW_N_R → BINDSW_N_FPM

V_5P0: C6G2 470 PF 5% 50 V NPO 402; C6F9 100 UF 20% 16 V ELEC TH
V_3P3STBY: C5G2 470 PF 5% 50 V NPO 402; C6F10 100 UF 20% 16 V ELEC TH
V_3P3STBY: R6F14, R6F17, R5F20 100 KOHM 1% CH 402

V_5P0: R5U10 1 KOHM 1% EMPTY 402
LED_NEXUS: R6F13 100 KOHM 1% CH 402; EG6F1 X882235-001 EMPTY 402

J6G1 CONN 2X8HDR_SODIUM X900666-001 TH

1	LED_NEXUS
2	SPARE
3	IR_BLAST
4	PWR_SW_N
5	EJECT_SW_N
6	INT_N (NC)
7	V_3P3STBY
8	V_5P0
9	ACC_RESET
10	IR_RX
11	GND_1
12	USB2_DN
13	USB2_DP
14	GND_2
15	SMBUS_CLK
16	SMBUS_DATA

BINDSW_N_FPM, FRONT_P_IR_OUT_R, PWRSW_N_R, EJECTSW_N_R, ACC_RESET_N, IR_DATA, USB2_ACC_DN, USB2_ACC_DP, SMBUS_CLK, SMBUS_DATA

R6G2 10 KOHM 1% EMPTY 402

EG5G1 X882235-001 EMPTY 402
EG5G2 X882235-001 EMPTY 402
EG6G3 X882235-001 EMPTY 402
EG6G4 X882235-001 EMPTY 402
EG6G2 X882235-001 EMPTY 402
EG6G6 X882235-001 EMPTY 402
EG6F3 X882235-001 EMPTY 402
EG6F2 X882235-001 EMPTY 402
EG6G1 X882235-001 EMPTY 402
EG6G5 X882235-001 EMPTY 402

V_5P0 → R6U2 0 OHM 5% 805 CH → V_SPKR_AMP

AUDIO_PWM_OUT → R6U4 200 OHM 1% 402 CH → AUDIO_PWM_FILT → C6U1 1 UF 10% 10 V X5R 402 → AUDIO_PWM_FILT_C → R6U3 6.19 KOHM 1% 402 CH → AUDIO_PWM_AMP_IN

R4E37 10 KOHM 1% CH 402
C6U2 0.1 UF 10% 25 V X7R 603

AUDIO_AMP_VREF: C6U3 10 UF 10% 16 V X5R 805

U6U1 IC ISD8104 X936772-001 S09
VREF, VIP, VIN, CE, VDD, VOUTP, VOUTN, GND, MPAD
SPKR_AMP_OUT_VP, SPKR_AMP_OUT_VN

V_SPKR_AMP: C6U4 0.1 UF 10% 25 V X7R 603; C6U5 10 UF 10% 16 V X5R 805
R6G7 10 KOHM 1% 402 CH — AUDIO_AMP_CE
C6U6 1000 PF 10% 50 V X7R 402

V_SPKR_AMP: R4F19 47 KOHM 1% 402 EMPTY
R4F22 47 KOHM 1% 402 EMPTY

U4F3 SPKR X913539-001 (POS, NEG)

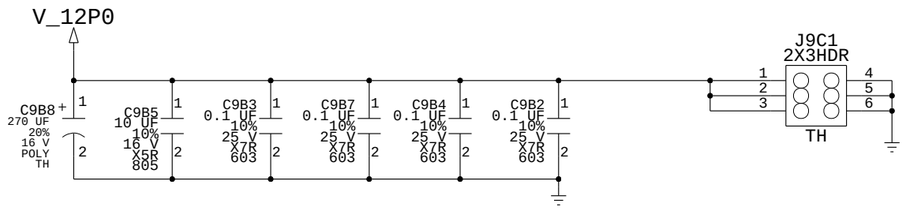
V_12P0_GATED
V_3P3: R9F11 4.7 KOHM 5% CH 402
C9F19 1 UF 10% 16 V X5R 603
C9F11 100 UF 20% 16 V ELEC TH
C9F18 100 UF 20% 16 V EMPTY TH
C9F13 4700 PF 10% 16 V X7R 603

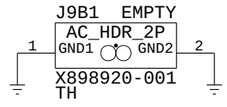
FAN_TACH_IN → R9F13 0 OHM 5% 402 CH → FAN_TACH_IN_R
FAN_PWM → R5T32 33 OHM 1% 402 CH → FAN_PWM_R

J9F1 1X4HDR TH (1 GND, 2 +12V, 3 FAN_TACH_IN_R, 4 FAN_PWM_R)

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	41/72	41/72	G	1.01

CONN: POWER

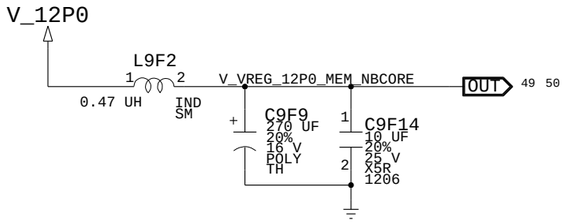




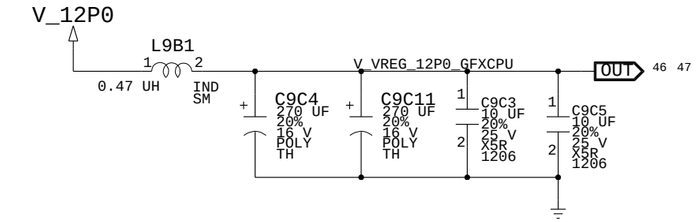
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	42/72	42/72	G	1.01

VREGS: INPUT & OUTPUT FILTERS

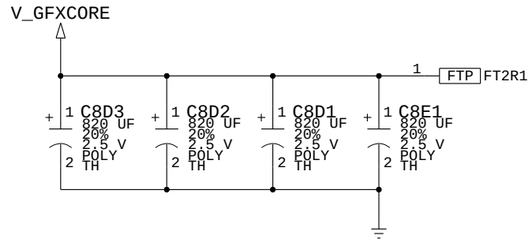
IO/NBCORE INPUT FILTER



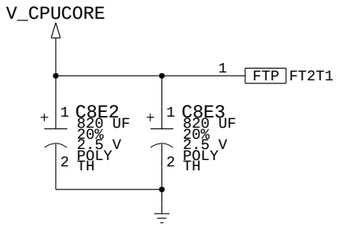
GFX/CPU INPUT FILTER



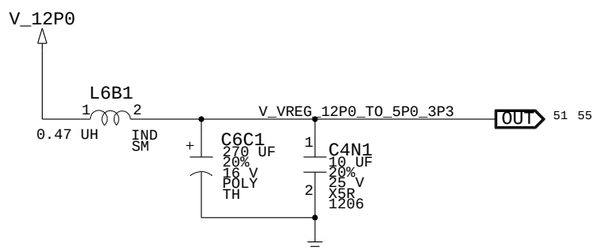
V_GFXCORE OUTPUT FILTER



V_CPUCORE OUTPUT FILTER

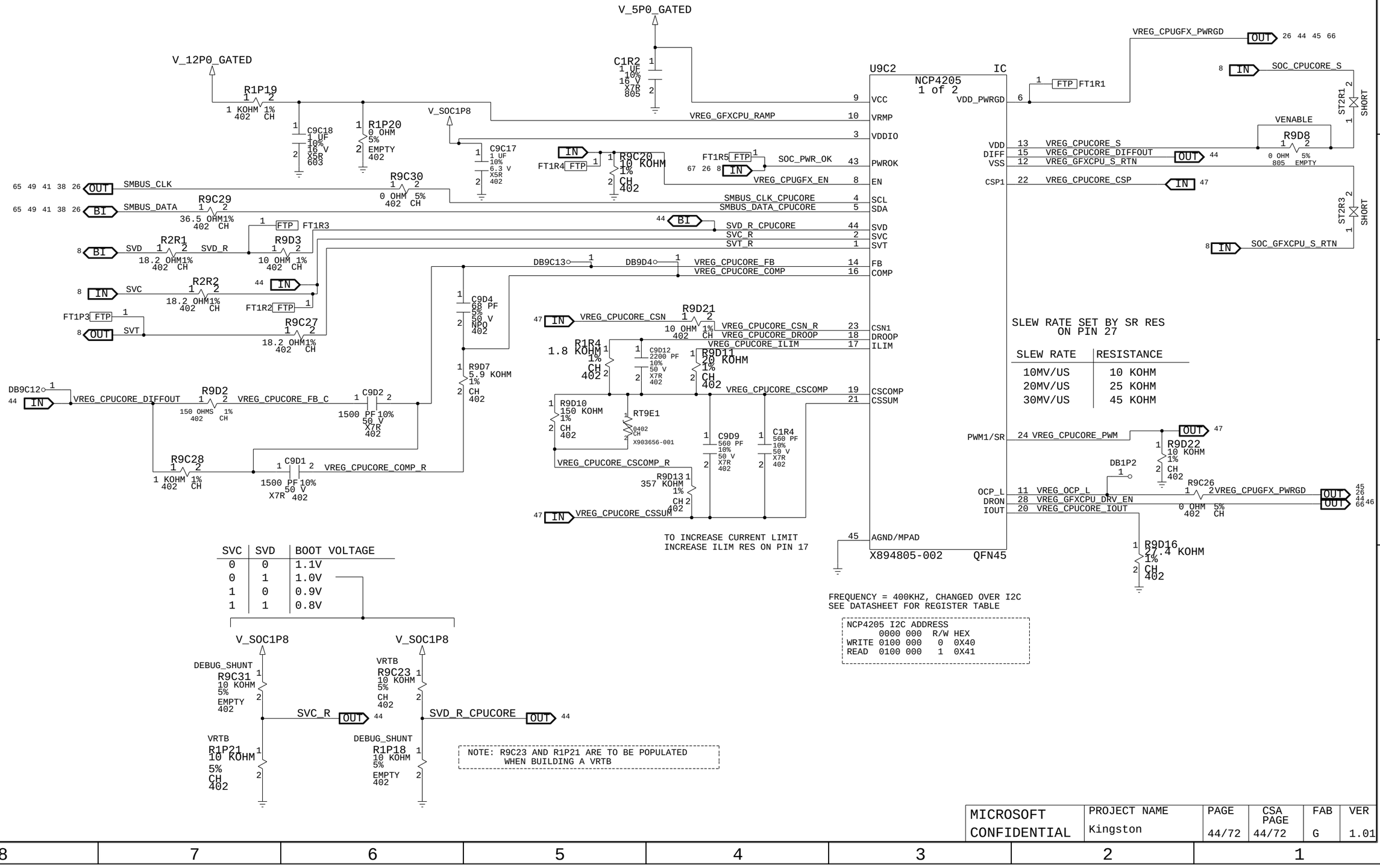


V_5P0 AND V_3P3 INPUT FILTER



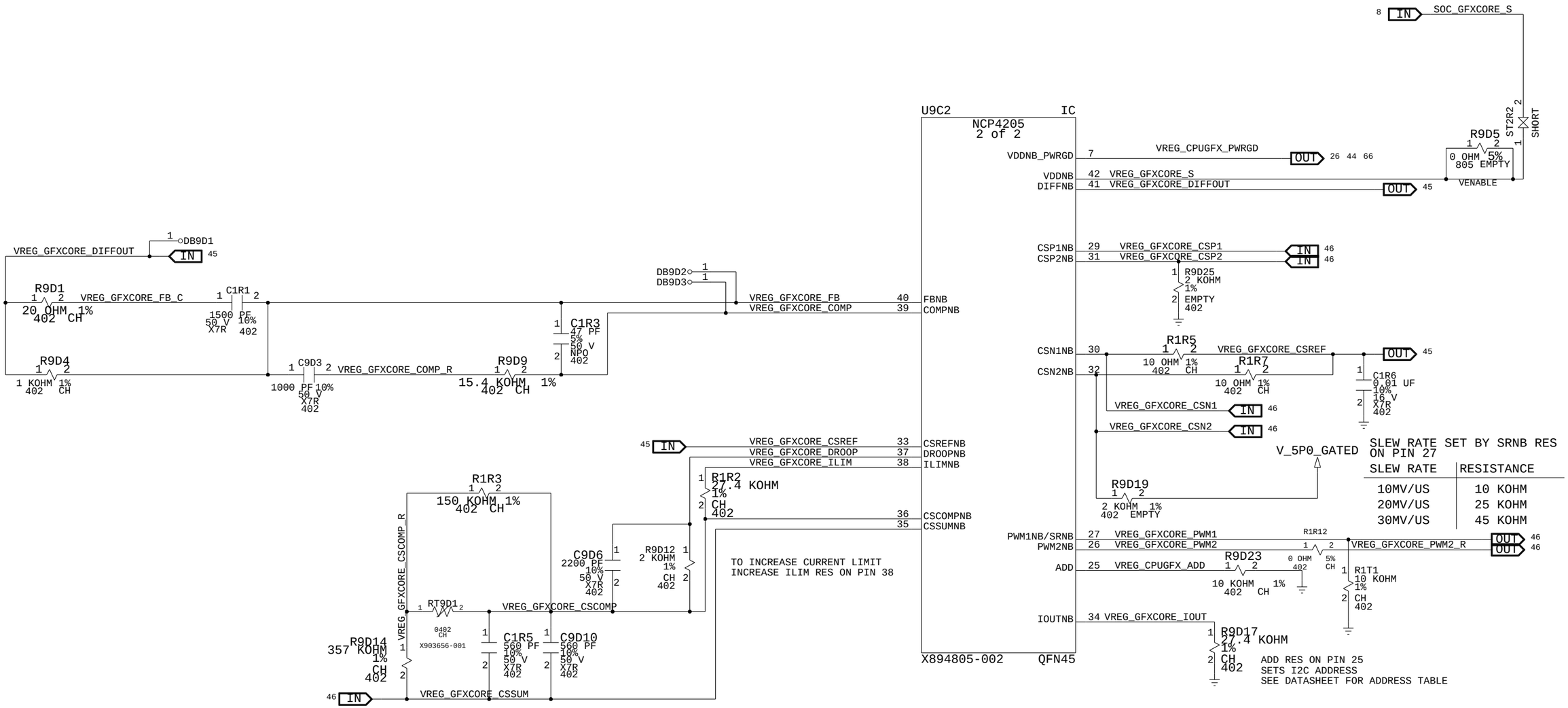
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	43/72	43/72	G	1.01

VREGS: CPUCORE



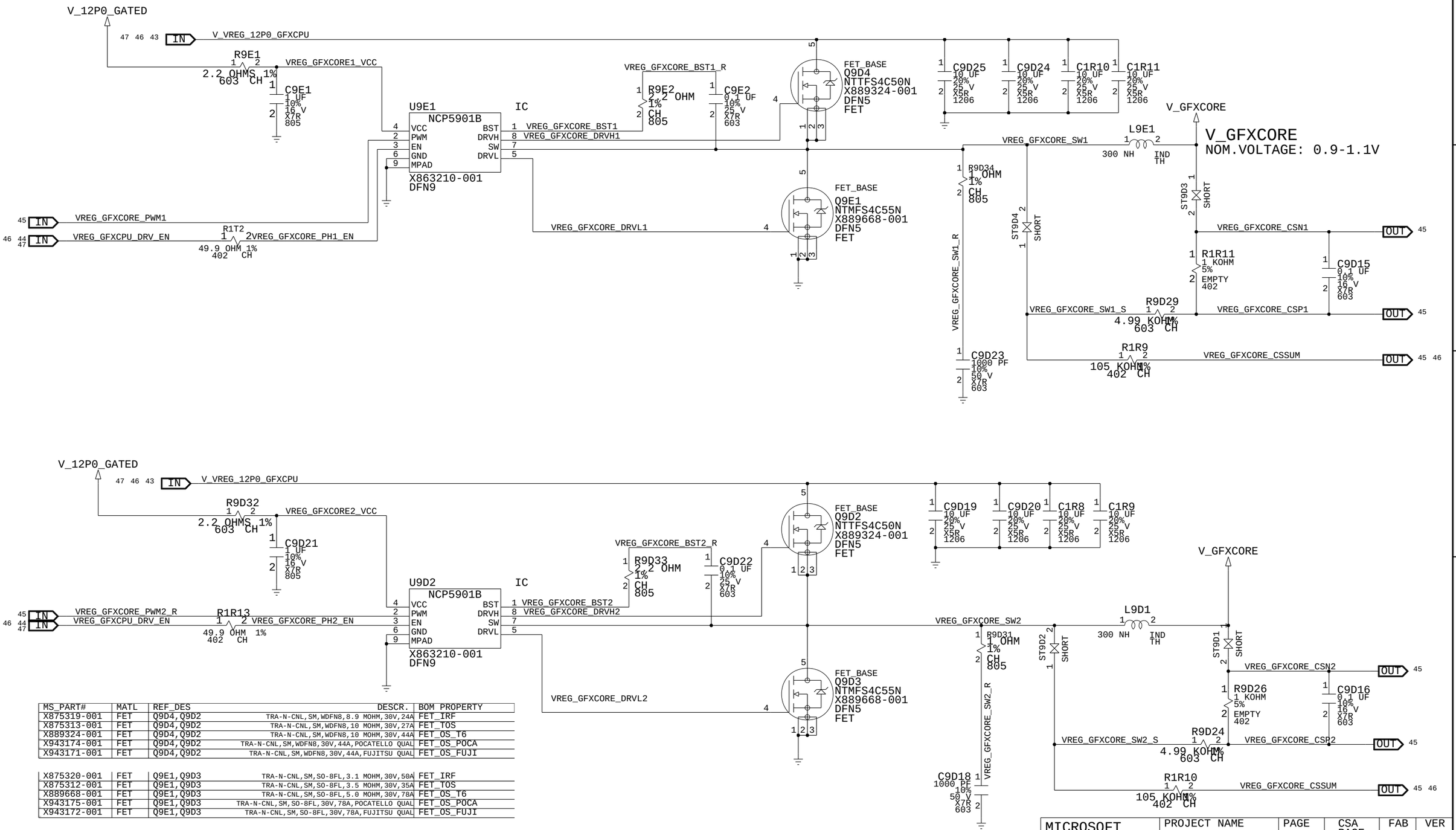
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	44/72	44/72	G	1.01

VREGS: GFXCORE



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	45/72	45/72	G	1.01

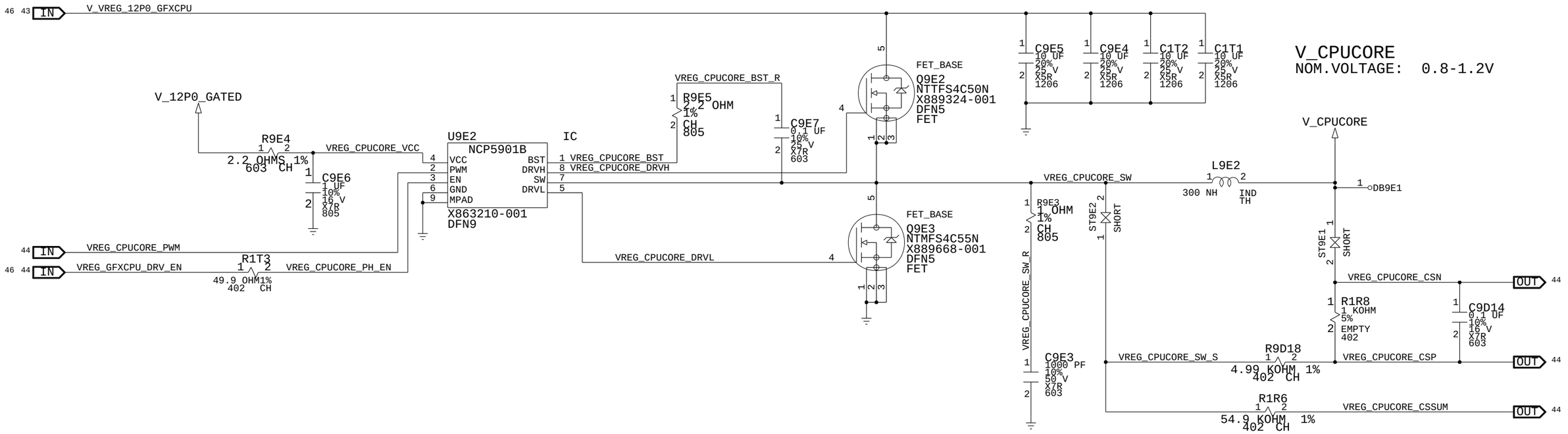
VREGS: GFXCORE OUTPUT PHASE 1 & 2



MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875319-001	FET	Q9D4,Q9D2	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30V,24A	FET_IRF
X875313-001	FET	Q9D4,Q9D2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,27A	FET_TOS
X889324-001	FET	Q9D4,Q9D2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,44A	FET_OS_T6
X943174-001	FET	Q9D4,Q9D2	TRA-N-CNL,SM,WDFN8,30V,44A,POCATELLO QUAL	FET_OS_POCA
X943171-001	FET	Q9D4,Q9D2	TRA-N-CNL,SM,WDFN8,30V,44A,FUJITSU QUAL	FET_OS_FUJI
X875320-001	FET	Q9E1,Q9D3	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30V,50A	FET_IRF
X875312-001	FET	Q9E1,Q9D3	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30V,35A	FET_TOS
X889668-001	FET	Q9E1,Q9D3	TRA-N-CNL,SM,SO-8FL,5.0 MOHM,30V,78A	FET_OS_T6
X943175-001	FET	Q9E1,Q9D3	TRA-N-CNL,SM,SO-8FL,30V,78A,POCATELLO QUAL	FET_OS_POCA
X943172-001	FET	Q9E1,Q9D3	TRA-N-CNL,SM,SO-8FL,30V,78A,FUJITSU QUAL	FET_OS_FUJI

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	46/72	46/72	G	1.01

VREGS: CPUCORE OUTPUT PHASE

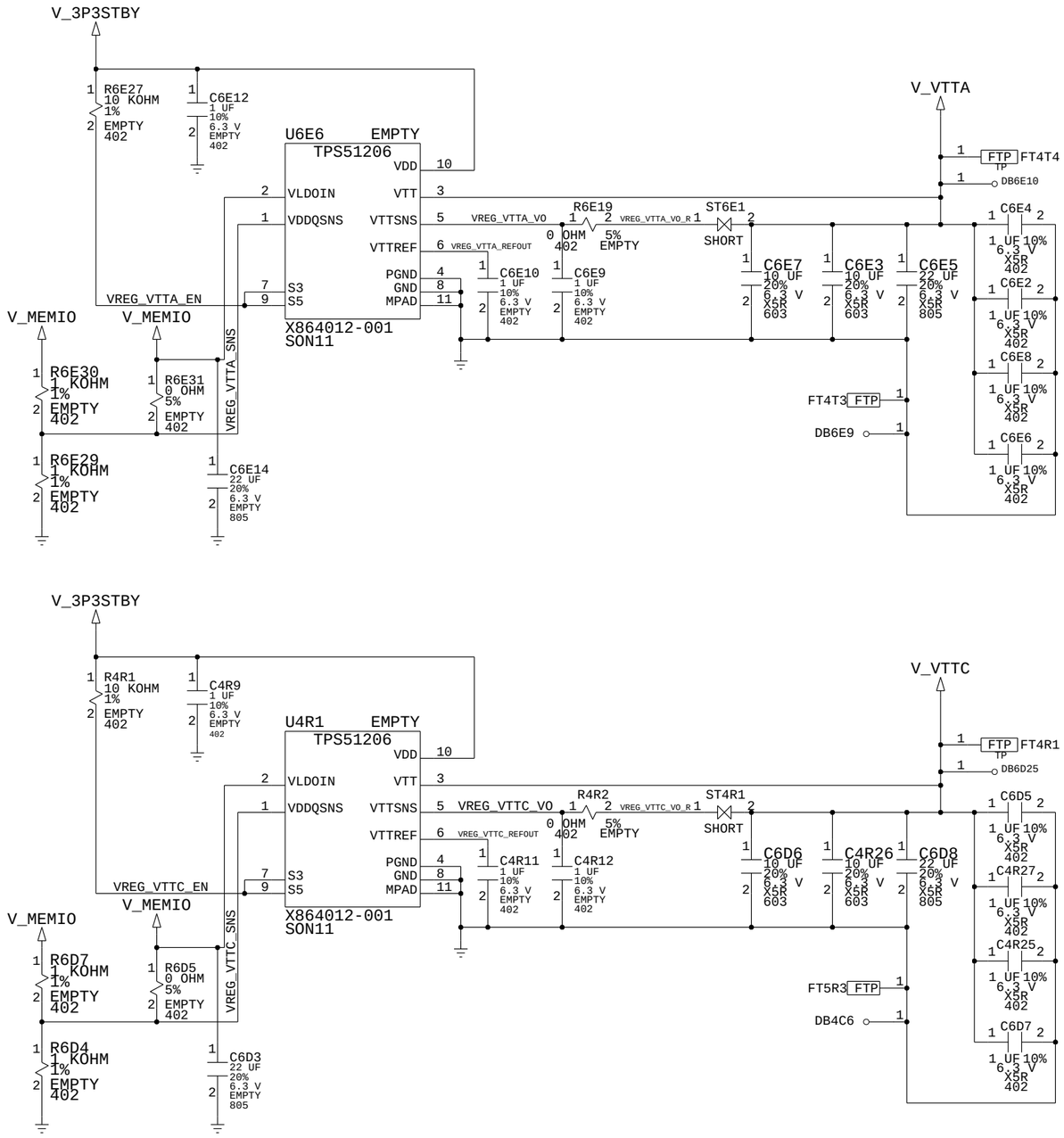


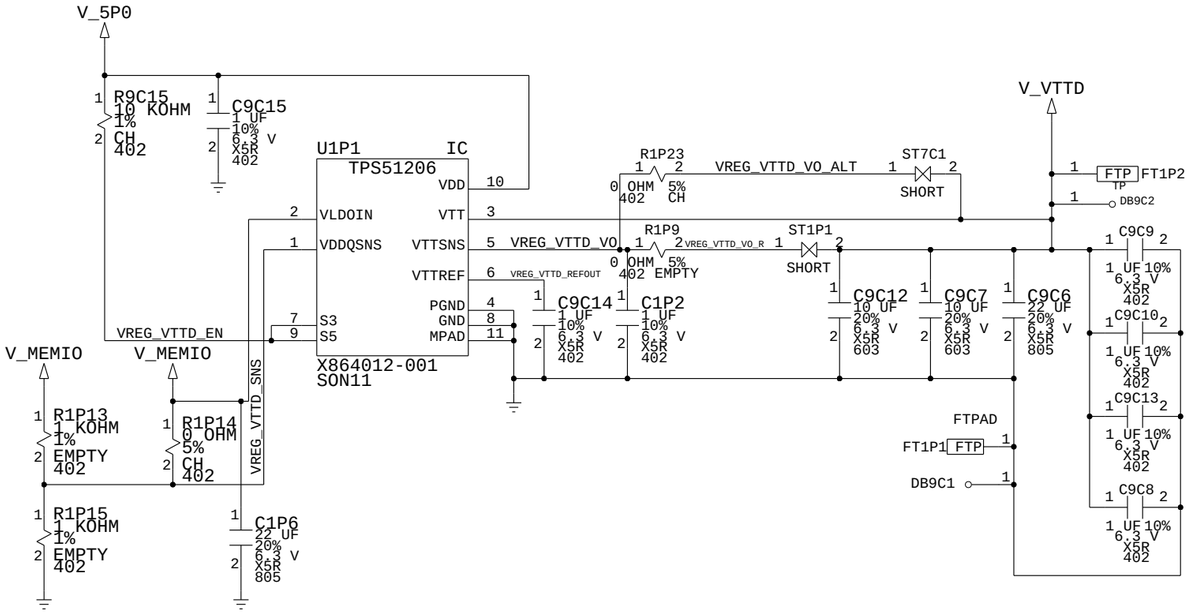
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875319-001	FET	Q9E2	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30V,24A	FET_IRF
X875313-001	FET	Q9E2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,27A	FET_TOS
X889324-001	FET	Q9E2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,44A	FET_OS_T6
X943174-001	FET	Q9E2	TRA-N-CNL,SM,WDFN8,30V,44A,POCATELLO QUAL	FET_OS_POCA
X943171-001	FET	Q9E2	TRA-N-CNL,SM,WDFN8,30V,44A,FUJITSU QUAL	FET_OS_FUJI

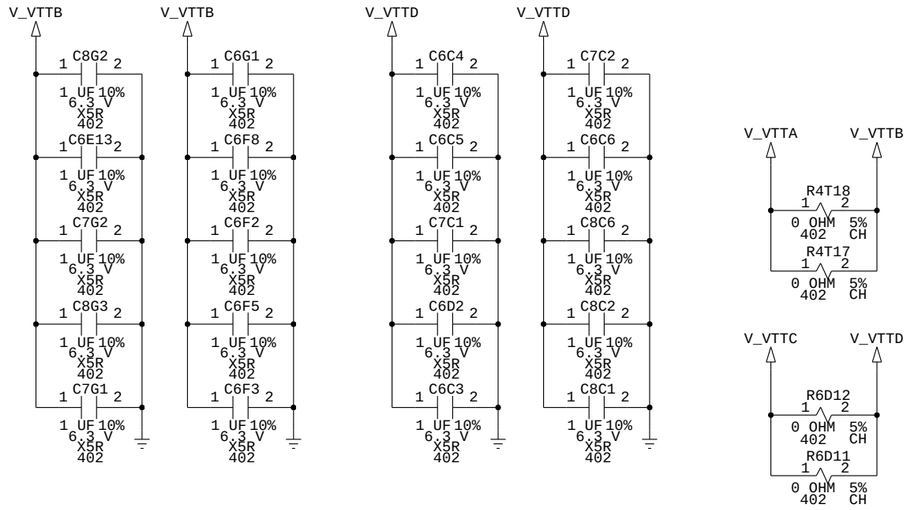
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875320-001	FET	Q9E3	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30V,50A	FET_IRF
X875312-001	FET	Q9E3	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30V,35A	FET_TOS
X889668-001	FET	Q9E3	TRA-N-CNL,SM,SO-8FL,5.0 MOHM,30V,78A	FET_OS_T6
X943175-001	FET	Q9E3	TRA-N-CNL,SM,SO-8FL,30V,78A,POCATELLO QUAL	FET_OS_POCA
X943172-001	FET	Q9E3	TRA-N-CNL,SM,SO-8FL,30V,78A,FUJITSU QUAL	FET_OS_FUJI

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	47/72	47/72	G	1.01

VREGS: VTT TERMINATION

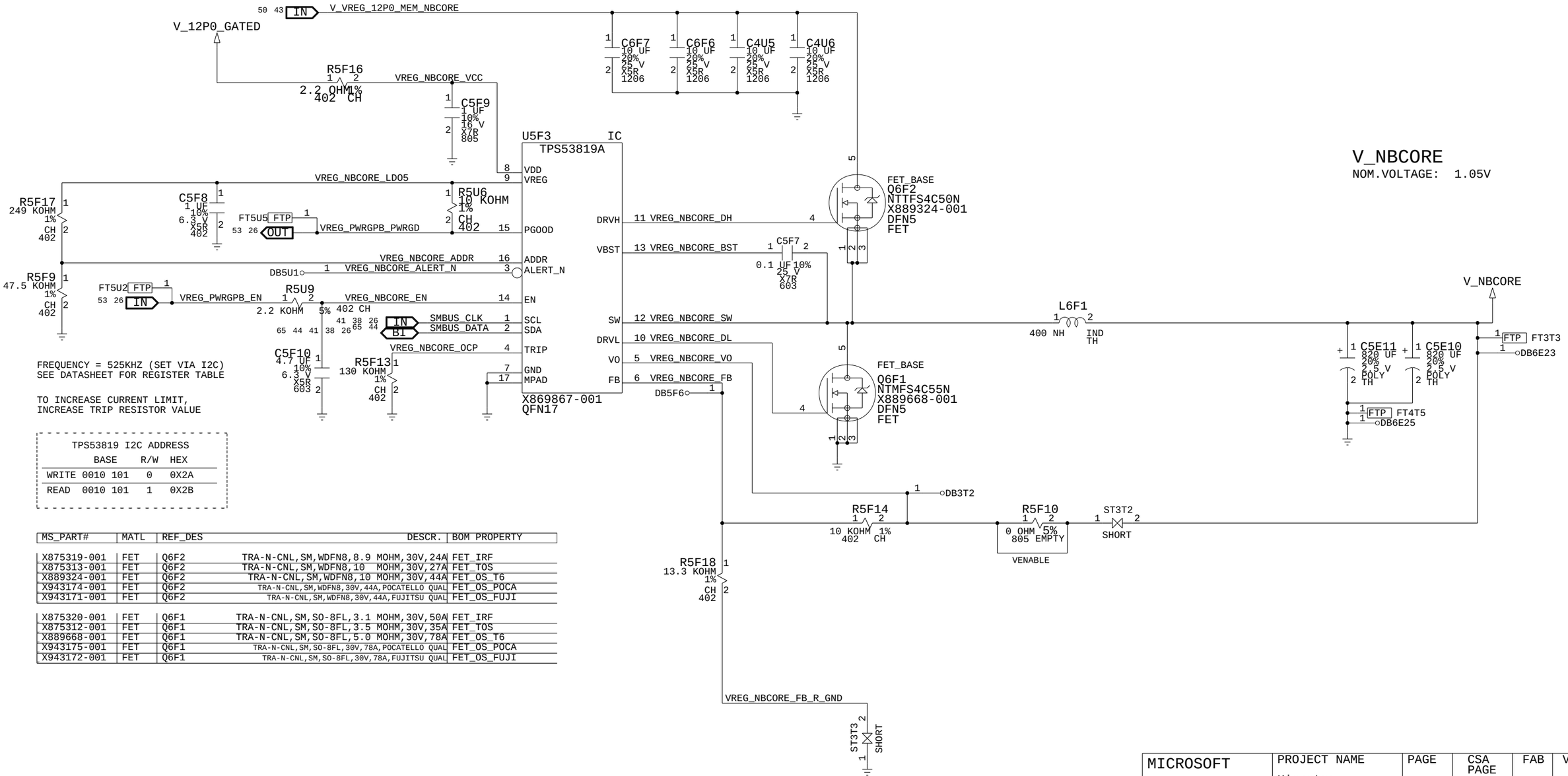






MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	48/72	48/72	G	1.01

VREGS: NBCORE



FREQUENCY = 525KHZ (SET VIA I2C)
SEE DATASHEET FOR REGISTER TABLE

TO INCREASE CURRENT LIMIT,
INCREASE TRIP RESISTOR VALUE

TPS53819 I2C ADDRESS

	BASE	R/W	HEX
WRITE	0010 101	0	0X2A
READ	0010 101	1	0X2B

MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875319-001	FET	Q6F2	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30V,24A	FET_IRF
X875313-001	FET	Q6F2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,27A	FET_TOS
X889324-001	FET	Q6F2	TRA-N-CNL,SM,WDFN8,10 MOHM,30V,44A	FET_OS_T6
X943174-001	FET	Q6F2	TRA-N-CNL,SM,WDFN8,30V,44A,POCATELLO QUAL	FET_OS_POCA
X943171-001	FET	Q6F2	TRA-N-CNL,SM,WDFN8,30V,44A,FUJITSU QUAL	FET_OS_FUJI
X875320-001	FET	Q6F1	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30V,50A	FET_IRF
X875312-001	FET	Q6F1	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30V,35A	FET_TOS
X889668-001	FET	Q6F1	TRA-N-CNL,SM,SO-8FL,5.0 MOHM,30V,78A	FET_OS_T6
X943175-001	FET	Q6F1	TRA-N-CNL,SM,SO-8FL,30V,78A,POCATELLO QUAL	FET_OS_POCA
X943172-001	FET	Q6F1	TRA-N-CNL,SM,SO-8FL,30V,78A,FUJITSU QUAL	FET_OS_FUJI

V_NBCORE
NOM.VOLTAGE: 1.05V

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	49/72	49/72	G	1.01

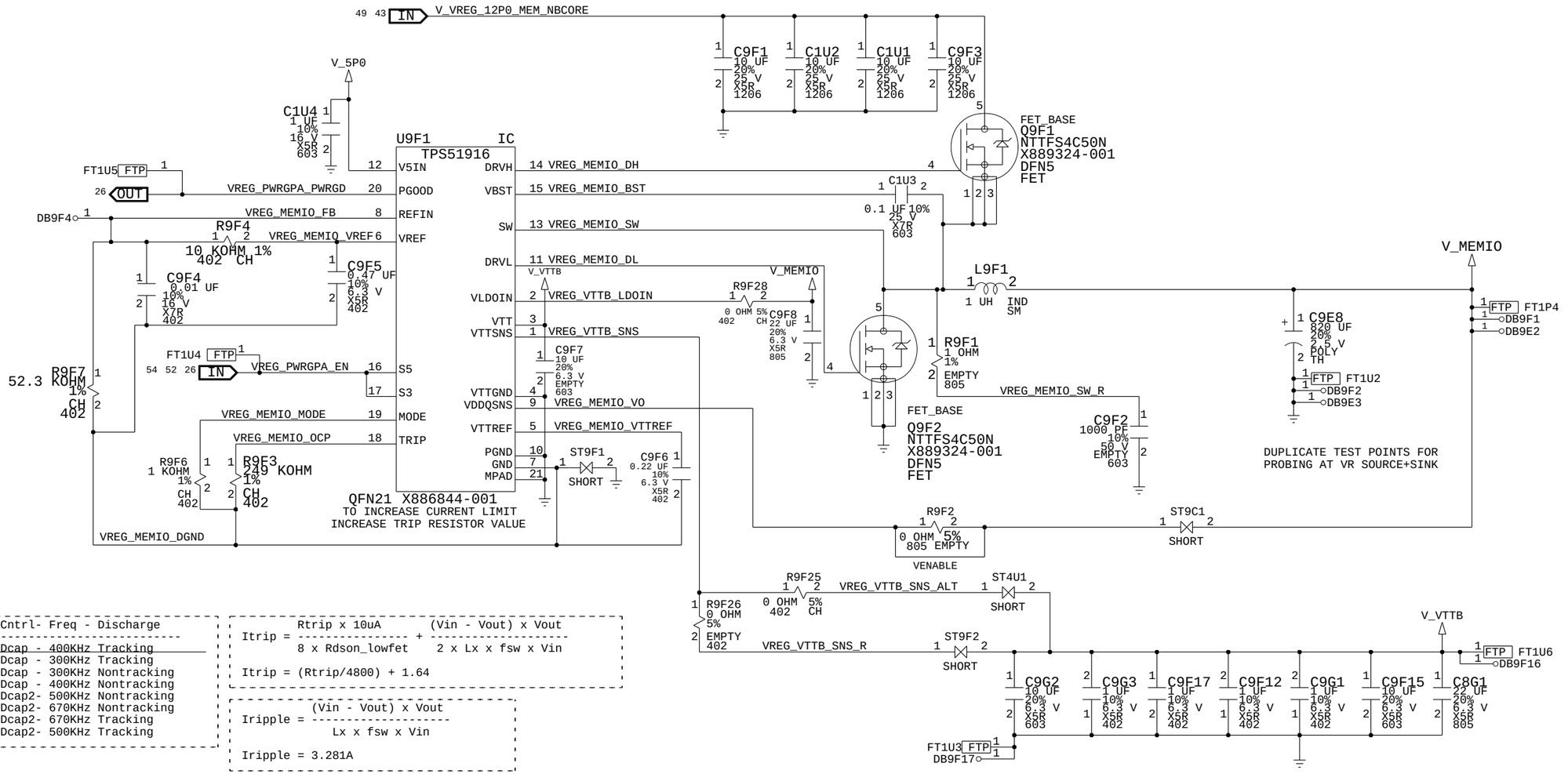
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875319-001	FET	Q9F1	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	FET_IRF
X875313-001	FET	Q9F1	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	FET_TOS
X889324-001	FET	Q9F1	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,44 A	FET_OS_T6
X943174-001	FET	Q9F1	TRA-N-CNL,SM,WDFN8,30V,44A,POCATELLO QUAL	FET_OS_POCA
X943171-001	FET	Q9F1	TRA-N-CNL,SM,WDFN8,30V,44A,FUJITSU QUAL	FET_OS_FUJI

MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X875319-001	FET	Q9F2	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	FET_IRF
X875313-001	FET	Q9F2	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	FET_TOS
X889324-001	FET	Q9F2	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,44 A	FET_OS_T6
X943174-001	FET	Q9F2	TRA-N-CNL,SM,WDFN8,30V,44A,POCATELLO QUAL	FET_OS_POCA
X943171-001	FET	Q9F2	TRA-N-CNL,SM,WDFN8,30V,44A,FUJITSU QUAL	FET_OS_FUJI

Q9F2 MATCHES Q9F1

VREG: MEMIO

V_MEMIO
NOM. VOLTAGE: 1.50



Rmode	Cntrl- Freq - Discharge
200K	Dcap - 400KHz Tracking
100K	Dcap - 300KHz Tracking
68K	Dcap - 300KHz Nontracking
47K	Dcap - 400KHz Nontracking
33K	Dcap2- 500KHz Nontracking
22K	Dcap2- 670KHz Nontracking
12K	Dcap2- 670KHz Tracking
1K	Dcap2- 500KHz Tracking

$$Itrip = \frac{Rtrip \times 10uA}{8 \times Rdson_lowfet} + \frac{(Vin - Vout) \times Vout}{2 \times Lx \times fsw \times Vin}$$

$$Itrip = (Rtrip/4800) + 1.64$$

$$Iripple = \frac{(Vin - Vout) \times Vout}{Lx \times fsw \times Vin}$$

$$Iripple = 3.281A$$

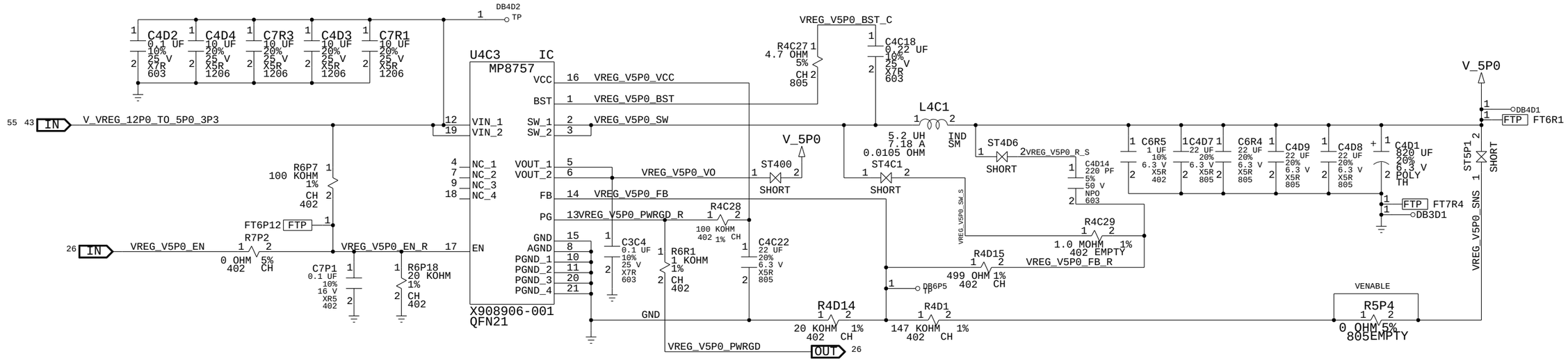
$$Vout = \frac{1.8 \times R2}{R1 + R2} + \frac{Iripple \times ESR}{2}$$

$$Vout = 1.8\ R2/(R1+R2) + 0.010$$

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	50/72	50/72	G	1.01

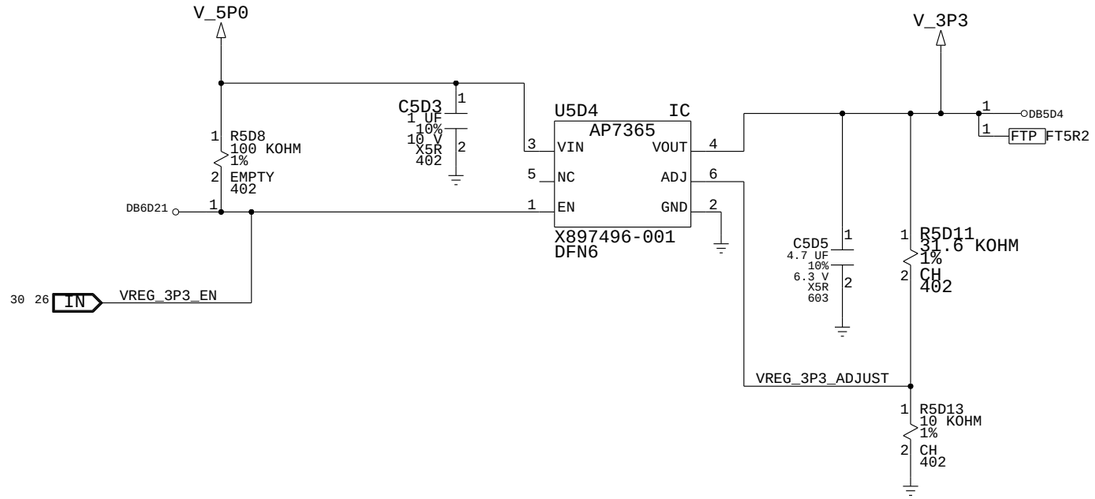
VREGS: V5P0

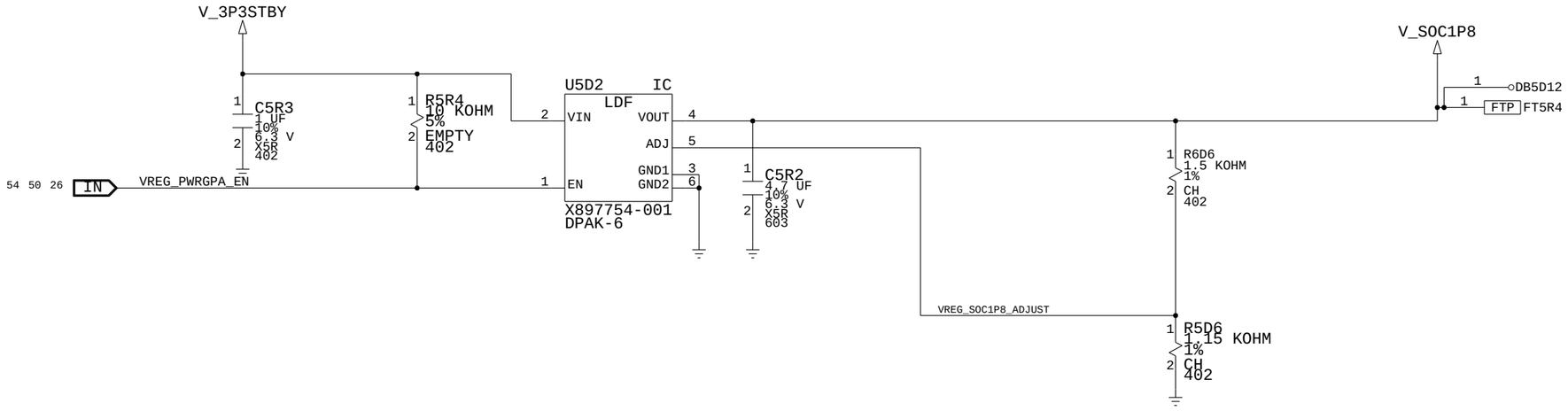
V_5P0
NOM. VOLTAGE: 5.09



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	51/72	51/72	G	1.01

VREGS: V3P3, VSOC1P8

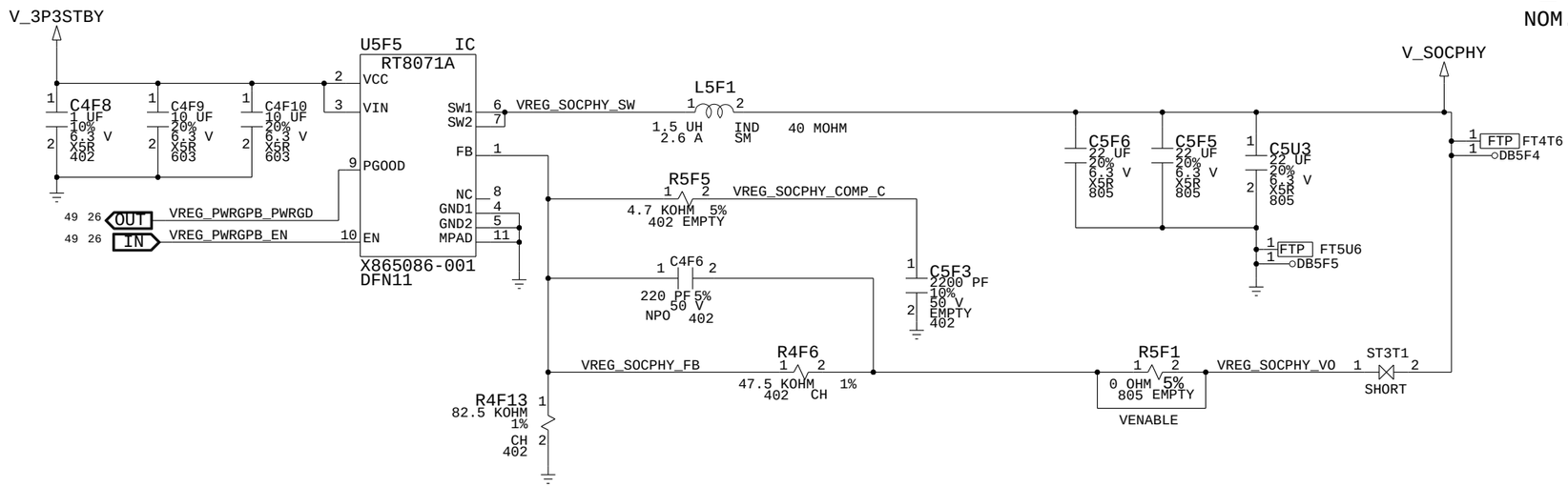


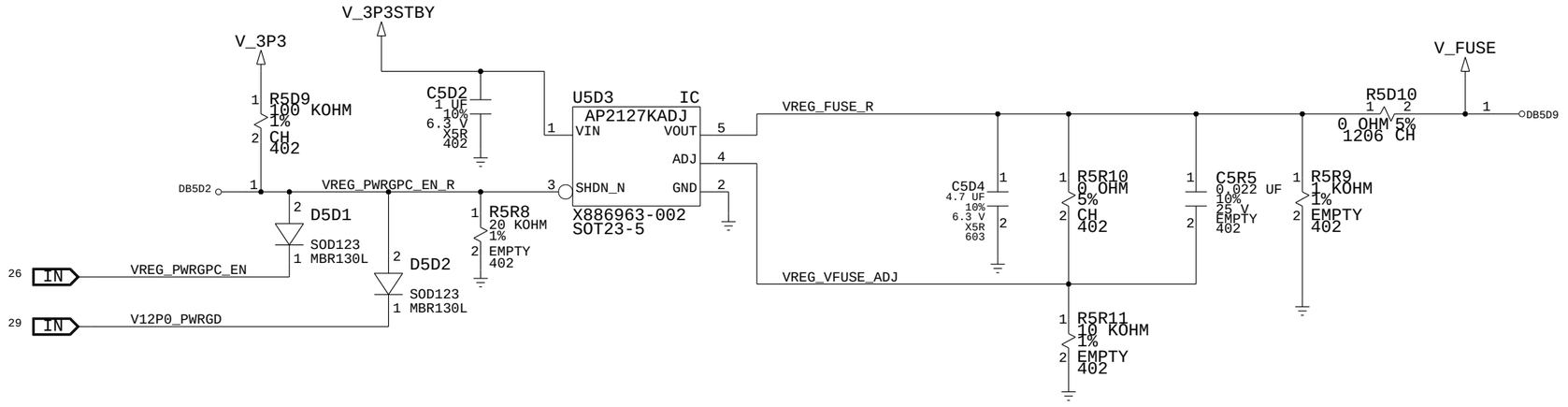


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	52/72	52/72	G	1.01

VREGS: VSOCPHY/VFUSE

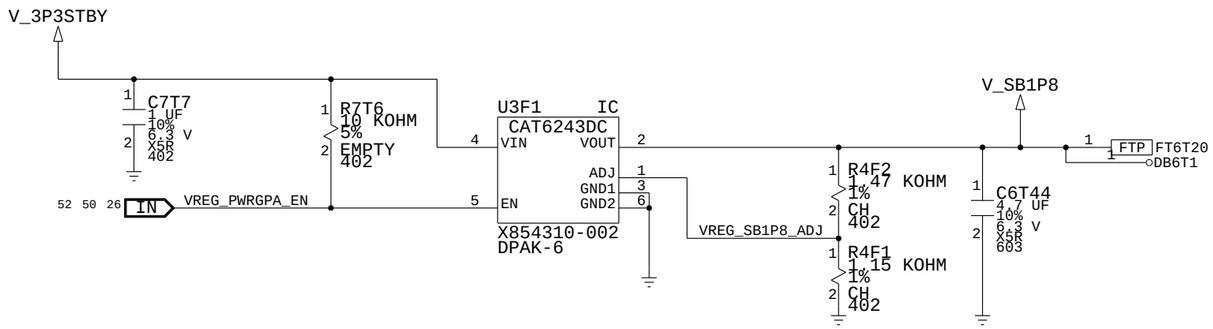
V_SOCPHY
NOM. VOLTAGE: 0.95

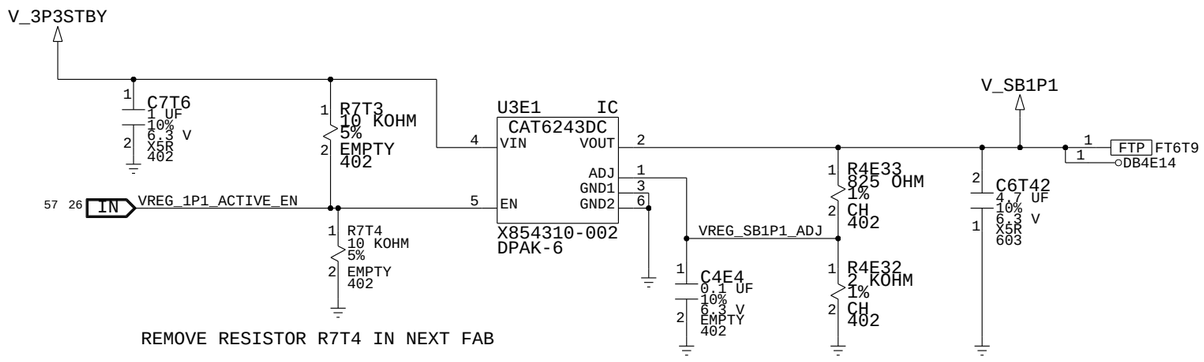




MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	53/72	53/72	G	1.01

VREGS: V_SB1P8, V_SB1P1



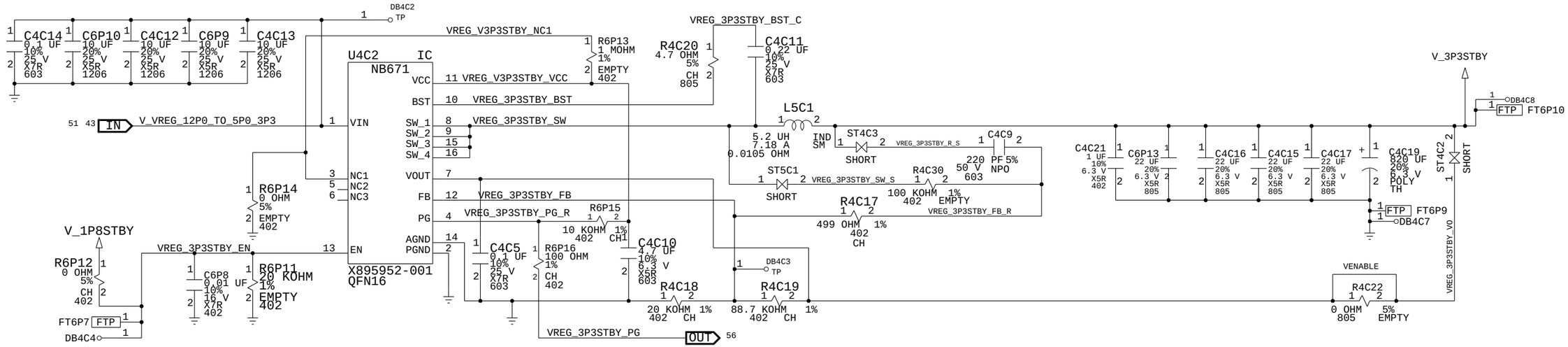


REMOVE RESISTOR R7T4 IN NEXT FAB

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	54/72	54/72	G	1.01

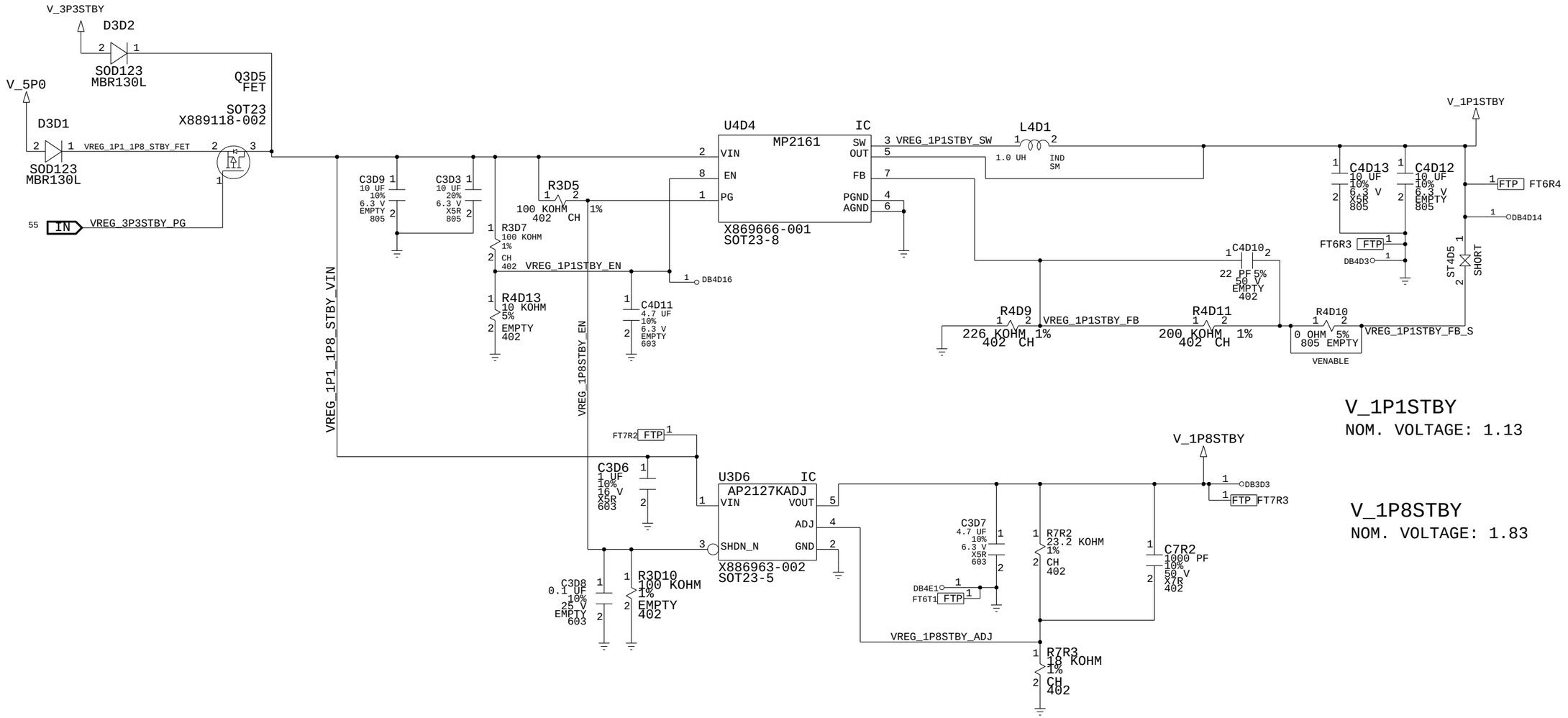
VREGS: V3P3 STANDBY

V_3P3STBY
NOM. VOLTAGE: 3.31



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	55/72	55/72	G	1.01

VREGS: V1P1 STANDBY, V1P8 STANDBY

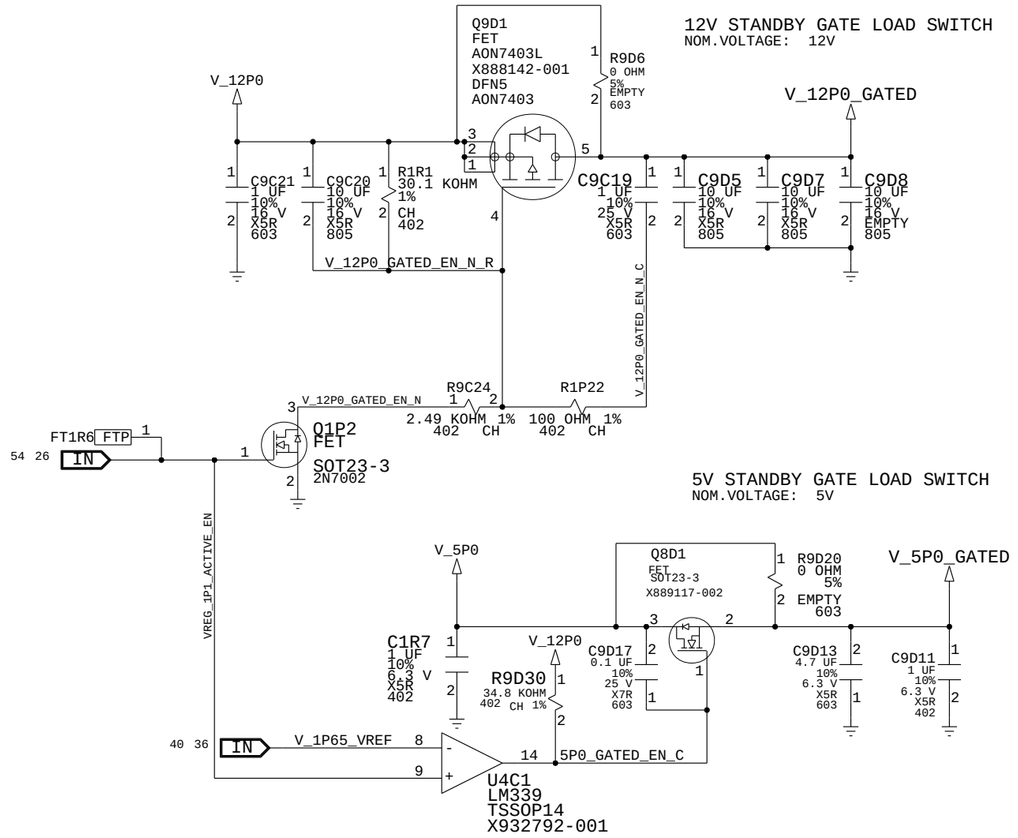


V_1P1STBY
NOM. VOLTAGE: 1.13

V_1P8STBY
NOM. VOLTAGE: 1.83

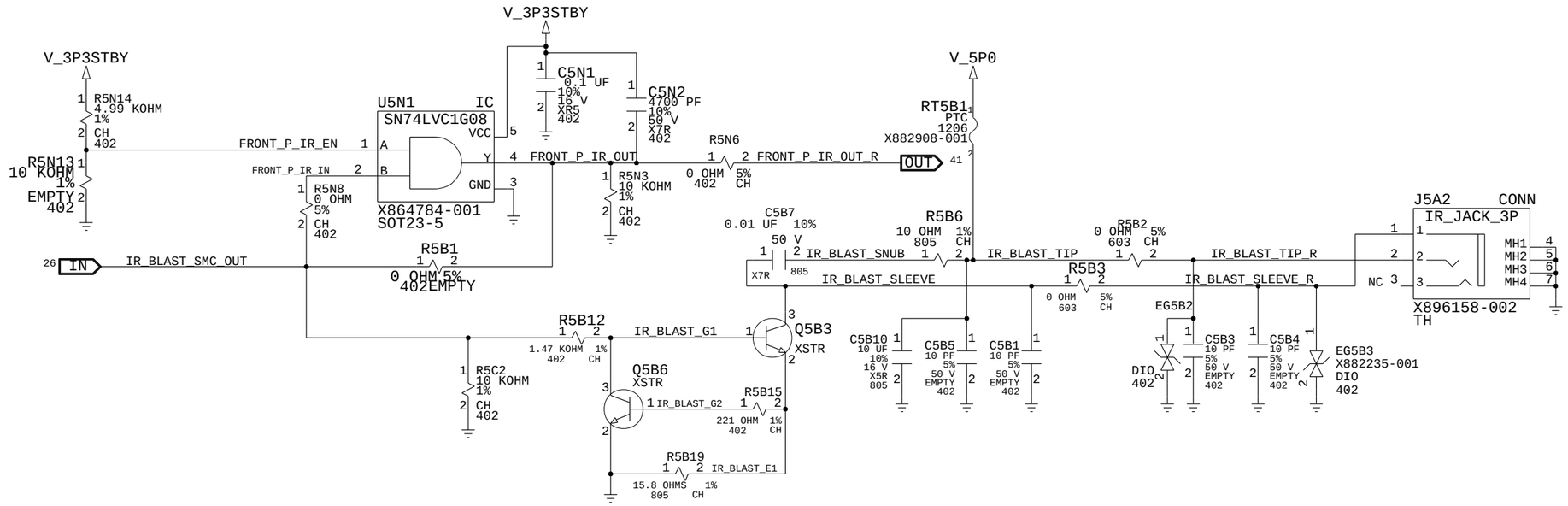
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	56/72	56/72	G	1.01

STANDBY GATES



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	57/72	57/72	G	1.01

IR BLASTER



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	58/72	58/72	G	1.01

I2C

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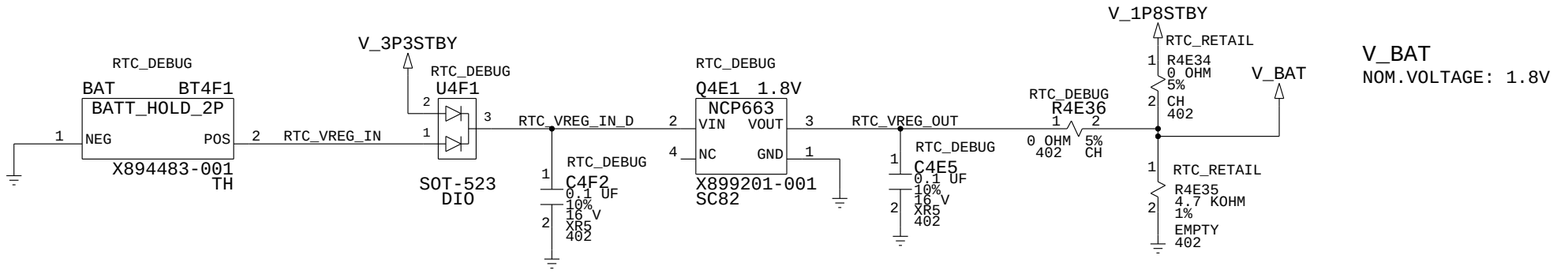
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	59/72	59/72	G	1.01

MARGIN: SOCPHY,SOC1P8,MEMIO,NBCORE

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MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	60/72	60/72	G	1.01

V_BAT



MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X895557-001	BAT	COIN_CELL1	BAT-COIN,OTH,3 V,OTHER,OTH,2032(VARTA QUAL)	BAT_VARTA

VARTA MAXIMUM OPERATING TEMP 70C

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	61/72	61/72	G	1.01

MONITOR: NBCORE,MEMIO

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MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	62/72	62/72	G	1.01

MONITOR: VSOC1P8,VSOCPHY

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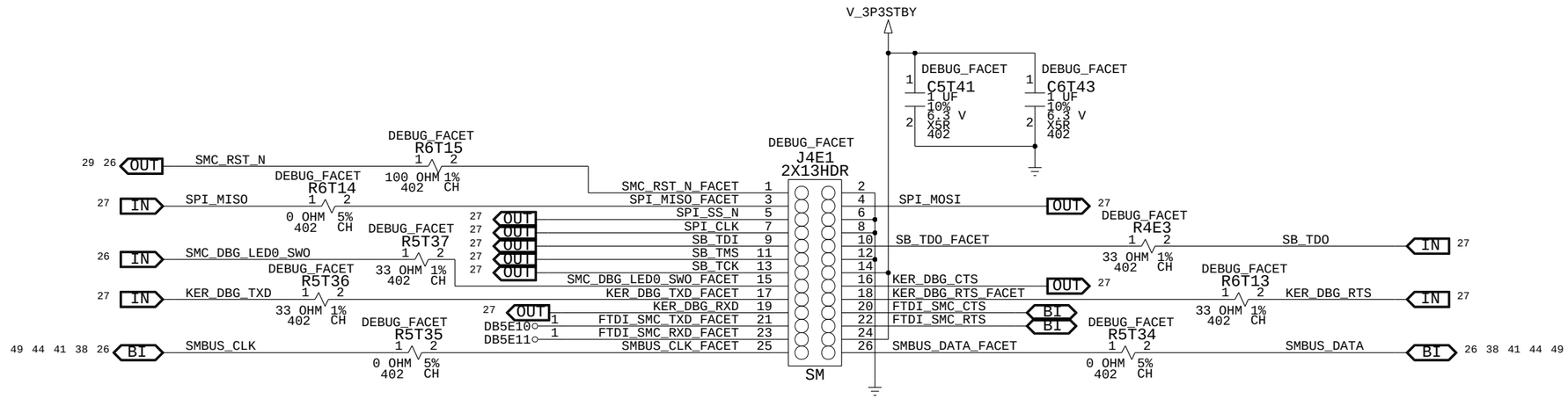
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	63/72	63/72	G	1.01

MONITOR: V12P0

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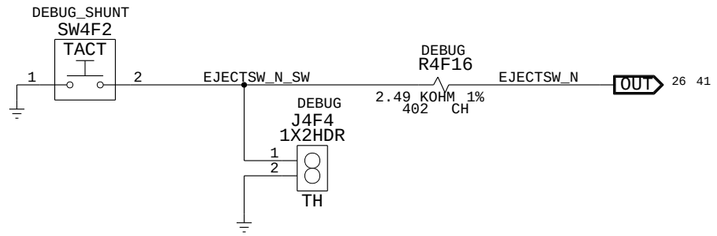
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	64/72	64/72	G	1.01

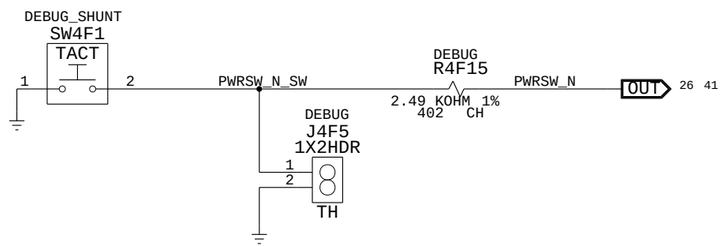
CONN: FACET BOARD

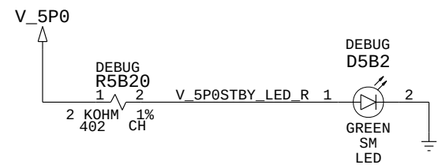


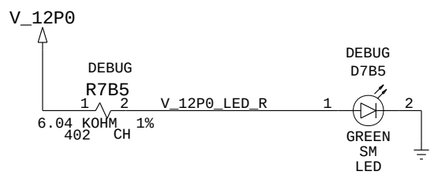
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	65/72	65/72	G	1.01

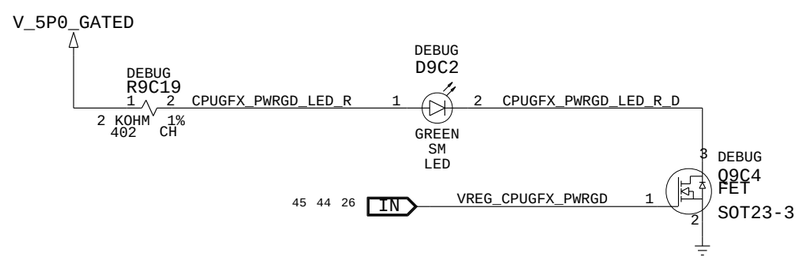
CONN: SWITCHES







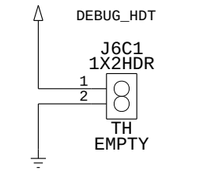


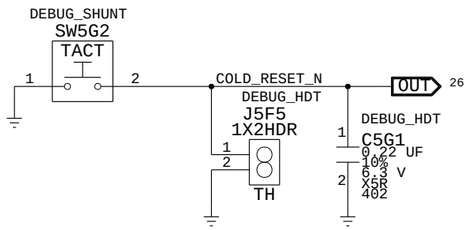


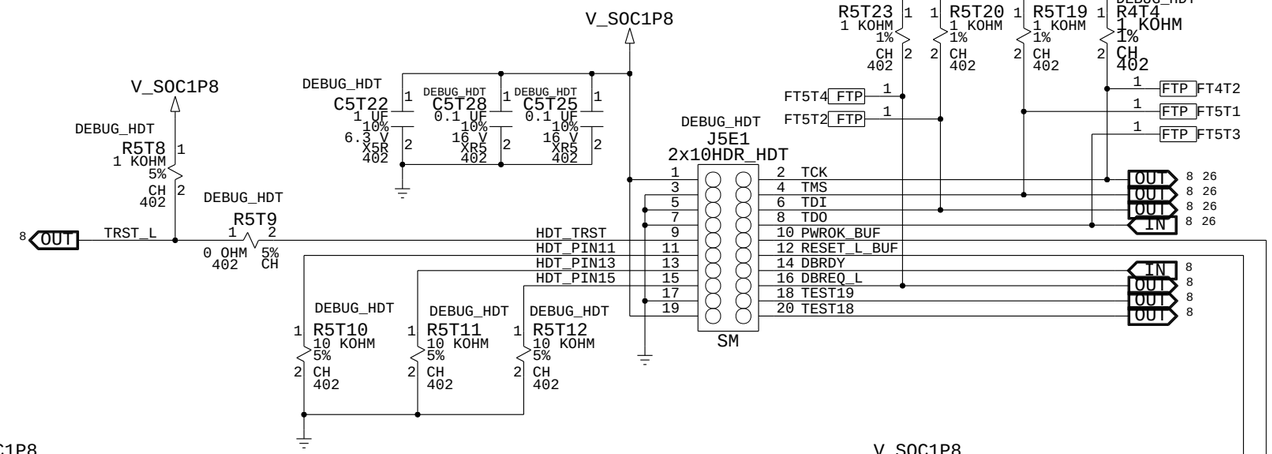
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	66/72	66/72	G	1.01

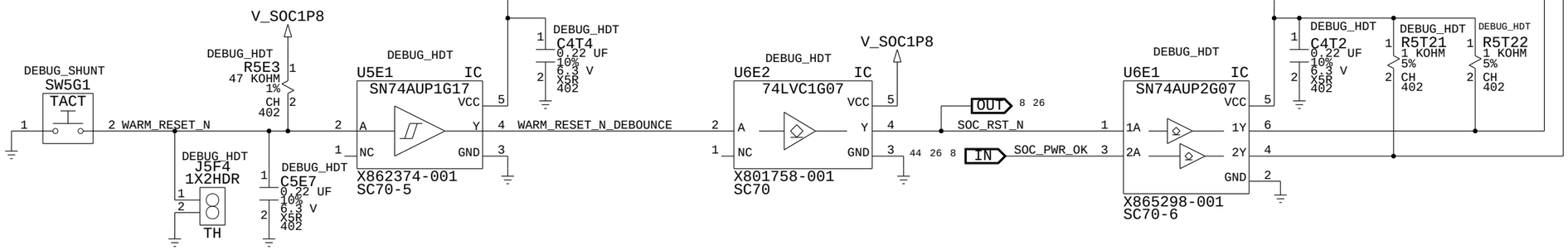
CONN: HDT

NOTE: INSTALL HEADER IF NEEDED
TITAN POWER CONNECTOR



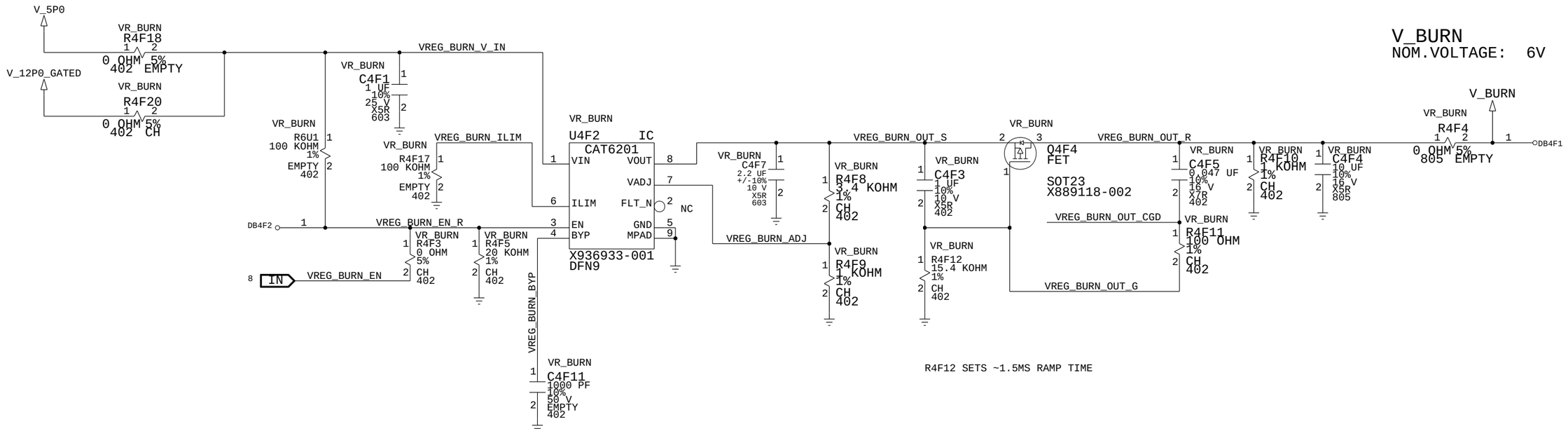






MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	67/72	67/72	G	1.01

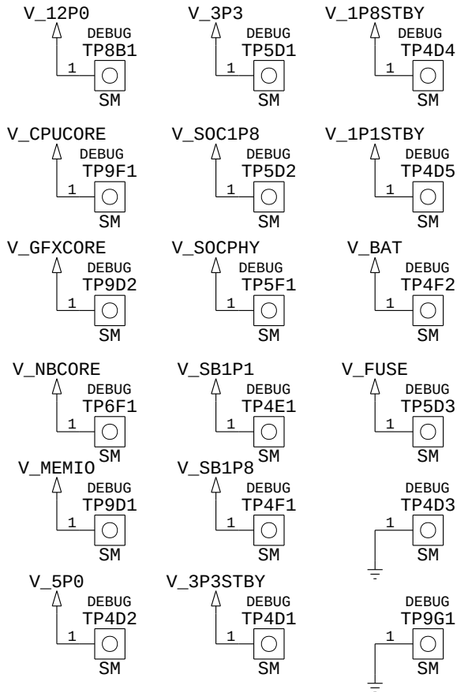
DEBUG: V_BURN



R4F12 SETS ~1.5MS RAMP TIME

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	68/72	68/72	G	1.01

DEBUG: VR HEADERS & TEST POINTS



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	69/72	69/72	G	1.01

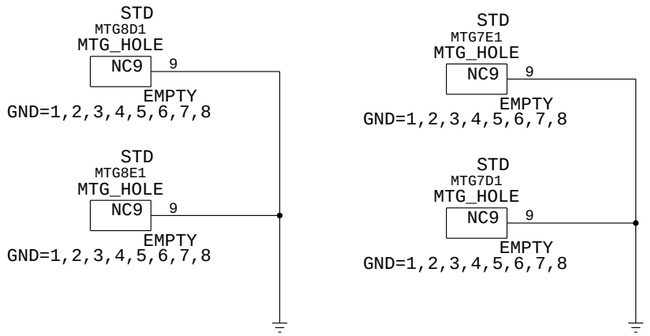
DEBUG: CONNECTORS

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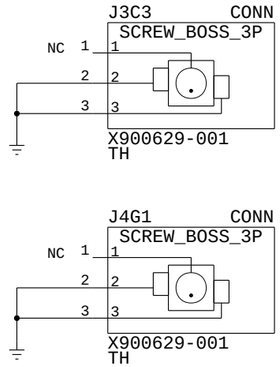
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	VER
	Kingston	70/72	70/72	1.01

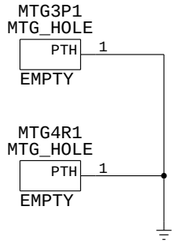
LABELS AND MOUNTING

HEAT SINK MOUNTING HOLES



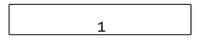
SCREW BOSS





INTELLIGENT SERIAL NUMBER TARGET

LB5G1
X801181-001
LABEL



1375X250_TARGET

MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
X947889-001	FR4	PCB1	PCB-RIGID,RETAIL,FAB G,6L,FR4,OSP	PCB_OSP

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	VER
	Kingston	71/72	71/72	G	1.01

BOM DEFINITIONS

BOM	DEFINTION
AUX	HDMI STUFFING OPTION. NEVER USED IN PRODUCTION. REWORK PURPOSES ONLY
COMMON	ALL COMPONENTS WITH NO BOM PROPERTY
DDC	HDMI STUFFING OPTION. ALWAYS USED IN PRODUCTION
DDR_BASE	DUMMY PLACE HOLDER FOR DDR3 DEVICES. NEVER USE THIS IN THE RECIPE FILE. SELECT ONE OF THESE 3 INSTEAD: DDR_HYNIX, DDR_MICRON, DDR_SAMSUNG
DDR_HYNIX	HYNIX DDR3
DDR_MICRON	MICRON DDR3
DDR_SAMSUNG	SAMSUNG DDR3
DEBUG	COMPONENTS REQUIRED FOR BRING UP & DEBUG
DEBUG_HDT	HDT-RELATED DEBUG COMPONENTS
DEBUG_HDMI	DEBUG HDMI CONNECTOR USED 8L DEBUG BOARDS
DEBUG_SHUNT	COMPONENTS WHICH ARE ON DEBUG BOARDS, BUT ARE REMOVED/SHORTED ON RETAIL
EMMC_BASE	DUMMY PLACE HOLDER FOR EMMC DEVICE & RESISTORS. NEVER USE THIS IN THE RECIPE FILE. SELECT ONE OF THESE INSTEAD: EMMC_HYNIX_20NM, EMMC_HYNIX_5P0, EMMC_HYNIX_1XNM, EMMC_SAMSUNG AND EMMC_TOSHIBA
EMMC_HYNIX_5P0	HYNIX EMMC V5.0 EMMC DEVICE
FET_BASE	DUMMY PLACE HOLDER FOR HIGH AND LOW FETS. NEVER USE THIS IN THE RECIPE FILE. SELECT ONE OF THESE INSTEAD: FET_AOS, FET_OS_T6, FET_STM, OR FET_TOS
FET_IRF	INTERNATIONAL RECTIFIER FETS USED FOR VOLTAGE REGUALTORS
FET_OS_T6	ON-SEMI T6 FETS USED FOR VOLTAGE REGULATORS
FET_STM	STMICROELECTRONICS FETS USED FOR VOLTAGE REGULATORS
FET_TOS	TOSHIBA FETS USED FOR VOLTAGE REGULATORS
GARFIELD	CONTAINS GARFIELD (SOC) RELATED PASSIVE/ACTIVE COMPONENTS
KIC_BASE	DUMMY PLACE HOLDER FOR KIC. NEVER USE THIS IN THE RECIPE FILE. USE ONE OF THESE INSTEAD: KIC_DEV OR KIC_RETAIL
KIC_DEV	DEBUG VERSION OF KRAKEN
KIC_RETAIL	RETAIL VERSION OF KRAKEN
MEM_FIXED	SETS V_MEMIO TO A FIXED VOLTAGE (NON-MARGINED). MUST BE USED IN CONJUNCTION WITH NOT MEM_MM
MEM_MM	ALLOWS V_MEMIO TO BE MARGINED FOR M&M BOARDS. MUST BE USED IN CONJUNCTION WITH NOT MEM_FIXED
PANTHER	CONTAINS PANTHER (SOC) RELATED PASSIVE/ACTIVE COMPONENTS
PANTHER_SOC	PANTHER SYSTEM-ON-CHIP (SOC)
PANTHER_SOC_LP	PANTHER SYSTEM-ON-CHIP (SOC) LOW POWER VERSION
PCB_GI	FAB TYPE: GOLD
PCB_OSP	FAB TYPE: ORGANIC SOLDERABILITY PRESERVATIVE
RTC_RETAIL	RTC CIRCUIT IMPLEMENTATION FOR RETAIL BOARDS
RTC_DEBUG	RTC CIRCUIT IMPLEMENTATION FOR DEBUG BOARDS
SOC_BASE	DUMMY PLACE HOLDER FOR SOC
VR_FIXED	SET ALL VRS TO FIXED VOLTAGES (NON-MARGINED). EXCLUDES V_MEMIO. MUST BE USED IN CONJUNCTION WITH NOT VR_MM
VR_MM	ALLOWS MOST VRS TO BE MARGINED FOR M&M BOARDS. EXCLUDES V_MEMIO. MUST BE USED IN CONJUNCTION WITH NOT VR_FIXED
VRTB	BOOT STRAPPING RESISTOR ONLY TO BE POPULATED WHEN BUILDING VOLTAGE REGULATOR TEST BOARD WHICH CONTAINS NO SOC
WW	CAPACITORS WHICH NEED TO BE NO-STUFFED WHEN AI CPU/GPU SOCKET IS INSTALLED